Northern Nut Growers Association Report of the Proceedings at the Thirty-Seventh Annual Report

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Affiliated with The American Horticultural Society

37th Annual Report

CONVENTION AT WOOSTER, OHIO

SEPTEMBER 3, 4, 5 1946

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Lewis, H. W., c/o Ann Cangero, Roslyn
Little, George, Ripley
Lowerre, James D., 1121 Bedford Ave., Brooklyn 16
* MacDaniels, Dr. L. H., Cornell University, Ithaca
MacEwen, Harold, R. D. 5, Fulton
Maloney Brothers Nursery Co., Inc., Danville
Mevius, William E., E. Church St., Eden
Miller, J. E., R. D. 1, Naples
Mitchell, Rudolph, 125 Riverside Drive, New York 24
Mitchell, Thomas, 16 E. 48th St, New York
* Montgomery, Robert H., 1 E. 44th St., New York
Mossman, Dr. James K., Black Oaks, Ramapo
Newell, P. F., 53 Elm St., Nassau
Oeder, Dr. Lambert R., 551 Fifth Ave., New York
Ohliger, Louis H., R. D. 2, New City
Page, Chas. E., R. D. 2, Oneida
Penning, Tomas, R. D. 3, Box 158, Saugerties
Price, Jacob, Price Theatre Co., 352 West 44th St., New York 18
Price, J., 385 Arbuckle Ave., Cedarhurst, L. I.
Rasmussen, Harry, R. D. 1, 85 Frederick St., E. Syracuse
Rebillard, Frederick, 164 Lark St., Albany 5
Salzer, George, 169 Garford Rd., Rochester
Schlegel, Charles P., 990 South Ave., Rochester
Schlick, Frank, Munnsville
Schmidt, Carl W., 180 Linwood Ave., Buffalo
Schwartz, Mortimer L., 1243 Boynton Ave., Bronx
Sheffield, Lewis F., c/o Mrs. E. C. Jones, Townline Rd., Orangeburg

Slate, Prof. George L., Experiment Station, Geneva
Smith, Gilbert L., State School, Wassaic
Smith, Jay L., Chester
Steiger, Harwood, Red Hook
Stern, Otto, Stern's Nurseries, Geneva
Stern-Montagny, Hubert, Erbonia Farm, Gardiner
Szigo, Alfred, 77-15 A. 37th Ave., Jackson Heights, New York
Timmerman, Karl G., 123 Chapel St., Fayetteville
Waite, Dr. R. H., Willowwaite Moor, Perrysburg
Weis, John F., Jr., R. D. 1, Carter Rd., Fairport
Wichlac, Thaddeus, 3236 Genesee St., Cheektowaga 21
Wilson, Mrs. Ida, Candor
Windisch, Richard P., W. E. Burnet & Co., 11 Wall St., New York
* Wissman, Mrs. F. de R., 9 W. 54th St., New York

NORTH CAROLINA

Dunstan, Dr. R. T., Greensboro College, Greensboro
Finch, Jack R., Bailey
Malcolm, Van R., Celo P. O., Yancey County
Parks, C. H., R. D. 2, Asheville

OHIO

Barden, C. A., 215 Morgan St., Oberlin
Bitler, W. A., 322 McPheron Ave., Lima
Bungart, A. A., Avon
Chapman, Floyd B., 1944 Denune Ave., Columbus 3
Cinadr, Mrs. Katherine, 13514 Coath Ave., Cleveland 20
Clark, R. L., 1184 Melbourne Rd., East Cleveland 12

Clay High School, R. D. 5, Toledo 5
Cole, Mrs. J. R., 163 Woodland Ave., Columbus 3
Cook, H. C., R. D. 1, Box 125, Leetonia
Cranz, Eugene F., Mount Tom Farm, Ira
Crawford, L. E., Sylvarium Gardens, 5499 Columbia Rd., N. Olmsted
Davidson, John, 234 E. 2nd St., Xenia
Davidson, Wm. J., Old Springfield Pike, Xenia
Diller, Dr. Oliver D., Dept. of Forestry, Experiment Sta., Wooster
Dubois, Miss Frances M., 4623 Glenshade Ave., Cincinnati 27
Elliott, Donald W., Rogers
Emch, Frank, Genoa
Evans, Maurice G., 335 S. Main St., Akron 8
Fickes, Mrs. W. R., R. D. 1, Wooster
Foraker, Major C. Merle, 152 Elmwood Ave., Barberton
Foss, H. D., 875 Hamlin St., Akron 2
Franks, M. L., R. D. 1, Montpelier
Frederick, Geo. F., 3925 W. 17th, Cleveland 9
Garden Center of Greater Cleveland, 11190 East Blvd., Cleveland
Gardner, Richard F., 1474 Wagar Ave., Cleveland 7
Gauly, Dr. Edward, 1110 Euclid Ave., Cleveland
Gerber, E. P., Kidron
Gerhardt, Gustave A., 3125 Jefferson Ave., Cincinnati
Gerstenmaier, John A., 13 Pond S. W., Massilon
Goss, C. E., 922 Dover Ave., Akron 2
Gray, G. A., 3317 Jefferson Ave., Cincinnati 20
Hamlin, Howard E., 1945 Waltham Rd., Columbus 8
Haydeck, Carl, 3213 West 73rd St., Cleveland 2
Headapohl, Miss Marjean, R. D. 2, Wapakoneta
Hill, Dr. Albert A., 4187 Pearl Rd., Cleveland
Hoch, Gordon F., 6292 Glade Ave., Cincinnati

Holley, Dr. C. J., 11 Elm St., Bridgeport
Hunt, Kenneth W., Yellow Springs
Irish, Charles F., 418 E. 105th St., Cleveland
Jacobs, Homer L., Davey Tree Expert Co., Kent
Jacobs, Mason, 3003 Jacobs Rd., Youngstown
Jacque, John V., 13722 N. Drive, Cleveland 5
Kappel, Owen, Bolivar
Kintzel, Frank M., 2506 Briarcliffe Ave., Cincinnati 13
Kirby, R. L., Box 131, R. 1, Sharonville
Kratzer, George, Kidron
Krok, Walter P., 925 W. 29th St., Lorain
Laditka, Nicholas G., 5322 Stickney Ave., Cleveland 9
Lashley, Chas. V., 216 S. Main, Wellington
Lehmann, Carl, Union Trust Bldg., Cincinnati
Livezey, Albert J., Barnesville
Madson, Arthur E., 13608-5th Ave., E. Cleveland 12
McBride, William B., 2398 Brandon Rd., Columbus 8
Meikle, William J., 730 Thornhill Dr., Cleveland
Metzger, A. J., 724 Euclid Ave., Toledo 5
Miller, Arthur R., R. D. 4, Wooster
Mutchler, Glenn M., Box 10, Massillon
Neff, Wm., Martel
Nicolay, Chas., 2259 Hess Ave., Cincinnati 11
Oches, Norman M., R. D. 2, Brunswick
Olney High School, R. D. 1, Eggleston Rd., Toledo 5
Osborn, Frank C., 4040 W. 160th St., Cleveland
Pomerene, W. H., Coshocton
Poston, E. M., Jr., 2640 E. Main, Columbus
Rowe, Stanley M., R. D. 1, Box 83, Cincinnati 27
Scarff's Sons, W. N., New Carlisle

Schaufelberger, Hugo S., R. D. 2, Sandusky
Shelton, Dr. E. M., 1468 W. Clifton Blvd., Lakewood 7
Sherman, L. Walter, Mahoning Co., Exp. Farm, Canfield
Shessler, Sylvester M., Genoa
Silvis, Raymond E., 1725 Lindbergh Ave., N. E., Massillon
Soliday, E. C., 834 Madison Ave., Lancaster
Southart, Dr. A. F., 24-1/2 South Main St., Mt. Gilead
Smith, Sterling A., 630 W. South St., Vermilion
Spring Hill Nurseries Co., Tipp City
Stocker, C. P., Lorain Products Corp., 1122 F St., Lorain
Sylvarium Gardens, L. E. Crawford, 5499 Columbia Rd., N. Olmsted
Thomas, W. F., 406 S. Main St., Findlay
Toops, Herbert A., 1430 Cambridge Blvd., Columbus
Urban, George, 4518 Ardendale Rd., South Euclid 21
Van Voorhis, J. F., 215 Hudson Ave., Apt. B-1, Newark
Walker, Carl F., 2851 E. Overlook Rd., Cleveland
Weaver, Arthur W., 318 Oliver St., Toledo 4
* Weber, Harry, R. Esq., 123 E. 6th St., Cincinnati
Weber, Mrs. Martha R., R. D. 1, Morgan Rd., Cleves
Weibel, A. J., 4130 Florida Ave., Cincinnati 23
Whitmer High School, 5530 Whitmer Drive, Toledo 12
Willett, Dr. G. P., Elmore
Wischhusen, J. F., 15031 Shore Acres Dr., N.E., Cleveland 10
Yates, Edward W., 3108 Parkview Ave., Cincinnati 13
Yoder, Emmet, Smithville

OKLAHOMA

Hirschi's Nursery, 414 N. Robinson, Oklahoma City
Hubbard, Orie B., Kingston

Hughes, C. V., 5600 N. W. 16-R No. 2, Box 564, Oklahoma City 8
Jarrett, C. F., 2208 W. 40th, Tulsa
Meek, E. B., R. D. 2, Wynnewood
Pulliam, Gordon, 407 Osage Ave., Bartlesville
Ruhlen, Dr. Chas. A., 114 W. Steele, Cushing
Swan, Oscar E., Jr., 1226 E. 30th St., Tulsa 5

OREGON

Borland, Robert E., 219 Mill St., Silverton
Carlton Nursery Co., Forest Grove
Dohanian, S. M., P. O. Box 246, Eugene
Flanagan, George C., 909 Terminal Sales Bldg., Portland
Miller, John E., R. D. 1, Box 312-A, Oswego
Russ, E., R. D. 1, Halsey
Schuster, C. E., Horticulturist, Cervallis

PENNSYLVANIA

Allaman, R. P., R. D. 1, Harrisburg
Anundson, Lester, 2630 Chestnut St., Erie
Banks, H. C., R. D. 1, Hellertown
Barnhart, Emmert M., R. D. 4, Waynesboro
Beard, H. G., R. D. 1, Sheridan
Blair, Dr. G. D., 702 N. Homewood Ave., Pittsburgh
Bowen, John C., R. D. 1, Macungie
Breneiser, Amos P., 427-N. 5th St., Reading
Brenneman, John S., R. D. 6, Lancaster
Brown, Morrison, Carson Long Military Academy, New Bloomfield
Buckman, C. M., Schwenkville

Catterall, Karl P., 734 Frank St., Pittsburgh 10
Clarke, Wm. S., Jr., Box 167, State College
Creasy, Luther P., Catawissa
DeHaven, Edwin, 404 Wall Ave., Pitcairn
Dewey, Richard, Box 41, Peckville
Dible, Samuel E., R. D. 3, Shelocta
Diefenderfer, C. E., 918 Third St., Fullerton
Driver, Warren M., R. D. 4, Bethlehem
Ebling, Aaron L., R. D. 2, Reading
Etter, Fayette, P. O. Box 57, Lehmasters
Gardner, Ralph D., Box 425, Colonial Park
Gebhardt, F. C., 140 E. 29th St., Erie
Gorton, F. B., 4110 Emmet Dr., Erie
Heasley, George S., R. D. 2, Darlington
Heckler, George Snyder, Hatfield
Heilman, R. H., 2303 Beechwood Blvd., Pittsburgh
Hershey, John W., Nut Tree Nurseries, Downingtown
Hewetson, Prof. F. N., Fruit Research Lab., Arendtsville
Hostetter, C. F., Bird-In-Hand
Hostetter, L. K., R. D. 5, Lancaster
Hughes, Douglas, 1230 East 21st St., Erie
Jackson, Schuyler, New Hope
Johnson, Robert F., R. D. 5, Box 56, Crafton
Jones, Mildred M., 301 N. West End Ave., Lancaster
Jones, Dr. Truman W., Coatesville
Kaufman, M. M., Clarion
Kirk, DeNard B., Forest Grove
Knouse, Chas. W., Colonial Park
Leach, Hon. Will, Court House, Scranton
Long, Carleton C., 138 College Ave., Beaver

Losch, Walter, 133 E. High St., Topton
Mathews, Mrs. Geo., R. D. 2, Cambridge Springs
Mattoon, H. Gleason, 258 South Van Pelt St., Philadelphia 3
McCartney, J. Lupton, Rm. 1, Horticultural Bldg., State College
Mercer, Robert A., 435 E. Phil-Ellera St., Philadelphia 19
Miller, Elwood B., c/o The Hazleton Bleaching & Dyeing Works, Hazleton
Miller, Robert O., 3rd & Ridge St., Emmaus
Moyer, Philip S., Union Trust Bldg., Harrisburg
Niederriter, Leonard, 1726 State St., Erie
Reidler, Paul G., Ashland
Rial, John, 528 Harrison Ave., Greensburg
* Rick, John, 438 Pennsylvania Sq., Reading
Robinson, P. S., Gettysburg
Rupp, Edward E., Jr., 57 W. Omfret St., Carlisle
Sameth, Sigmund, Grandeval Farm, R. D. 3, Kutztown
Schaible, Percy, Upper Black Eddy
Schmidt, Albert J., 534 Smithfield St., Pittsburgh
Sheibley, J. W., Star Route, Landisburg
Shelly, David B., R. D. 2, Elizabethtown
Smith, Dr. J. Russell, 550 Elm Ave., Swarthmore
Stewart, E. L., Pine Hill Farms Nursery, R. D. 2, Homer City
Stewart, John H., Yule Tree Farm, Akeley
Stoebener, Harry W., 6227 Penn Ave., Pittsburgh
Theiss, Dr. Lewis E., Bucknell University, Lewisburg
Twist, Frank S., Northumberland
Waggoner, Charles W., 432 Harmony Ave., Rochester
Washick, Dr. Frank A., 501 Cottman Ave., Philadelphia 11
Weinrich, Whitney, 134 S. Lansdowne Ave., Lansdowne
Wicks, Dr. A. G., 227 Baywood Ave., Mt. Lebanon

* Wister, John C., Scott Foundation, Swarthmore College, Swarthmore
Wood, Wayne, R. D. 1, Newville
Wright, Ross Pier, 235 West 6th St., Erie
Zimmerman, Mrs. G. A., R. I, Linglestown

RHODE ISLAND

+ Allen, Philip, 178 Dorance St., Providence
R. I. State College, Library Dept., Green Hall, Kingston

SOUTH AMERICA

Pereda, Celedonia V., Arroyo 1142, Buenos Aires, Argentina

SOUTH CAROLINA

Bregger, John. T., Clemson

SOUTH DAKOTA

Bradley, Homer L., LaCreek National Wildlife Refuge, Martin

TENNESSEE

Howell Nurseries, Sweetwater
McDaniel, Dr. J. C, Tenn. Dept. of Agriculture, 403 State Office Bldg. Nashville 3
Meyer, James R., Agronomy Dept., University of Tenn., Knoxville
Rhodes, G. B., R. D. 2, Covington

Richards, Dr. A., Whiteville
Roark, W. F., Malesus
Zarger, Thomas G., Norris

TEXAS

Florida, Kaufman, Box 154, Rotan
Price, W. S., Jr., Gustine

UTAH

Jeppesen, Chris., Wildwood Hollow Farm Nursery, Provo City
Oleson, Granville, 1210 Laird Ave., Salt Lake City 5
Peterson, Harlan D., 2164 Jefferson Ave., Ogden

VERMONT

Aldrich, A. W., R. D. 3, Springfield
Ellis, Zenas H., Fair Haven. Perpetual Membership "In Memoriam"
Farrington, Robert A., Vermont Forest Service, Montpelier
Foster, Forest K., West Topsham
Ladd, Paul, Hilltop Farm, Jamaica

VIRGINIA

Acker, E. D., Co., Broadway
Brewster, Stanley H., "Cerro Gordo", Gainesville
Burton, George L., 728 College St., Bedford
Case, Lynn B., R. D. 1, Fredericksburg
Dickerson, T. C, 316-56th St., Newport News

Gibbs, H. R., McLean
Johnson, Dr. Walter R., Garrisonville
Morse, Chandler, Valross, R. D. 5, Alexandria
Nix, Robert W., Jr., Lucketts
Pertzoff, Dr. V. A., Carter's Bridge
Peters, John Rogers, P. O. Box 37, McLean
Pinner, H. McR., P. O. Box 155, Suffolk
Powell, Frank, Stuart
Stoke, H. F., 1420 Watts Ave., Roanoke
Stoke, Dr. John H., 408-10 Boxley Bldg., Roanoke
Thompson, H. C., Short & Thompson, Inc., Hopewell
Variety Products Co., 5 Middlebrook Ave., Staunton
Virginia Tree Farm, Woodlawn
Webb, John, Hillsville
Zimmerman, Ruth, Bridgewater

WEST VIRGINIA

Cannaday, Dr. John E., Charleston General Hospital, Charleston 25
Cross, Andrew, Ripley
Frye, Wilbert M., Pleasant Dale
Glenmont Nurseries, Arthur M. Reed, Moundsville, W. Va.
Golden Chestnut Nursery, Arthur A. Gold, Cowen
Gross, Andrew, Ripley
Holcomb, Herbert L., Riverside Nurseries, P.O. Box 5, S. Charleston 3
Hoover, Wendell W., Webster Springs
Margolin, Abe S., University of West Virginia, Morgantown
Slotkin, Meyer S., 629-10th Ave., Huntington

WASHINGTON

Altman, Mrs. H. E., 2338 King St., Bellingham 9
Barth, J. H., Box 1827 R. D. 3, Spokane 6
Bartleson, C. J., Box 25, Chattaron
Biddle, Miss Gertrude W., 923 Gordon Ave., Spokane 12
Carey, Joseph E., 4219 Letona Ave., Seattle
Clark, R. W., 4221 Phinney Ave., Seattle
Denman, George L., 1319 East Nina Ave., Spokane
Ferris, Major Hiram B., P. O. Box 74, Spokane 1
Jessup, J. M., Cook
Kling, William L., R. D. 2, Box 230, Clarkston
Latterell, Ethel, Greenacres
Linkletter, F. D., 8034-35th Ave., N.E., Seattle 5
Lynn Tuttle Nursery, The Heights, Clarkston
Martin, Fred A., Star Route, Chelan
Naderman, G. W., R. D. 1, Box 370, Olymphia
Shane Bros., Vashon

WISCONSIN

Bassett, W. S., 1522 Main St., La Crosse
Brust, John J., 135 W. Wells St., Milwaukee 3
Dopkins, Marvin, R. D. 1, River Falls
Downs, M. L., 1024 N. Leminwah St., Appleton
Johnson, Albert G., R. D. 2, Box 457, Waukesha
Koelsch, Norman, Jackson
Ladwig, C. F., 2221 St. Lawrence, Beloit
Mortensen, M. C., 2117 Stanson Ave., Racine
Zinn, Walter G., P. O. Box 747, Milwaukee

WYOMING

Greene, W. D., Box 348, Greybull

* Life Member
\+ Contributing Member
** Honorary Member

CONSTITUTION

ARTICLE I—NAME

This Society shall be known as the Northern Nut Growers Association, Incorporated.

ARTICLE II—OBJECT

Its object shall be the promotion of interest in nut-bearing plants, their products and their culture.

ARTICLE III—MEMBERSHIP

Membership in this society shall be open to all persons who desire to further nut culture, without reference to place of residence or nationality, subject to the rules and regulations of the committee on membership.

ARTICLE IV—OFFICERS

There shall be a president, a vice-president, a secretary and a treasurer, who shall be elected by ballot at the annual meeting; and a board of directors consisting of six persons, of which the president, the two last retiring presidents, the vice-president, the secretary and the treasurer shall be members. There shall be a state vice-president from each state, dependency, or country represented in the membership of the association, who shall be appointed by the president.

ARTICLE V—ELECTION OF OFFICERS

A committee of five members shall be elected at the annual meeting for the purpose of nominating officers for the following year.

ARTICLE VI—MEETINGS

The place and time of the annual meeting shall be selected by the membership in session or, in the event of no selection being made at this time, the board of directors shall choose the place and time for the holding of the annual convention. Such other meetings as may seem desirable may be called by the president and board of directors.

ARTICLE VII—QUORUM

Ten members of the Association shall constitute a quorum but must include two of the four officers.

ARTICLE VIII—AMENDMENTS

This constitution may be amended by a two-thirds vote of the members present at any annual meeting, notice of such amendment having been read at the previous annual meeting, or copy of the proposed amendment having been mailed by any member to each member thirty days before the date of the annual meeting.

BY-LAWS

ARTICLE I—COMMITTEES

The Association shall appoint standing committees as follows: On membership, on finance, on programme, on press and publication, on exhibits, on varieties and contests, on survey, and an auditing committee. The committee on membership may make recommendations to the Association as to the discipline or expulsion of any member.

ARTICLE II—FEES

Annual members shall pay two dollars annually. Contributing members shall pay ten dollars annually. Life members shall make one payment of fifty dollars and shall be exempt from further dues and shall be entitled to the same benefits as annual members. Honorary members shall be exempt from dues. "Perpetual" membership is eligible to any one who leaves at least five hundred dollars to the Association and such membership on payment of said sum to the Association shall entitle the name of the deceased to be forever enrolled in the list of members as "Perpetual" with the words "In Memoriam" added thereto. Funds received therefor shall be invested by the Treasurer in interest bearing securities legal for trust funds in the District of Columbia. Only the interest shall be expended by the Association. When such funds are in the treasury the Treasurer shall be bonded. Provided: that in the event the Association becomes defunct or dissolves then, in that event, the Treasurer shall turn over any funds held in his hands for this purpose for such uses, individuals or companies that the donor may designate at the time he makes the bequest or the donation.

ARTICLE III—MEMBERSHIP

All annual memberships shall begin October 1st. Annual dues received from new members after April first shall entitle the new member to full membership until October first of that year and a credit of one-half annual dues for the following year.

ARTICLE IV—AMENDMENTS

By-Laws may be amended by a two-thirds vote of members present at any meeting.

ARTICLE V

Members, shall be sent a notification of annual dues at the time they are due and, if not paid within two months, they shall be sent a second notice, telling them that they are not in good standing on account of non-payment of dues and are not entitled to receive the annual report.

At the end of thirty days from the sending of the second notice, a third notice shall be sent notifying such members that, unless dues are paid within ten days from the receipt of this notice, their names will be dropped from the rolls for non-payment of dues.

Proceedings of the Thirty-seventh Annual Convention

Report of the Proceedings of the Northern Nut Growers Association at its thirty-seventh Annual Convention, held at Wooster, Ohio, September 3, 4, 5, 1946, in the auditorium of the Ohio Agricultural Experiment Station.

The convention was called to order at 10 A.M. with the President, Carl Weschcke, in the chair.

Address of Welcome

By Dr. J. H. Gourley, of the Wooster Experiment Station

The thing that would strike me particularly about this meeting we are having is to see people come from so far away; a group that is on fire with interest in a fruit which has no great economic importance, in a place like the central west, in comparison with other fruits. Another thing that is interesting, as contrasted with other fruit groups, would be this; that the extent to which nuts become of great economic importance in these places lies very largely with you. It seems to me that without the insistent desire of a very small minority of people an industry like this would not get very far.

Ohio has not done as much as she should. You may have come to Ohio to give us a shot in the arm. On behalf of the Director, I want to extend to you a cordial welcome to the Experiment Station and to Wooster. This Station has 3600 acres of land and one-third is at Wooster—1200 acres. We have 15 district and county farms, 63,000 acres in state forests and parks.

This station has introduced a number of varieties of wheat. Sixty to seventy-five per cent of all wheat in Ohio is grown from varieties that originated at this station.

This station was organized in 1882 at Columbus. The Federal Hatch Act permitting this type of organization was passed in 1887; thus Ohio was five years ahead of the Federal Act. In 1892, the station was moved from Columbus to Wooster. The state act provided that an experiment station should be located within fifty miles of Columbus, but later it was permitted to extend the distance to 100 miles. They settled on Wooster, which is 90 miles.

The tendency is to work more and more closely with the State University. The trend seems to be so they will function as one agricultural institution.

I would like to extend the keys of the Station to you, but the keys may not unlock the fruit storage.

I trust you will have a most profitable time while you are with us.

Response

By John E. Cannaday, M. D., Charleston, West Virginia

It is a pleasure to meet here under such favorable auspices and to be received with these hospitable words by Dr. Gourley. In recent years, Ohio has gone far in nut growing under his leadership and that of his staff. Pennsylvania also has done a great deal to put nut growing on its feet. My own state, West Virginia, is also making good headway.

In the early 1900's I got the 'bee', but I lost two or three of my first few trees. In 1917 I imported some chestnuts from Japan for planting and tried out various schemes in nut growing. In my opinion, chestnuts are the most important nuts for human food that grow in the temperate zone. It is interesting to observe how chestnuts follow true to seed in many respects. I have been advised that all of the chestnuts grown in China are from selected seed.

Every foot of steep mountain land in some sections of Italy is said to be completely covered with chestnut trees. In my state, the weevil is the scourge of chestnuts; I had hoped that after the chestnut blight destroyed our native chestnuts, the Chinese and Japanese chestnuts would be free from that pest. Where it came from I do not know, unless it came from the chinkapin. West Virginia has chinkapins and these, being blight resistant, apparently have kept up the supply of weevils. Occasionally, shortly before the chestnuts begin to ripen, a few decay from some type of rot.

I took a census of my chestnut trees recently and found 80 trees of bearing age. Some of the largest are 22 to 24 feet in height, with a trunk diameter of 5 inches or more. None have been pruned but have maintained their normal branch formation and grow low. The timber tree must be yet to come. I have read interesting statements to the effect that in parts of China and Burma, there are chestnut trees of timber shape and size. Chestnut trees are likely to become of extreme importance in our future economy. The nuts fill a very significant place in our dietary needs. We should continue to plant chestnut trees and take care of them. I have also from 350 to 400 younger trees that are coming on, and I want to plant additional chestnut trees every year. The black walnut and hickory nut are very important, but the chestnut crop is the corn crop of the nuts.

Carl Weschcke, St. Paul, Minnesota

Our last convention at Hershey, Pa., in September 1941, was a very outstanding one. Not only was it successful because of good attendance, excellent addresses and the places of interest we visited, particularly the home of Mildred Jones, our Secretary, at Lancaster and of the late Dr. G. A. Zimmerman at Linglestown, but it was important because it marked the beginning of a long period during which we had to forego our conventions. The death of Dr. Zimmerman shortly before that meeting dampened our usually jovial spirits when we were entertained at his home, but his wife did much to alleviate this.

To me, the last convention we held was by far the most important since the very first one at New York in November, 1910, because at it I received the honour of being chosen president for the ensuing year. This was during the era when presidents were usually re-elected for a second term, but I assure you that I have not served as president for this long period because I have been seeking to emulate other presidents, but only because the war years prevented our holding the annual meetings at which our officers are elected.

In mentioning any part of the history of our group, we should always remember that we owe its existence to Dr. Deming, who is now Dean of the Association.

Now it is not my province to make a long speech about the N. N. G. A., because a number of other people will talk to you about it. I believe that the growth of our society in recent years has fulfilled the fondest dreams of Dr. Deming, since we have almost doubled our membership since 1941. We now have approximately over 600 members. People all over the United States are becoming aware of the value of nuts as food important to men. It is too bad that nuts have not been available on a competitive price basis with other foods, and that luxury prices have limited interest in nuts among the women buyers. A better understanding of the uses and comparative value of nuts is gradually coming about which will result in a tremendous demand on the nut-growing industry, which of course, includes the nurserymen who develop and grow all varieties of nut trees.

It is unfortunate for our newer members that they will never have the opportunity of knowing those men who were among our earliest and most valued associates whom death has recently taken from us and that they are thus deprived of the pleasure and knowledge they might have gained through personal contact with the wisdom and friendliness these men displayed. Let us all take advantage of every opportunity we have to meet with and learn from the senior members of our group who are with us today. They are the salt of the earth, I assure you.

To those of you who have come long distances from your homes to attend this annual meeting of the N.N.G.A., to our hosts and to all of my good friends here, may I express my great pleasure at meeting again with you after so long a time.

Secretary's Report

Mildred M. Jones, Lancaster, Pennsylvania

In addition to the regular routine duties of answering inquiries about the Association, sales of reports, giving information about nut trees, where they may be obtained, and sources of additional reading material and reference material about nut tree work, a large part of the time I could devote to Association affairs this year was in preparation for this meeting.

Because of travel restrictions, and the fact that the Canadian National Exhibition would not be held this fall, and assurance from the Toronto Convention and Tourist Association, Inc. that the Exhibition would be resumed in the fall of 1947, and that it would be a newer and greater show, it seemed advisable to place these facts before the members, and allow them to vote on their preference for a meeting place this fall. In addition to responses from the officers, I received 63 votes from members, 37 of which were for Wooster, Ohio, 24 for Beltsville, Maryland, and 3 for Canada. Since the letter asking for votes carried the understanding that we were putting the Canadian meeting off for a year by voting for a place in the U. S. this year, and were not canceling the Canadian invitation, this would explain the small vote for Canada.

Our program committee this year was comprised of three members and myself—Mr. C. A. Reed, whose many years of Association work and wide acquaintance made him an invaluable source of suggestions; Dr. Oliver Diller, who took charge of the tremendous task of handling local arrangements; and Mr. A. A. Bungart, who helped greatly in procuring speakers. These men helped so splendidly that I should like here to voice my thanks and appreciation.

Much new data for the revision of the 4-page pamphlet giving information about the Association, sources of seeds, nut tree nurserymen, and reference material for reading has been gathered for printing. Since I accepted the secretaryship in time for the first convention after the war, it seemed advisable to me to hold this material until it could be turned over to my successor who will be elected at this meeting, rather than put the Association to the expense of printing only a small number of circulars.

A good many inquiries were received during the year for sources of certain varieties of nuts. It would help the secretary, and also the members, to have a list of those who have nuts for sale.

Treasurer's Report

For Period from October 1, 1945 to September 30, 1946

RECEIPTS:

Annual Membership	$871.00	
Contributing Membership—		
Philip Allen	10.00	
Sale of Reports	154.80	
Zenas H. Ellis Legacy	950.00	
Miscellaneous	4.00	
		$1,989.80

DISBURSEMENTS:

Subscriptions to Fruit Grower	$ 79.40	
Supplies	12.52	
Secretary's Expense	60.52	
Treasurer's Expenses	41.94	
Miscellaneous	10.00	
		204.38

Excess of Receipts over Disbursments $1,785.42

Balance on Hand—October 1, 1945 1,474.46

Total Balance—September 1, 1946 $3,259.88
Deposited in Walker State Bank $3,236.07
Cash on Hand 23.81

$3,259.88

Notes on the Annual Meeting

A telegram was sent to Dr. Deming in reply to one of greeting from him, and various committees were appointed.

Mr. Corsan suggested that an exhibit of nuts be placed on display in the Royal York Hotel, Toronto, Canada.

Mr. Hirschi said that for killing trees by poison he uses two pounds white arsenic, one pound caustic soda and one gallon of water.

A member stated that a few drops of mercury would answer the same purpose.

Mr. Hirschi stated that he found the Niblack pecan an almost perfect cracker, bringing a premium price.

Mr. Wilkinson stated that while the Niblack pecan had never been a prolific bearer, the nut has few equals. Perhaps intensive cultivation would improve the bearing.

It was voted to leave the date of the next meeting to the executive committee.

Mr. Spencer Chase, of the TVA, invited the members to meet in Tennessee at an early date.

The President: "We should consider this a fine invitation for 1948. For 1947 we should honor our commitments and go to Canada."

A free discussion occurred on the suggestion to establish a nut journal and on the proposal to raise the dues.

The President suggested that the way to get the work of association done promptly would be to pay for it.

Dr. McKay expressed doubt about the inadvisability of raising the dues.

Mr. Walker thought that if the dues were raised it should be to the extent of a dollar on account of the inconvenience of sending fractional currency. The treasurer suggested the advisability of getting out a mimeographed letter to record progress. Mr. Slate emphasized the importance of producing a good report to hold the members.

Mr. Hershey also approved the idea of getting out a news letter or progress report. The President suggested that one thousand members would settle the whole question. Mr. Jay Smith stated he thought the Association should advertise in some way, especially in sportsmen's magazines.

A motion on the part of Mr. Stoke to raise the dues by fifty cents per year was lost.

The nominating committee made the following nominations for officers for the ensuing year, 1946-47:

Clarence A. Reed, President
Dr. L. H. MacDaniels, Vice-President
Miss Mildred M. Jones, Secretary
D. C. Snyder, Treasurer

The nominating committee also, through its chairman, Mr. Weber, recommended that appropriate steps be taken at the next annual meeting to amend the Constitution to consolidate the offices of treasurer and secretary so that they can be filled by one person, and that the remuneration of the secretary-treasurer be fixed at fifty cents per member.

Mr. Stoke moved that the report of the nominating committee be approved, and that the nominees be declared elected. Motion was seconded and carried.

Mr. D. C. Snyder offered the following resolution:

> "Because of the great and enduring service that Dr. William C. Deming has rendered the Association, I move that he be named Dean of the Association and be given an honorary life membership, without payment of dues."

The motion was seconded, and carried with applause.

On being called to the chair, the newly-elected President, Clarence A. Reed, spoke as follows:

> "I take this as a very great honor; it is an equally great responsibility. All I can say is that I appreciate it deeply, and that I will give you the best service I have in me."

The Ohio Section of the Northern Nut Growers Association, Inc., submitted a copy of its Constitution containing a provision that it affiliate with the Northern Nut Growers' Association by having its accredited members become also members of both Associations.

After an open discussion by officers and members of both Associations, a resolution was adopted by the Northern Nut Growers' Association expressing appreciation to the Ohio organization for their offer of affiliation, and accepting such affiliation on the terms stated.

It was also brought out as the sense of the meeting that the Executive Committee work out any necessary details in connection with this and any subsequent affiliation on the part of any district or state Association, the same to be submitted to the next annual meeting of the Northern Nut Growers' Association for ratification.

It was also recommended that the President appoint a member of each affiliating Association to the Executive Committee of the Northern Nut Growers' Association.

This statement is made in lieu of an accurate transcript of the proceedings, or a verbatim report of the resolution as adopted, neither of which is available.

Aims and Aspirations of the Ohio Nut Growers

A. A. Bungart, Avon, Ohio

In one of the previous bulletins of the NNGA, there appeared an eighteen-point program formulated by the Ohio Nut Growers. No doubt you are wondering what has been done and is being done to make this program function. We have eliminated one point, the one on the pollen bank. At the time our program was being prepared we assumed that nut pollen could be stored for several weeks or months: Since nut pollen does not remain viable in storage, we shall substitute a point on the use of lime, fertilizers of various formulas and the use of trace elements in nut culture.

The Ohio Forestry Association on January 18, 1944, passed a resolution approving our eighteen-point program.

As you are well aware, the war put a damper on many activities, nut and otherwise. Here in Ohio, the nut crops of 1944 and 1945 were virtually failures; even the crop of 1946 is decidedly spotty. Yet in spite of the war and adverse weather conditions, the Ohio growers are looking forward, and planning for the future. As a group we are directing our efforts to the attainment of two specific objectives.

In the first place, we have almost $300 collected as prize money for State nut contests. I take this opportunity to announce a donation of $105 from Mr. John Davidson, of Xenia, Ohio. With the aid of such a generous contributor, we are able to offer a first prize of $50; second prize of $25; third prize of $15; fourth prize of $10; fifth prize of $5; and five one-dollar prizes for black walnuts.

In three or five years we intend to have another contest; either a sweepstakes of $110, or a repetition of the amounts offered this year. We may keep the contest open next year and the year after for those wishing to enter nuts for the final awards. In this way, too, we include black walnuts which are not bearing this year.

Our follow-up will work something like this: We intend to keep a record over the years of the performance of each of the ten prize winners and the ten honorable mentions of the 1946 contest. To that end we have made a score card. The first section of this card will contain information useful to the Department of Forestry and to nut culture in general, but it will not be a factor in selecting the prize winner unless a virtual tie might result in the sweepstakes contest. This section will include:

1. Location—owner, County, rural route, village, town, state route, etc.

2. Location of Tree—isolated, moderately crowded, in dense woods, farm, pasture, city lot, fence row, general ecology; types of other trees in neighborhood, air drainage, exposure.

3. Size of Tree—circumference 4-1/2 ft. from the ground, probable age, height, limb spread; shape, tall, short; symmetry or lack of it.

4. Type of Soil—bottom land, slope and direction, upland; clay, loam, alluvial; presence or absence of humus; acidity; sod or cultivated, mulch or not; depth and kind of subsoil.

5. Moisture Conditions—presence of stream or tile drain, proximity to to stream, lake, pond, etc.

6. Fertility Conditions—wild natural state, near barnyard, fertilized or not with manure or commercial fertilizers, application of lime, etc.

The second section will contain information that will aid in awarding the final prizes. Superior rating under this head might, in the final judging, make an "honorable mention" of the 1946 contest the best all around performer three or five years hence. This section will include:

1. Resistance to disease and insect pests 2 points

2. Bearing habits over the given period; annual, biennial, occasional 7 points

3. Length of growing season; rate of growth; time of blossoming (staminate and pistillate flowers), time of leafing out, time of nut ripening, time of leaf fall 4 points

4. Size of nut clusters, range in number of nuts, per cluster, number of pounds of immature nuts 2 points

5. Size of crop in proportion to tree 5 points

Total 20 points

Some formula will have to be worked out for the last, i.e., size of crop in proportion to the size of tree. Perhaps we might say the crop equals (pounds of nuts) / (r squared x h) in which "r" would represent the radius or half the limb spread and "h" the height, measured from the top to lowest branches.

For example, if a tree that yielded 100 pounds of nuts had a limb spread of 20 feet and was twenty feet high, it would have a value of 100 / (10 squared x 20) or 1/20. The fraction, of course, could be eliminated if the number of nuts were substituted for pounds. It is hardly likely that such a formula would be used for all the trees, probably only in instances where scores in other respects were close.

The third section of the score card will record the rating of the judges on the cracking qualities and other characteristics of the nuts themselves. Any form accepted and approved by the NNGA will be satisfactory.

We plan to use this system for hickory, butternut and other nut contests. Without a Mr. Davidson, however, we shall be compelled to reduce our prizes for the other contests.

I should like to take this opportunity to thank Mr. C. A. Reed for originating this plan. He told us we ought to know more about the trees from which the prize nuts were taken. Our score card aims at a complete record.

Our second aim is to secure a full time research worker in nut culture under the Horticultural Department of Ohio. We have the promise of Director Secrest that he will include in his biennial budget an appropriation for such a specialist. We have the encouragement of Dr. Gourley, the head of the department. But both men will expect us to do our part. Both expect us to speak for our group and our project when the time comes. We accept that responsibility.

Our group has already contacted the members of the finance committee that passes on the budget, and we expect to have our representatives present when the budget is discussed in committee. At present, to be sure, we cannot furnish or even promise an endowment in money. Sixty Nut Grower members can scarcely compete with such powerful groups as the Apple Growers, the Hybrid Corn Breeders, the Poultrymen and others. We can, however, furnish an endowment of men. Among our members we have such men as Mr. Davidson, Mr. Shessler, Mr. Cranz, Mr. Smith and Mr. Weber, along with many others who have done a great deal with nut trees.

A research worker could draw upon their advice, their experience, their technique. He would have as his assistants men who were actuated by no mercenary or selfish motives, and would give of their time and trees to make this dream a reality. Certainly much of the experimental work such as the crossing of varieties could well be performed on the trees of individual members.

The need of such an expert is obvious. The job of getting ahead in nut culture is too big for any one of us. We all know, frequently to our regret, that nut growing is a slow and at times a discouraging business. If we are honest with ourselves we have to admit failures again and again; yet the work is creative and fascinating. We always plan to eliminate some blunder, to perfect some method, next year.

Sometimes a man has a green thumb, or a magic touch, or whatever it takes to make grafts grow, or buds take, or hunches to succeed. Such a man was Mr. Otto Witte, of North Amherst. As a nonagenarian, he was ever looking ahead to another year with his beloved trees, but he died in his nineties. Some of his prize trees have been cut down and probably others will be. What has happened to the experiments of 60 years? Another such man was Mr. Ross Fickes, of Wooster, whose skill in grafting nut trees was at once our envy and our admiration. When his farm is sold, will the new owner sense the hand of the master and watch carefully over the walnuts and hickories, or will he cut them down?

I suppose that death brings an end to many a business, but the nut business is a new one, and a slow one, too. It is regretted that a life time of patient care and painstaking research is lost to us and to nut culture.

True, a nut specialist will not keep death from the door of nut growers, nor will he save their groves from destruction, but he can keep a record of each grower's trees. He can plant his trees and lay out his plantings on state land where there would be more assurance of permanency. Once a nut department is established there is good reason to suppose that the work would go on until certain objectives were attained.

Well, what should our specialist specialize in? May I suggest a few activities? Such a specialist would be the proper person to keep the score cards of the prize-winning black walnuts, hickories, butternuts and English (Persian) walnuts of nut contests held in the state. He would have the time and space for grafting scions from such trees for further observation and study.

In the second place, he could plant and study other varieties under identical conditions and observe their performance. By correlating these data with the records of individual growers he ought to be able to recommend certain varieties of nut trees for various sections of the state.

In Ohio, we have chapters of the Izaak Walton League; we have Friends of the Land; we have sportsmens clubs; we have extensive tracts of municipal and state land. We have the problem of doing something constructive with strip mining areas; we have, and will have under contour farming, little crazy-quilt blocks of land unsuitable for cultivation. All these agencies and all these needs tie in with the intelligent use of trees, particularly nut trees, because they serve a fourfold purpose; lumber, food, erosion control, and a balanced wild life. Here is where the nut specialist would enter the scene. By collecting data, by pooling the results of the individual growers, and especially by selection and breeding, he should be able to recommend the proper varieties of nut trees for specific needs.

It seems to me, however, that the main job of such a worker should be to produce a streamlined black walnut, a thin-shelled, good-cracking, fast-growing walnut.

The black walnut is, indeed, a regal tree. It grows all over the State. Here is a tree of almost infinite variation. What an opportunity for the genetic scientist! What an opportunity for the nut specialist!

In connection with the improvement of the black walnut as a nut and timber tree, the specialist might well investigate the English or Persian walnut. What about the possibilities of Circassian walnut lumber? What is to prevent the growers and the specialist from planting the English walnut for timber? Here in Northern Ohio, English walnut trees have been cut for timber. There are probably several hundred English walnut trees scattered through the northern counties of Ohio. Some of them are from 10 to 18 inches in diameter. A few are second generation. As these trees seem to be fairly rapid growers it would seem reasonable that nuts from these hardy trees would grow into valuable timber, apart from the value of the nuts.

Perhaps all these aspirations and aims seem Utopian. Probably such a program would keep a dozen workers occupied. In cooperation with the Forestry Department, however, students might be assigned to study certain phases of nut culture. A Ph.D. dissertation might well be written on the variation of the Thomas walnut in Ohio.

In conclusion, the Ohio growers will try to produce better nut trees. Through prize contests they hope to find what nature has produced. Through the services of a scientist they hope to find what man can produce. The two aims dovetail. We are reasonably certain of the prize contests; we are not yet certain of securing the nut scientist.

Ohio is host to the NNGA this year. May the Ohio growers ask you for your moral support in this venture? The NNGA is the mother organization. Through the efforts of the officers, past and present, the association is in a flourishing condition with prospects of a very bright future. Whatever we do in Ohio, whatever will be done in other states and countries will be a monument to the NNGA. The groping years, the hard years, are behind. The spade work has been done. We want you to feel that the aims and aspirations of the Ohio growers sprang from your advice, your experiments and enthusiasm.

I would like to add a final word about the unique advantage we enjoy here in Ohio. We have the cooperation of a powerful and excellent farm paper, "The Ohio Farmer." Through its pages our contests get a wide publicity. Mr. Ray Kelsey has furnished us with 5000 folders announcing the contest and the purpose behind it. We have the cooperation of the Experiment Station here at Wooster and its affiliated agencies. Drs. Secrist and Gourley have been kind, encouraging, helpful. Dr. Oliver Diller, of the Forestry Department, and Mr. Walter Sherman, of the Mahoning Farm, have helped and worked with us in a hundred ways. We feel the NNGA ought to know about this harmonious and whole-hearted team work.

Nut Growing Under Semi-Arid Conditions

A. G. Hirschi, Oklahoma City, Oklahoma

The pecan is the major nut crop in Oklahoma. The timber growth along the rivers and creeks contains enough pecan trees, if they were properly distributed, to make one continuous pecan grove entirely across the state.

Pecan improvement work is only in its beginning. The Oklahoma Pecan Grower's Association was organized in 1926. It is devoted to the general improvement of the pecan, and to the dissemination of information gained by the members from their experience and observation. Dr. Frank Cross, head of the Department of Horticulture of our A & M College at Stillwater, is very active in nut improvement and is giving us much valuable assistance. Early in the history of our association we began to graft the large improved varieties on our seedling trees. True, many mistakes were made. I recall when all our trees producing small and inferior nuts, were cut down level with the ground, and the sprouts growing from the roots, were budded or grafted to paper-shells. This meant a long wait for production. We soon realized it was better to stub back the limbs and graft these, or permit the sprouts to develop and bud them, plus saving most of the framework of the trees, which gives us heavy production of grafted pecans in a short time.

Competing growth, that is underbrush and all kinds of trees other than pecans, rob the grove of moisture, sunlight, and plant food. This growth was formerly removed by hand grubbing, but now with a large bulldozer it is pushed right out of the ground into piles where it is burned. Now the ground is clean, no stumps to grub out, and ready for a cover-crop or clean cultivation. Nothing remains but pecan trees, some elm, hackberry and oak, too large for the bulldozer. These are poisoned and burned right where they stand the following winter. For poisoning a mixture of two pounds of white arsenic and a pound of caustic soda to a gallon of water, if applied from an oilcan with a spout in an open circle chopped in the bark so as to girdle the tree, will usually deaden it in a short while. Within the year nothing is left but pecan trees. These are watched carefully for production and shelling quality and, if not desirable, or standing too thick, are removed for greater spacing for permanent trees.

Today, only the smaller pecan trees are top worked, either to named varieties or to selected seedlings. Due to changed conditions of market and labor, the native pecan has come into its own. The pecan sheller buys 90% of the native pecans. He will pay only a few cents more for the big paper-shell. The native pecan is as staple as butter and eggs. Every produce man buys them for the shelling plants. This leaves the big paper-shell to seek a special market at an advertising cost. Due to the small differential in the wholesale price of the native and the paper-shell, the larger native trees are no longer top-grafted but are encouraged in every way for heavy production.

Thus, when creek and river bottom thickets are opened up to sunshine and air, nothing left but pecan trees properly spaced, and this on land usually considered worthless, these trees will quadruple in production and pay a handsome return on a $200 per acre valuation. This is a real and altogether possible two-story agriculture to those who are fortunate enough to own undeveloped pecan timber land. Many of our members have beautiful groves redeemed from the wild with bounteous crops of nuts overhead and cattle grazing on enriching cover crops underneath. The pecan means a lot to the farmers of Oklahoma. The average yearly tonnage is about 16,000,000 pounds, with a peak production of 30,000,000 pounds. This amounts to an average of $2,000,000 annually, with a peak of $5,091,000.

I want to show you what it means to some of our members to develop their native pecans, either as natives or grafted to improved varieties. The proceeds from one lone pecan tree in Mr. Skorkosky's cotton patch paid the taxes on his farm several different years. Thus encouraged, he redeemed a small thicket, added a few nursery trees of paper-shells, about ten acres in all, which now often makes a return equal to the rest of the farm. Mr. Kramer paid $1,000 for 10 acres, with part in seedling pecans. He sold $1,000 worth of pecans several different times, and the rest of the farm makes a good return in pasture and hay. He also has 51 acres that often makes a return of $50 per acre in pecans, besides pasturing 20 Herfords. Mr. Kramer destroys trees by girdling. Mr. Pfile makes it a business to buy farms on which there are pecan thickets. One farm has 70 acres, all top-grafted to improved varieties. Trees were small and no production for five years, supporting production for the next four years. Tenth year grossed $8,500; eleventh year, $5,400; twelfth year, $1,800, and this year his conservative estimate is $10,000. Mr. Camp has 600 acres in pecans, 90% improved varieties. He planted 50 acres on upland sandy land on terraces, with pecan trees 40 feet apart and an apple tree between each two pecan trees. The tenth year he produced 8,000 pounds of paper-shells and 4,000 bushels of apples. More recently he planted 125 acres on upland, but planted the pecans 60 feet apart on terraces with an apple tree between. In this orchard he produces 3/4 of a bale of cotton per acre and plants vetch in the fall between cotton rows. In October he had four crops on this land, cotton, vetch, apples and pecans. He says apple trees alternated with pecans on terraces are OK. Cotton, potatoes and sweet potatoes between the terraces for the first ten years are OK, but vetch as a winter cover crop to improve the soil must not be neglected. Grover Hayden has the largest native pecan grove in the world—1,800 acres fenced hog tight. He started 31 years ago as a farm hand. He had rather have 500 acres of pecans than 1,000 acres of alfalfa. Now after 30 years he owns the place at a purchase price of $90,000, not counting improvements and equipment. His average production is about 300,000 pounds per year. In 1935, he produced 400,000. He held back his 1941 crop and together with his 1942 crop, he sold both for $61,000. Think of the faith a man must have in pecans in Oklahoma to go in debt for $71,000 as Mr. Hayden did! He rode a pony that was mortgaged for all it was worth from Arkansas to this ranch.

Those of us who do not have native or seedling pecan trees to work with, must develop orchards from nursery trees. I was raised on a poor farm in Missouri. I always had a desire to take a poor piece of land and see what I could do to improve it. Consequently, I planted 225 improved pecan trees of 25 different varieties and all other kinds of nut trees, fruit trees and a variety of berries on a piece of worn-out upland, pronounced by our county agent to be the poorest piece of ground in our county, and predicted it would be a complete failure.

I planted the pecan trees 60 feet apart, and interplanted with other nut and fruit trees. The trees were planted on the contour with youngberries and many others planted in rows between the tree rows, making a perfect soil conservation arrangement. Barnyard fertilizer was used to start the trees. Every September, vetch and rye were sown as a cover-crop and soil-builder and disked into the soil the following spring. Clean cultivation is practiced during the summer to conserve moisture. This procedure has been adhered to most rigidly without a single crop failure. At 12 years most of the trees are producing $25 worth of paper-shells. The youngberries and plants sold have paid the expense of the orchard and a handsome profit besides, until the trees needed all the room. This project has proved to my satisfaction that profitable nuts and fruit crops can be grown on upland, if soil conservation and improvement are practised. The limiting factors of nut and fruit production are plant food and moisture, and if these are supplied, good production is assured.

Black Walnuts

The native black walnut of Oklahoma is small and of little value. Most pecan growers have a few native black walnut trees they graft to the improved varieties. I have Thomas, Stabler, Ohio, Mintle, Myers, and others. Thomas has been used most extensively, but does not fill well on upland. However, in deep sand and low bottoms it fills perfectly. It is an alternate bearer and is subject to sunscald in our hot dry summers. Ohio and Stabler have not proven satisfactory. Mintle is a fine nut, splendid cracker, fills well, but is an alternate bearer. I like Myers very much, a consistent bearer, has thinnest shell of all, vegetates after frost in spring, has abundant foliage and twigs, holds leaves until late autumn. Myers is my choice of all varieties at present. However, as with pecans, what varieties to use is each grower's individual problem. We will be looking for better varieties 50 years from now. For five years I am offering $25 annually for the best seedling black walnut. Write to our A & M College, Horticulture Department, Stillwater, Oklahoma, for rules and regulations of the contest.

How to Make Money with Black Walnuts

I believe I have discovered the best way to market black walnuts. I have not had much success selling them either husked or unhusked, "too hard to crack." Then someone remarked, "If you would crack them and put in some horseshoe nails to pick out the meats, they might sell." There it is: the secret is discovered. The lowly and almost extinct horseshoe nail will sell cracked black walnuts. I have the reputation among local hardware dealers of having more horses than any man in Oklahoma. Black walnuts and horseshoe nails are reminiscent of the good old days down on the farm. The big fat meats of improved cracked walnuts peering through the sides of one or two pound cellophane bags pinned shut with a couple of horseshoe nails is a temptation few people can resist. Leave a few packages with your grocer or druggist and try it. I get 25c per pound for the whole walnuts, and 35c for those cracked. I sell several thousand pounds every season, and since the black walnut does not become rancid we sell them all the year. I have a down-town shop window to display nuts and fruits. We husk our walnuts by running them thru an ordinary corn-sheller, or by jacking up the rear wheel of an automobile, put on a mud chain, with a trough underneath, place car in gear and scoop walnuts into trough in front of the wheel. This will husk them rapidly and well. We should promote the growing of more improved black walnuts. Most catalog nurseries still list seedling walnuts. We sold 3000 Thomas and Myers black walnut trees to one mail order nursery, and they could have used more.

English Walnuts

I tried all the California varieties, but these lacked hardiness. The Wiltz Mayette grew into a big fine tree but the 1940 Armistice Day freeze proved fatal. Breslau, Broadview, Schafer, and several others with some 25 Carpathian seedlings are just coming into bearing. Some give promise of adaptation here. I am determined to find a prolific and adapted variety. In the meantime we must content ourselves to grow this most attractive tree with its large waxy leaves and beautiful light-colored bark as a useful novelty.

Heartnuts

Here is a surprise nut and tree to Oklahoma people. Both are unlike anything ever seen here. When they see this most unusual tree, with its tropical leaves and taste the delicious nuts they want a tree for their yard. Visitors stare in amazement at the immense catkins, and the grape-like clusters of nuts that develop later. This is a heartnut year. In most all varieties, ten to fifteen nuts to the cluster hang from the terminal of each twig. The leaves sun-burn easily. In spite of this we had a heavy crop of well-filled nuts. Of the several varieties I have, Stranger is the most prolific. Fodemaier, and Walters bloom late enough to escape our late spring freezes, are larger nuts, and should prove to be the best eventually.

Butternuts

Butternuts grow native in Missouri and Arkansas. Our section is most too hot and dry for them. However, I have a few grafts of Buckley and Weschecke bearing nicely, grafted on native black walnuts.

Hickories

The wooded hills and river bottoms contain several kind of hickories. I have several pecan trees grafted to the Pleas and McCallister hybrids, but they are light producers in Oklahoma. I have 80 acres of river bottom hickory nuts in southwest Missouri that bear abundantly.

Oriental Persimmons

Persimmons grow native here. The Early Golden, an American variety, is very productive and ripens in September long before frost. Of the Orientals I have Tamopan, Eureka, Fuyu, Data Maru, Tanenashi. Most all bear heavily, in fact usually overbear. They stand our dry weather better than does the native persimmon. The very large fruit usually in colors of yellow and red attract much attention from visitors who think they are oranges. The persimmon belongs to the ebony family. The fruit contains as high as 40% sugar and in the Orient is a national dish. We propagate them by grafting our native stock.

Pawpaw

The Pawpaw is native in Missouri and Arkansas and in the eastern part of Oklahoma. It is a beautiful tree and very productive. We shade the small trees here until they get started, after which they do quite well. The fruit is a favorite with many.

Chestnuts

I think the greatest tragedy that ever befell American horticulture was the chestnut blight. Not so long ago every hill and mountain-side east of the Mississippi River, from near the Gulf of Mexico to the Canadian border was covered with native chestnut trees producing millions of pounds of food for man and beast. Today all has been devastated by this terrible blight and nothing remains save leafless trunks, like tombstones, in memory of a grand food tree.

In 1889, Tom and Mary Jones left their Kentucky mountain home to establish a new one in Oklahoma. As with all pioneers they brought seeds of many species with them, including chestnuts. I now own the farm they homesteaded. On it today there is an American chestnut tree 4 feet in diameter with a limb-spread of 50 feet. This grand tree has been an inspiration to me, surviving our hot dry summers and outliving two generations of fruit trees by its side. This beautiful tree, now nearly 60 years of age, was proof-sufficient that chestnuts would grow in Oklahoma. I began to plant chestnuts. I planted all the Riehl varieties—Progress, Dan Patch, Van Fleet and others. I also had Boone, an American and Japanese hybrid, brought about by Endicott, also of Illinois. These have borne well. Being isolated and outside of the native chestnut range, they have not blighted.

Since 1906, the Government has imported many thousand seed chestnuts from China. Later, it distributed little trees among the nut growers in an effort to re-establish chestnut growing in this country. This Chinese chestnut is blight-resistant. The best Chinese seedlings have been selected for propagation and have been named; of these I have Stoke (a hybrid), Hobson, Carr and several others. They are very prolific and often set burs the same year set out. Mr. Stoke sent me scions of the newer varieties this spring—Colby's hybrid, and Stoke seedling's Nos. 1 and 2. I grafted these on Chinese stocks; they set burs and matured nuts the same year grafted. The named varieties of Chinese Chestnut are the most precocious bearers of all the nut trees, are adapted and worthy of planting over a wide area. It should be the duty of every man who is interested in food trees to lend a hand to help re-establish chestnut growing in this country, now that we have blight-resistant varieties.

Almost within the shadow of our State Capitol, on a main highway leading from our fair city, I have planted 2-1/2 acres of blight-resistant Chinese chestnut trees, as a living memorial to our only child, Harold, who gave his life to our country in a Jap prison camp in the Philippines. We shall devote the rest of our days to this Living Memorial, and leave means for its continuance, so that passers-by in generations to come may be reminded of the world's greatest tree tragedy, and to demonstrate that chestnuts which once grew native over half the nation, and were laid low by a terrible disease, may again be grown.

In conclusion, let me warn you to improve your soil continually. NO TREE CAN BE BETTER THAN THE SOIL IN WHICH IT GROWS. No man is a greater exponent of soil improvement than one of Ohio's most illustrious sons, Louis Bromfield. In his book, "IN PLEASANT VALLEY," he says, "What we need is a new kind of pioneer, not the sort which cut down the forest, and burned off the prairies, and raped the land, but the kind which creates new forests and heals and restores the richness of the country God has given us."

Weather Conditions versus Nut Tree Crops

By J. F. Wilkinson, Indiana

Nut tree crops, like other crops, are dependent on heat, light and moisture in proper amount at the right time to produce a crop of nuts of normal quality; soil conditions also to be taken into consideration.

These conditions are probably more essential to a nut crop than most of us have realized; even the weather of the preceding season of late summer and fall may affect or determine next seasons nut crop.

The size of the nut depends on the weather in Spring and early Summer, for when the shell is once formed and hardened little more growth can be expected under any conditions, while plumpness of kernel depends on favorable conditions in late Summer and Fall.

After the shell is formed it fills with water which gradually changes to kernel, beginning at outer part next to shell, and unless there is plenty of heat, light, and moisture, kernel may not be filled, which will cause kernel to shrink, and not be plump, neither will it have normal germinating vitality, flavor, or weight.

In the past there have been seasons when nuts were not up to normal quality, but I did not realize then just what caused this condition, until a few years ago, I heard a party remark that a dry season was an indication of a good nut crop the following year.

Recalling back several seasons this, as a rule, has been true, especially where there was no unusually early Fall freezes, and Spring weather was favorable.

The season of 1944 here was one of the driest on record. Up until the middle of August, nut trees were showing signs of going dormant. Late in June, sap was getting so low that I did all my budding late in June, a month earlier than usual.

This early dry weather caused the nuts that year to be very small, especially on trees growing under less favorable conditions. Trees that were well cultivated produced nearer normal sized nuts.

About the middle of August rains began, and these nuts were well filled. The rains of August brought new life into the trees causing them to hold their foliage unusually late, and not being thoroughly dormant before cold weather, at which time no doubt many of the fruit buds were killed, with the result that a very light crop of nuts was set in Spring of 1945.

Spring opened very early with a bright warm March starting growth before usual time, even some trees set Pistillate bloom by the first of April; then later in April it began raining, and rains continued most of the Summer with much cool, cloudy weather with the result that most of the nuts failed to properly fill, or mature. This was true of hickory nuts, walnuts and pecans of both the named varieties and native seedlings.

While the 1945 nut crop was very light of both pecans and walnuts, I had a few trees with fair crops, though none of the nuts had well filled plump kernels.

Some of the first nuts to ripen seemed to have fairly well filled kernels after gathering and kernels got dried out, they shriveled and lacked flavor.

Walnuts seemed to suffer even worse than the pecans. I was not able to find a walnut tree in this section that produced good planting nuts; even farm crops suffered, especially corn of which much of the crop was not of normal quality.

The spring of 1946 began very much as in 1945 with a very warm March, again causing trees to start growth unusually early, and this spring, pistillate bloom was visible on some pecan trees in the last days of March. This weather condition remained until about the middle of April when cool rainy weather set in lasting for a month with frosts and light freezes as late as May 10th, which took all the nut crop in this section with the exception of a very few walnuts, and these were of very poor quality.

Another very peculiar thing happened in Spring of 1946. The Posey and Giles varieties, both of which are usually heavy bloomers of Stamen bloom, failed to set a single catkin this spring, while trees of other varieties growing near them set heavy crops of catkin bloom.

The behavior of nut trees in this section in the past two years, both of which have been unusual seasons, is evidence that nut crops are subject to weather conditions, not only of the present, but of previous season as well.

Nut Tree Notes from Southwestern Ohio

Harry R. Weber, Cincinnati, Ohio

Influence of Stock on Scion

At my farm home in the northwestern part of Hamilton County, Ohio, at about 800 feet elevation, on clay soil, the Carpathian walnuts commence growth too early in spring for their own good and my comfort, well knowing what lurking Jack Frost can do to them. These Carpathian walnuts are uninfluenced by their black walnut understocks, the Schafer variety alone excepted. I also have two Schafer trees that came grafted apparently on Carpathian understock; but these start as early as the others.

The Schafer exception, to which I refer, is grafted on a native black walnut stock to which the Broadview variety also had been grafted. (The Schafer variety is patented. I had permission to use the graft as I did.) With these two hardy varieties in the same tree, which itself is a late starter in the spring, I unwittingly laid the foundation for an unanticipated result. This became apparent after a severe early spring frost in 1945 caused me to examine all my hardy (Persian) walnut trees to note the effects of that freeze. The new growth of Broadview on the same tree with the Schafer was frozen, while the Schafer with the rest of the tree was dormant. The new growth of the other two Schafer trees; of Breslau top-worked on two trees; of Broadview on another tree; of an unknown variety on still another tree; all trees being native black walnut, all were frozen. The same was true as to Breslau seedlings and also a Kremenetz on Minnesota black walnut. Of course, all these trees staged come-backs with no bad after effects.

In April, 1945, we had a severe hail storm that clipped clean the second new growth from these trees. The topworked Schafer was still dormant, while its companion Broadview in the same tree suffered like the rest. The spring of 1946 showed the topworked Schafer still dormant, while all the others were active. The Broadview on the same tree with the Schafer was almost in full leaf before the Schafer and the rest of the understock showed signs of growth. A number of persons thought the rest of the tree was dead.

The Keystone Black Walnut

I have a cut leaf black walnut tree, of value as an ornamental, which originated in Pennsylvania. Although it had catkins for several seasons, not until the past season did it produce, and then only one lone nut. The husk of that nut had a smooth exterior similar to that of a Persian walnut; but it lacked the characteristic black walnut odor. In fact, it had none. If this tree has any Persian walnut blood in its makeup, that hybrid strain may have manifested itself in the foliage; in any event, there was an influence of some kind that caused the change in the usual type of foliage. I was more interested in planting the nut to see what kind of foliage the seedling will have rather than in cracking it for examination to determine its value as a nut.

Throp Walnut

The parent Throp tree stood bordering a road along the Ralph Throp farm in Indiana, 40 miles from my home. About six years ago, with the permission of Mr. Throp, and being a very old tree, it was cut down as its branches interfered with overhanging wires. When I last saw the stump early in 1942, it had staged a come-back by throwing numerous suckers. However, the main point in mentioning this tree is to register the fact that it bears two kinds of nuts, single-lobed, or peanut type, and doubled-lobed, with the peanut type predominating. A Throp tree of mine showed this variation, and on my next visit to the Throp farm, in the presence of Mr. G. A. Gray, one of our members, Mr. Throp definitely confirmed the fact that the parent tree bore the two kinds of nuts aforesaid and that the peanut type predominated.

I am prompted to make this statement for the reason that one of our prominent members, well versed in the performance of our best varieties of northern nut trees, had not been aware of the dual performance of the Throp tree, until I called it to his attention.

Black Walnut Nursery Studies

S. B. Chase, Tennessee Valley Authority, Norris, Tennessee

Briefly summarized, here are the results of a series of black walnut nursery studies undertaken in 1940 and 1941 by the Tennessee Valley Authority. The object was to develop nursery practices which would yield the large uniform seedlings most desirable as understocks for grafted or budded trees.

Germination and Stratification

It is known that either fall- or spring-planted walnuts germinate readily if the nuts are viable and if those planted in the spring are properly stratified over winter. To find out just what effect spring and fall planting has on germination and to compare various methods of stratification, three seedlots were given the following treatments on November 28, 1940:

1. Planted in seedbeds 5. Stratified at 65-75 deg. F
2. Stratified outdoors 6. Stored dry at 45-50 deg. F
3. Stratified at 38-40 deg. F 7. Stored dry at 45-50 deg. F
4. Stratified at 45-50 deg. F subsequently soaked in
water prior to planting

Nuts from the three seedlots were kept separate and planted in random plots in three seedbeds. Each treatment was therefore represented nine times with a total of 162 nuts in each treatment.

To determine whether time of outdoor stratification has any effect on germination and emergence, three other seedlots were treated as follows:

1. Planted November 28, 1940 5. Stratified January 30, 1941 2. Stratified November 28, 1940 6. Stratified February 20, 1941 3. Stratified December 19, 1940 7. Stratified March 13, 1941 4. Stratified January 9, 1941

These three seedlots were also planted in three seedbeds with a total of 135 in each treatment.

With one exception, all nuts in the two tests were planted April 3, 1941. One of the two lots stored dry was soaked in water for five days, then planted April 7. Seedbeds were equipped with screen wire cloth at a depth of 10 inches.

~Results~: In both tests, fall nut planting resulted in the best germination. Germination was higher for nuts planted in the fall than for nuts stratified on the same day for spring planting, although the difference was significant only in the second test. Outdoor stratification produced the best results, followed in order by indoor stratification at 38-40 deg. F and 45-50 deg. F. None of the nuts stored dry germinated. Time of stratification proved to be important. Any delay after November 28 resulted in reduced and retarded germination and consequently smaller seedlings.

Depth of Planting and Seed Orientation

The effect of planting depth on germination and on seedling size was investigated by planting black walnuts one, two, three, and four inches deep. Other nuts were planted in three positions: (1) radicle end up, (2) on side, and (3) radicle end down.

~Results~: Germination was unaffected by any of these treatments. The emergence of the seedlings was retarded by deep planting and hence the final diameter of seedlings was smaller. There was little difference in seedlings from nuts planted one and two inches deep but they were noticeably larger than those planted 3 and 4 inches deep. Planting nuts with the radicle end down invariably produced seedlings with undesirable crooks in the root-stem region which made them unsuitable for grafting. Planting nuts radicle end up produced straighter seedlings than planting them on their side. The latter method was the most economical for nursery practice.

Seed Size

To study the effect of kernel size on size of seedling produced in the nursery, nuts from nine wild trees and Thomas nuts were planted. Kernel weights ranged from 1.21 to 5.61 grams; nut weight from 6.5 grams to 24.3 grams.

~Results~: With one exception where germination was poor, nuts with small kernels produced small seedlings and nuts with large kernels produced large seedlings. Under nursery conditions the need for uniformly large seedlings for budding and grafting is apparent. The results of this study indicated the desirability of using seed nuts with large kernels for production of understocks.

Seedbed and Budding Studies

Density of stand in seedbeds influences seedling size. As size of seedling is important in budding and grafting black walnut, information on the most desirable spacing in seedbeds was needed. In three seedbeds Thomas nuts were planted in three nut spacings: 4 x 4 inches, 5 x 5 inches, and 6 x 7 inches. In other plots nuts were planted 4 x 4 inches and after emergence the stand was thinned. All seedlings from the thinning test were set out in nursery rows the following spring and those large enough were budded in the summer.

~Results~: Increasing the spacing produced seedlings of larger girth and shorter height—a desirable characteristic in black walnut budding stocks. The most desirable spacing appeared to be 6 x 7 inches. Even though the number of seedlings resulting from this spacing was approximately half the number produced at 4 x 4 inches spacing, more usable seedlings were produced at the wider spacing.

Thinning seedlings spaced 4 x 4 inches resulted in larger girth of those remaining—very similar in size to seedlings spaced 5 x 5 inches. Seedlings from the thinned and unthinned plots averaged 0.62 cm. and 0.55 cm. in diameter, respectively. In the nursery row 73 percent of the larger transplanted seedlings were large enough for budding the following summer, while only 59 percent of the smaller seedlings attained proper size. Bud survival was 22 percent on the larger stocks indicating the desirability of using large stocks.

My Experiments, Gambles and Failures

John Davidson, Xenia, Ohio

In reading the past reports of this Association, I find one thing lacking. One becomes interested in a report dated, let us say, 10 or 20 years ago, which contains an account of a project then started. It had great possibilities. What was the outcome? We do not know. No mention of it has appeared since. Did it fail? Let us say it did. Why? The answer to this final query is almost, if not quite, as important as would be an account of the means employed to make it successful—if it succeeded.

I should like to know, for example, whether anything remains of the Neilson-Post project in Michigan and what its history has been. I should like to hear more, also, about the outcome of many of Mr. Gerardi's intensely interesting and original experiments, such as his method, described in the 29th Annual Report, of asexual propagation of heartnut trees on their own roots; or his method of artificially creating beautifully marked burls on black walnut logs by systematically and repeatedly scoring the bark. These and many others. Which experiments were successful and which were not? Mr. Gerardi's original and adventurous mind is the sort that should be probed for the benefit of those who come after us.

My report today is my own short and tentative contribution to such a resume.

In the 1938 Report, on page 73, you will find my ambitious and optimistic "Farm Plan for Nut Tree Planting." In it I tried to outline a plan which could be used by any practical farmer with but slight sacrifice of useful land. Its last sentence reads as follows: "Meantime, I shall have kept practically all my land in profitable use all the time." Well, that depends upon what is interpreted as "profitable use." Tree growth is surely profitable.

The plan, in substance, was as follows: First, plant 20 acres in a modified forest formation to selected seed, mostly black walnut, the trees to stand 8 feet apart in rows 22 feet apart. Use the space between the rows first for truck gardening and later for an interplanted row of some fast-growing species for timber. No grazing permitted. Second, plant another 20 acres to a nut orchard using grafted trees of named varieties spaced 80 feet apart. Protect from livestock and permit grazing. Finally, plant seed in another 30 acres, spaced 80 feet apart, the seedlings to be eventually topworked to the wood of promising discoveries from the first plot. Protect and cultivate or graze.

What has been the outcome of this plan to date? The proposed plan worked very well in a 20-acre plot where a meadow was planted to an orchard of grafted trees, mostly pecans and Carpathians, which were protected by cattle guards, but was not completed in the seedling 20-acre plantation where the trees stood 8 feet apart in rows 22 feet apart. No grazing was permitted there, but berries and truck crops were put out. I couldn't keep it up. The reason: a World War, and lack of help for the intensive type of farming required for the project. Finally, when I attempted to interplant the rows with fast-growing trees, weeds choked out most of them in spite of my own efforts. My own physical and time limitations defeated me in the interplanting undertaking.

This leads up to an enumeration of my mistakes. First, I did not start early enough in life. The elements of health and strength have their part in success. Then, too, let us see what might have been the result if I had started at the age of 20. Remember, in this first tract of 20 acres I planned a forest plantation of selected black walnut seedlings, some chosen for nut quality and some for large, straight timber growth. A tract of 20 acres planted 8 x 8 x 22 feet will hold about 4500 trees. Allow for thinning and other reductions. If only 1250 trees should reach log size in 50 years, that is, by today for me, at an average of $50 each, they would come to $62,500—a very tidy estate.

Just now there are perhaps 2500 well grown trees in the good portion of the ground in this 20 acres. Pleasantly enough, they do not now seem to need the interplanting of faster-growing trees in order to develop upright growth but are pushing each other up as they stand, 8 x 8 x 22 feet apart. In this planting, then, there is evidence of successful timber growth in the good ground but of almost complete failure in the poor ground.

Another failure is to be noted in my original plan for cattle guards. These guards were 12 feet in diameter, and about 6 feet in height. These were satisfactory for sheep after I had installed pipe for posts, but not for cattle. Trees grow horizontally as well as vertically. Cattle, reaching for these side shoots, reached over the guards and pushed in and under. I later reduced the guards to a 6-foot diameter of stronger woven fence-wire with 6-inch stays, not 12-inch, and raised the height to not less than 10 feet. The cattle may now nibble off the side shoots if they wish but the vertical growth is protected. Above 10 feet the trees can spread out without danger.

Others say, "Permit no grazing at all." This statement, I think, should be made with certain qualifications. Where bluegrass bottom is used for the orchard planting of pecans or black walnuts, there is a possible slight reduction in growth from lack of cultivation, but this loss will be nowhere nearly proportionate to a farmer's loss of pasturage. And even in my 8 x 8 x 22-foot planting of seedlings, though no grazing was permitted while the trees were young, now the older trees are large and strong enough in the good soil to take care of themselves. Some lower branches are rubbed off but they should be off anyhow. Also, thank heaven, the weeds are at last kept down by grazing, the grass is utilized and, most important of all, the hazard of grass fires is entirely wiped out. I know of a neighbor's planting destroyed in this way and I shall always fear fire. I should not permit grazing in a general purpose woods lot where young growth is constantly coming on.

Failure three: I have failed completely to interest my tenant in my project. Each mowing or clean-up job is just a chore to him. I can't blame him. Why should I expect anything else? With a World War on hand, and with his son in the army, and with two farms to care for, the immediate bread-and-butter jobs come first and my mowing suffers. However, the wonderful trees somehow continue to grow in spite of weeds and wars, perhaps a bit more slowly than they otherwise might, but I am in no hurry.

The last war casualty was my original plan to make a further orchard planting of seedlings in loco, ready to be top-worked to the wood of some outstanding find among the selected seedlings. It has not been done—period.

I think I do have one or two rather outstanding nuts among the seedlings, but this leads up to another casualty which must be faced by all of us—a temporary one, fortunately, namely, crop failures due to the weather. The larger trees began to bear at age seven. Then, three years ago we had a drouth. For the two years since then, we have had summer in March and winter in May. The catkins were mostly killed and the pistillate bloom was delayed in growth upon the new wood until most of it came too late for even such pollination as was so sparingly available. Thus we have had no generally good nut producing season for three years in our part of Ohio. As a result, my truly outstanding nut is still in hiding, and I am waiting for a good season to bring it out.

Another disappointment with me has been the Carpathians. They partially winter-kill each winter. Their trunks still live and send up shoots. I let them stand, hoping for an eventual hardening of the wood. I regard them not as failures but as not yet proven.

For purely experimental purposes I planted apple and peach trees close up to the walnuts. Whichever won out was to stay. Both are there yet. There is as yet no sign of the results of toxicity. They stand, literally, arm in arm.

One success I feel may safely be chalked up. In selecting seed for my original planting, some were chosen for better nuts, as stated, and some because of the magnificent growth of the parent trees. One such tree gave me seedlings that are definitely superior in growth to other trees which stand in equally good soil—in fact, in adjoining rows. This is noteworthy.

As for the seed selected for nut quality, because of the three poor producing seasons now past, the result is not so apparent. I can only say that, out of some score or more sources, the nuts produced upon such seedlings as have fruited tend to resemble the qualities of their parents. They all show some variations. Each nut tree is a new individual but with a family inheritance strongly enough marked to make the planting of seedlings, when done in large quantities, from the best parents, a sort of gamble in which the percentage is in favor of the gambler—which is, as you should know, unusual.

One utterly complete failure must be noted. I shall never again plant a black walnut seed or tree in any but good soil. Even the best inheritance cannot prevail against hardpan or worn-out soil.

I was unfortunate, when I made my first and largest planting of seed, in not knowing about the Northern Nut Growers Association. So I advertised for local nuts, paying double for the seed I accepted. So far as the seed which was selected because of the timber growth of the parent tree was concerned, I am well satisfied. But nut quality was only fair; far below the quality of our named varieties. Then, through the fine missionary work of Harry Weber, I was introduced to the NNGA. All my replanting since then has been from seed bought from the member's plantations. Next year I expect some of them to come into bearing.

Most of you are chiefly interested in grafted or budded trees, and this is as it should be. Where sure results and the best possible nuts are the aim, one would be utterly foolish to plant a seedling. Upon the other hand, where plantings are made in great quantities, as is the case with foresters, state or federal agencies, colleges and other institutions—and with occasional individuals like myself who find their greatest interest in this particular exciting gamble, I think it is fairly well demonstrated that the percentage of success can be turned in favor of the planter by intelligent selection.

But where can the best seed be found? The answer is as plain as the nose on your face. The best possible source is in existing plantations of named, proven varieties. As a farmer, I should not use a cross-roads maverick when I can use a registered Jersey, Hereford or Angus. As a planter of black walnuts, or any other nuts, either for timber or wood, I should not pick up my seed haphazardly from cross-roads trees. Every nut produced by planters of orchards of the best named varieties should be in active demand by state and national agencies for their own plantings, and the seedlings from them should be available for the widest distribution to the public. This urgent demand for better seed will make existing plantations of proved varieties more profitable and will fill our forests and farms with far better trees.

Nut Trees in Wildlife Conservation

By Floyd B. Chapman Ohio Division of Conservation & Natural Resources

Attesting to our great faith in the value of the nut trees for wildlife conservation and restoration, the Ohio Division of Conservation and Natural Resources has distributed free of charge, to cooperating landowners: 132,000 American hazelnut, 1000 European and American hazel hybrids, 1000 pecans, 1000 butternut, over one thousand shagbark hickory, 1500 Asiatic chestnut, 2000 black walnut trees, and more than 50 bushels of black walnut nuts for seed spotting. This program has only been in operation since 1942, and I think a great deal has been accomplished in spite of the war and difficulties in growing and shipping of nursery stock. This record would not be so impressive had we not been able to take advantage of a vast amount of surplus stock made available when the U. S. Soil Conservation Service closed out a large nursery in this region.

To show how dependent are certain wildlife species on an adequate supply of nut mast, I need only mention one group, the squirrels. Much information concerning the abundance of squirrels in the original forests is on record. It is also well known that nuts of several kinds were always plentiful: native chestnuts, walnuts, butternuts, hazelnuts, hickorynuts, and beechnuts. The supply was so large that an occasional crop failure was unimportant; much of the production from the preceding year was still available. Numerous wild animals, including squirrels, deer, rabbits, raccoons, and others fed on the native chestnut. It was such an important staple in the diet of many animals that its passing is one of the most devastating blows to befall the wildlife of this continent. In order to compensate for the loss of the chestnut, and at the same time restore some of the food and cover destroyed through pasturing of woodlots, and the removal of fencerow cover in clean farming, the Division of Conservation instituted its popular tree and shrub unit planting project four years ago. In this program, units of 100 or 200 pine trees and shrubs for food and cover are distributed free of charge to farmers who will plant them as suggested and protect them from fire, grazing, and cultivation. American hazelnut was extensively used in this project during the first two years. Since then we have been unable to obtain seedling plants in the large quantities that are needed.

The Division also has several other wildlife restoration projects in which the nut trees are utilized. These are a farm pond project, a small wildlife refuge program, and a fencerow cover restoration proposition. In the pond development program, a farmer is assisted in impounding a small body of water, from which livestock is fenced, if he will agree to permit hunting on a portion of his farm. The pond margins are seeded to a grass mixture to prevent soil erosion and silting, and several hundred trees and shrubs having value as wildlife food and cover are planted in the area. The land immediately surrounding the pond becomes a wildlife refuge where no hunting is permitted. Many Asiatic chestnuts have been planted on these sites, in addition to American hazelnuts, and considerable seed spotting with black walnuts has been accomplished.

In the small refuge plan, areas are selected, developed for wildlife by planting and other management measures, and are then closed to hunting for a period of years. Many hazelnuts, butternuts, some hickorynuts, walnuts and Asiatic chestnuts have been used in this work. Our own field men plant the seedlings or assist the landowner in planting them, then give advice on the culture of the plants.

In the third undertaking, which is research to determine the best methods of restoring or developing fencerow food and cover strips; nearly a thousand hazelnut hybrids have been planted. Among these hybrids are: Barcelona x European Globe, avellana x Italian red, Barcelona x purple aveline, Barcelona x Cosford, Barcelona x Italian red, Rush x Kentish Cob, and Barcelona x various other types. The better sorts of hazelnuts have been used in this project to familiarize the farmers with them so that they will have an incentive to grow something valuable in fencerows. We have found that most farmers will not listen to the argument of growing anything in fencerows purely for the benefit of wildlife. By using a more subtle, convincing, and practical approach, we are convinced that success will be attained and that wildlife will be benefitted in the end.

In addition to these projects which are of a statewide nature, the Division of Conservation owns some 14,000 acres of game lands on which experimental plantings of nut trees have been made. From plantings of Chinese chestnuts established in 1941, we are now beginning to realize definite returns. On the Zaleski State forest game area, one of these trees, now about 6 feet high, is bearing 21 burs this year. In connection with a squirrel research problem, one of our field men, Robert Butterfield, is carrying on some experiments in fertilizing nut and other trees which should yield some very valuable information. I recently saw a plot of Castanea mollissima which had been treated with a 33-1/2% nitrogen fertilizer. Planted on poor, acid, eroded soils in the hill country, these have barely survived. After treatment, the yellow, stunted foliage changed miraculously to a striking dark green, the leaves grew larger, and the entire plants showed every evidence of healthy growth. It has been suggested that interplanting chestnuts with black locust might have the same beneficial effect and we intend to try it.

None of us has ceased to hope that some day the blight which has stricken our native chestnuts can be conquered. We can be assured that whenever a resistant variety of chestnut does originate in the wild, squirrels will find it and give it widespread distribution. In Ohio, squirrels are still proficient in locating the few sprouts that are fruiting, burying the nuts and forgetting them in the woods each year, with the result that we always have a few seedling trees coming on. Last spring, I found several bushels of chestnut burs cached in a sandstone cave in southern Ohio by woodrats.

The States which are most interested in the nut trees from the standpoint of wildlife are usually those in which squirrels or wild turkeys are important game species. If those who are growing nut trees commercially would concentrate their efforts in these states which extend from Pennsylvania to Missouri and throughout the south, I think they would be helping themselves and contributing in an important measure to wildlife conservation and recreation. I think many States, and I know this is true of Ohio, would like to introduce some of the better named varieties of walnuts, hazelnuts, filberts, and other nut trees to the landowners of the State through conservation projects which I have described, but the cost is thus far too prohibitive for stock which is distributed by us free of charge. I am personally interested in the fine program of nut tree research which is being initiated in Ohio and elsewhere. The hill culture experiments are especially interesting and valuable. However, I believe every grower should give increasing attention to the possibilities of nut trees in conservation, to the end that better and more prolific varieties can be made available for this purpose. States which can use good nut tree stock in their conservation work should be solicited, their interest aroused in plantings for the dual purpose of home use and wildlife, and a few select varieties sold or given to them each year for experimental use. Some growers are already generous in releasing a few new and promising nut tree varieties for trial growing in various sections of the country.

Most Conservation departments are financed on an annual basis with funds from hunting and fishing licenses. This prevents our knowing from year to year exactly what our requirements are going to be in the line of planting material. Such stock cannot be contracted for even one year prior to purchase. We have no Division-owned nursery for propagating game food and cover plants, and nearly all hardwood stocks are purchased from commercial nurseries. Most states prefer to purchase nursery stock that is grown locally, and if nut growers could succeed in lining up their own state conservation departments, I am sure that they could expand their production to furnish the stock needed, both at a profit to themselves and at a price we could afford to pay.

Commercial Aspects of Nut Crops As Far North As St. Paul, Minnesota

By Carl Weschcke, St. Paul, Minnesota

For the benefit of those new members who are not familiar with my nut tree plantation at River Falls, Wisconsin, I wish to explain its geographical conditions. Situated in the 45th parallel, longitude 92-1/2 deg., about 860 feet above sea level, this is a very severe climate for growing most species of nut trees. Fortunately, I did not realize that fact 30 years ago, and I learned a great deal about the hardiness of many species and varieties and the difficulties of growing them before I was convinced of it. My optimism in those years so ruled me that I was influenced by it to try out such tender species as almonds, English walnuts, filberts, pecans and chestnuts, along with hardier types such as butternuts, black walnuts, hicans, hickories and hazels.

To give you a rough idea of the testing I did, I will mention some of my work among hickories. I was fortunate enough to have a forest of bitternut trees on my land. It is a well-known fact that, at least temporarily, these bitternut hickories lend themselves well as grafting stock for many superior varieties of hickories, hicans and pecans, although the last species seldom is considered permanently compatible with bitternut. The number of varieties I tested on bitternut stock is roughly about 75. During the years since I started such grafting, most of these have been lost by natural elimination, lack of hardiness or incompatibility. Those varieties which on my place have proved hardy and compatible with bitternut stock for at least ten years are: Bridgewater, Cedar Rapids, DeVeaux, Glover, Kirtland, and Weschcke. Those which have endured well on this stock for from 6 to 15 years are: Barnes, Davis, Fox, Leonard, Milford, Netking, Platman, and Taylor. Among hybrids which have stood for 10 years or more, there are: Beaver, Burlington, Laney, Pleas, and Rockville. Of pecan, there are Hope and Norton. There are a few other survivors of whose identity I am not certain, as they have not yet fruited. This does not mean that all of those listed have borne, but only that the identity of some of the survivors can not be established without such verification.

Preeminent among the hickories which have produced nuts, stands the Weschcke variety, which has borne the greatest quantity with the most regularity. This variety, grafted on bitternut in 1932, produced one nut that year. Its bearing record has been unbroken from then to 1946, when, on May 11, the temperature dropped to 26 deg.F and on May 12, a similar, low temperature was accompanied by four inches of snowfall. Pictures I have on display verify these statements. The frost at that time destroyed the whole crop in a nearby 30-acre orchard of apples, pears, plums, and nuts. Although the first growth of Weschcke was totally destroyed along with the crop, the second growth contained a fair distribution of pistillate flowers which probably would have produced nuts, had they been pollinated. The Weschcke produces no pollen, being one of those curious freaks of nature which aborts its staminate flowers before they reach maturity.

Other hickory hybrids and shagbarks which have borne satisfactory crops on my farm, with fair regularity, are the Beaver, Fairbanks, Bridgewater, Cedar Rapids, Kirtland, Siers and Laney, in the order of their worth. The remaining varieties that I mentioned have not yet fruited, although I hope they will do so.

The facts I have given are my reasons for recommending the Weschcke hickory as a tree suitable for commercial use in the north. I realize, of course, that farther south, where hardiness is not so essential a quality, other trees may be just as satisfactory. I might also mention that the size and cracking qualities of the Weschcke variety are also commendable. The quality of the kernel, which is practically 50% of the total weight of the nut, is praised by all who have tasted it.

It is with great regret that I admit that I have no black walnut varieties which I can recommend for commercial use this far north. However, I would place Ohio ahead of Thomas, because of its greater hardiness. The ease of hulling, the size and appearance of Thomas, plus its productivity, would certainly place it first were it not for the frequent winter-killing it suffers, to which Ohio, of course, is not completely invulnerable. Other varieties which have been fairly satisfactory but which are not as well-known, include Patterson and Rohwer. The fact remains, however, that not one black walnut I have tested has produced a regular and satisfactory crop, although they have been more productive than native butternuts. At present, I would rule out both species as apparently having no commercial value in the northern climate where my plantation lies, although they may be satisfactory for home orchards.

Before leaving the hickories, I do want to mention that I feel there is a good chance for growing pecans in this climate. I have seedling trees, now more than 20 years old, which are in bearing but do not mature their fruit. It is possible that some of these may become acclimated to an extent that their cycle of dormancy will reduce itself, bringing their period of flowering early enough in the spring to allow sufficient time and heat units for maturing the nuts.

Early in my experimental work, I tested chestnuts and chinkapins but met with poor results. Only in the last few years have experiments with them been successful enough to warrant their being mentioned as commercial possibilities in the north. At present, I have several Chinese and two American varieties, as well as one chinkapin, which have proved hardy and fruitful. Further testing is necessary before I can report anything definite about them.

I have grafted on native plum stock most of the almonds which have been considered hardy, including the hard-shelled varieties from Michigan and the Northland from the Pacific Coast. Some have flowered but none have set nuts. All proved too tender for our climate. I feel more hopeful for success with some of the many seedling hybrid plums I am growing. A number of these have edible kernels and the trees could be considered for their fruit as well as for the kernels of their seeds.

Among other species of walnut I have tested is the heartnut, which is a sport of the Japanese walnut. This is a worthy nut and has done well when grafted on black walnut stock. Only two varieties have proved hardy and only one of these, Gellatly, has produced good crops for a long time. Were it not for the insect pests which attack it and, worse still, the sapsucker, this tree might be considered for semi-commercial use in the north. The sapsucker is a woodpecker. It chips out bark right down to the wood, girdling large limbs and killing whole sections of a tree. This results in an excessive amount of succulent, tender growth which is subsequently winter-killed. Insects attack the new shoots, laying their eggs in the bud and stem portions, causing immature growth which stunts the tree and prevents its bearing. I have also found the heartnut difficult to graft, even on black walnut, which is a favorable combination.

I began testing Persian walnuts 30 years ago by grafting them on wild butternut stock. Although many grafts were successful, not one even lived through a winter. It was not until 1937, when I grafted hundreds of trees with thousands of grafts of the many varieties of Crath importations from the Carpathian Mountains, that I succeeded in getting any to survive our winters. A few eventually bore nuts, but the severity of our winters and the inroads of new insects during the war years finally proved fatal to them. I made strenuous attempts to save the varieties by regrafting, but I was wholly unsuccessful. Right now, I am not at all hopeful that Persian walnuts of any kind can ever survive very long this far north.

We now come to the last group of species mentioned at the beginning of this report, namely, filberts and their hybrids. In my opinion, these have potentialities of commercial value in the north. Even the frosts of May 11th and 12th this year (1946) did not wipe out the crops which had been set. With proper pollinization, I am certain that their production will become as reliable as the corn crop in this part of the country. At the banquet, I shall give each of you a sample of a new product made from these nuts.

The combination of qualities of the cultivated filbert from Europe and our wild Wisconsin filbert results in an extremely hardy plant, with characteristics sometimes like the former, sometimes like the latter. Many times, the hybrid combines the best of each. I am testing these for field culture, to be cared for much as corn is. I expect to have three experimental farms before very long, demonstrating the success of commercial orchards of these hybrids which I call "hazilberts." "Hazilberts" is a word I coined by borrowing from the names of its parents. It has been readily accepted by the lay public and is easily understood to refer to hybrids between hazels and filberts. Furthermore, I was able to obtain a U. S. trademark on this for application to these plants.

Hazilberts are all subject to the native hazel blight, ~Cryptosporella anomala~, a fungus infection. They are also susceptible to another blight similar to the bacteriosis of the Persian walnut. More serious than these, though, is the damage caused by a curculio, which cuts down heavily the production of nuts if measures are not taken to combat them. Breeding has demonstrated that some hybrids are so resistant to the inroads of this pest that they may almost be considered immune, especially when they are interplanted with other hazilberts which do attract curculios and so act as trap-plants. In this way, the insects are encouraged to concentrate in one place where they may be poisoned, thus protecting the main-crop plants. Since pollinators are required for filberts anyhow, the pollinators may be the trap-plants. This is actually the case in the initial plantings. Clean cultivation will also do away with many of the curculios, since they depend on unbroken soil in the fall for their metamorphosis.

The presence of blight makes it unwise to depend on a single-trunked tree and I find that great productivity can be maintained when the plant is allowed to grow in stools having from three to five trunks. The management of such plants is like that of raspberry bushes, except that instead of thousands of plants per acre to be cared for, with hazilberts there are only 145, 15 x 20 feet apart.

Judging by the number of nuts on small plants, one may reasonably expect crops to average one-half ton of nuts per acre. The hybrids I have grown so far have been self-husking. The size of their nuts is good, some measuring an inch in diameter. For commercial purposes, however, the large size is not particularly desirable nor necessary.

In conclusion, I want to say that there is a very promising situation developing for these nuts commercially. Not only are these hazel-filbert hybrids easily planted, but they are easy to propagate, since they are one of very few species of nut trees which are easily propagated by layers and root sprouts. Out of more than 600 hazilberts which I planted in the fall of 1945, only about a dozen were dead in June of 1946, which gives you a practical idea of the ease and safety of transplanting them.

The 1946 Status of Chinese Chestnut Growing In the Eastern United States

By Clarence A. Reed U. S. Plant Industry Station, Beltsville, Maryland

Introduction

The Chinese chestnut, Castanea mollissima, now dominates interest among well-informed chestnut planters of the eastern United States almost to the exclusion of other species. Since its introduction in 1906, it has had but one important competitor, the Japanese chestnut, C. crenata. Among the world's most important producers of tree chestnuts, only these two species are effectively resistant to blight. However, the Japanese chestnut lacks the palatability to which Americans are accustomed and for all practical purposes it has been rejected in this country. Many small plantings still survive; but this species serves better for shade and ornamentation than for food production.

Description of the Chinese Chestnut

The nut of this species is usually of good size, roundish in form, not pointed at the apex, and with the basal scar smaller than the lower end of the nut. A certain amount of gray down is on the surface. This down may be confined to a small area about the apex or it may cover much of the upper end of the nut, and it may be thick, thin, or scant. The nut may have good cleaning quality, meaning that the kernel and its pellicle are easily separated. Cleaning quality may be good from the time the nut falls from the tree or it may become so only after curing for a time. Once it develops it may remain good as long as the kernel is usable or it may last for a short while only. In texture and in palatability, the kernel of the Chinese chestnut is not excelled by any other true chestnut. Individual nuts are sometimes sweet from the first but the great majority become so only after being cured for a week or 10 days. Very few nuts of the pure species fail to be sweet when fully cured.

In the open the Chinese chestnut tree attains much the same size and general proportions as does the apple but it may become somewhat larger, more upright and considerably taller. Young seedlings vary greatly in form and are often ungainly and unsymmetrical; but others are all that could be desired with respect to symmetry. Early lack of symmetry tends to become less objectionable as the tree grows older and is seldom conspicuous after the first decade or so.

In fruitfulness, many of the seedling trees of bearing age are definitely disappointing. Also in many cases the nuts are small. To judge the species by the past fruiting performance of a majority of its representatives in this country would leave little justification for commercial hope. However, there are a good many individual trees about the country whose performance record is excellent and a large number of these are under careful observation as potential varieties.

The species has gained rapidly in popularity since the middle 'thirties when enough good-performing trees began bearing for a fair appraisal of the species to be possible. It was also at about that time that trees for planting began to be available from the nurseries. Before then trees could only be had in limited numbers from the Department of Agriculture. Today, they are listed in nursery catalogues of one or more firms in each of a half dozen or more states. The total number of trees yet planted is comparatively small and both nurserymen and planters up to this time have proceeded cautiously because of the newness of the industry and its uncertainties.

Environmental Requirements

The Chinese chestnut requires much the same conditions of climate soil, and soil moisture as does the peach, but there are indications that it will succeed somewhat farther both north and south. As with the peach air drainage must be good and frost pockets must be avoided, for while at the latitude of the District of Columbia, the flowering period is from late May until toward the end of June, growth begins early and may be badly damaged in April. This is especially true during such seasons as those of 1945 and 1946 in the middle Atlantic States when summer temperatures prevailed during a great part of March, and new shoot growth up to two inches had developed when sub-freezing temperatures killed all new growth and so injured the buds that at Beltsville, Maryland, and general vicinity there were no crops in either year. In some cases young trees were killed out-right as were occasional older trees that had become devitalized in some way.

Young trees are so sensitive to lack of soil moisture that sometimes whole plantings are killed by drought. Spring growth is rapid as long as the soil is moist but root development is shallow during the first few years and, unless watered, trees are likely to fare badly in case of prolonged drought. Another serious type of injury, especially to newly planted trees, is sunscald on the exposed sides of the trunks. Probably the best means of prevention is to head the trees low enough to provide for shading by the tops.

It is said[1] that at the altitude of 2200 feet in West Virginia, snow and ice frequently cause much injury to young trees. It is a notable characteristic of the species for young trees to retain their leaves during much of the winter. Unless these are removed soon after turning brown, they are apt to become heavily weighted with wet snow and to cause severe breakage. Hail and spring freezes also cause much damage in that locality. The last, however, is not peculiar to high altitude alone as frost injury is frequent at much lower elevations. It was generally in evidence in central Maryland during the springs of 1945 and 1946 as has already been mentioned. This type of injury is easily overlooked, but the cambium will be found dark if a cut is made through the outer bark. Recovery usually takes place rapidly if the injured trees are left undisturbed, but healing will be slow if they are dug up for transplanting or the tops are severely cut back in preparation of the stock for grafting.

[Footnote 1: Verbal statement by Mr. Authur Gold, of Cowen, W. Va., made during April, 1946.]

Bearing Ages

Young trees may bear a few nuts three or four years after being transplanted, but it usually takes from 10 to 12 years for tops to become large enough to produce profitable crops. While there are occasional trees that become profitable at these ages, there are many that do not. The only significant record of yields yet made public is one reported by Hemming.[2] His statement shows that 18 seedling trees planted in 1930 bore an average of 29.5 pound (green weight) during six of the eight years from 1937 to 1944, inclusive, when crops were large enough to be separately recorded for each tree. The range in total production per tree for the six years was from 106 to 277 pounds. At an arbitrary price of 25 cents a pound, the average gross return per tree would have been $7.39 for each of the six crops. The 1944 crop was a practical failure. That of 1946, amounted to about 1000 pounds, or an average of about 55 pounds per tree.

[Footnote 2: E. Sam Hemming, Easton, Md., "Chinese Chestnuts in Maryland," Ann. Rep't., Nor. Nut Growers Association, Incorporated, vol. 35, pp. 32-34. 1944.]

The Seedling Tree

The original planting stock of the Chinese chestnut as grown in the United States consisted wholly of seed nuts imported direct from the Orient. It was therefore, inevitable that a period of seedling development should follow. The great majority of the earliest trees grown proved unfit for use as potential varieties, although with some exceptions, they produced nuts that were sweet and palatable. Since the middle 'thirties, superior strains have been introduced, cultural and environmental requirements have become better understood, and the outlook for commercial orchards is much improved.

To a great extent the seedling has served as well as would a grafted tree for the pioneer experimental work that had to be done. It has been far better than no tree at all and even now it has its advantages. With it there is no expense for grafting, no problems of congeniality between stock and scion and those of cross pollination are held at a minimum. Moreover, it must not be forgotten that it is only from seedling trees that superior varieties are possible. In 1946, the year in which this paper is being written, very few grafted trees are available from any source.

The Grafted Tree

The first varietal selections were made in 1930. Quite unavoidably they were chosen solely by what could be judged from the nuts with no knowledge of the bearing habits of the parent trees. These were first grafted in 1932 and first catalogued in 1935. Already by 1946, some had been supplanted by others of greater promise. Few grafted trees have been brought into bearing and with minor exceptions, it has not been possible to obtain bearing records. It is, however, mainly with the grafted tree that the future of the industry is expected to be built up.

Individual Varieties—Abundance

This variety was first catalogued in 1941 by Carroll D. Bush, then a nurseryman at Eagle Creek, Oregon. Of the very few trees of this variety sold by him, one went to Mr. Fayette Etter, Lemasters, Penna., with whom it early became a favorite among 7 or 8 he had under test. During 1945, he sent a quantity of Abundance chestnuts to Dr. J. Russell Smith, Swarthmore, Penna., who in turn forwarded 12 specimens to the Plant Industry Station. These arrived October 11 and were immediately placed in a refrigerator. On October 22, they averaged 50 to the pound and ranged from 38 to 76. The appearance was very attractive as the color was a rich brown and there was very little down over the surface. The cleaning quality was also very good and the flavor excellent.

The Abundance has attracted considerable attention and, while it does not appear to be listed in any nursery catalogue, a number of leading growers are using it in top working seedling trees and it may soon be available through regular nursery channels.

Carr

The Carr chestnut originated as one of two seedlings sent by the Department of Agriculture in 1915 to the late R. D. Carr, Magnolia, N. C. Sixty-two nuts from Mr. Carr were received by the Department in 1930. These were not especially attractive as the surface was thickly coated with gray down. The lot averaged 58 per pound and the nuts were considered large. Cleaning quality was very good and the flavor was sweet and pleasing. The variety was immediately named in honor of Mr. Carr although propagation did not begin until 1932. It is believed to have been the first variety of the species ever grafted in this country. The work was performed by H. F. Stoke, Roanoke, Va. Later the Carr became available for several years from a number of nurseries. It was a strong grower but often failed to make good unions with its stock and is not now in general favor.

Hobson

This also originated as one of two seedling trees sent to a private grower by the Department. He was Mr. James Hobson, Jasper, Ga., in whose honor it was named in 1930. It was later taken up by commercial nurserymen and widely distributed for several years. It has much in its favor as it is easy to graft, precocious, prolific, annual in bearing, and the nuts are very sweet. Also, the cleaning quality is very good, but the nuts are too small to meet market requirements of this country to best advantage. Furthermore, being small, they are expensive and time consuming of labor at time of harvest. The average per pound for a lot of 110 nuts received in 1930 was 78. Others received during later years were even smaller. The variety rapidly lost favor with most nurserymen and its propagation was largely if not entirely discontinued. However, for home use, it is much to good to be abandoned at this time.

Reliable

Reliable was an introduction of H. F. Stoke, Roanoke, Va., by whom it was propagated for a short time only, beginning in 1938. It is not known to have been catalogued by any other nurseryman. Ten fresh nuts in 1939 averaged at the rate of 79 to the pound. Six days later, after further curing had taken place, the number became 101 to the pound. Aside from having a good bearing record, there appears to be little reason for continuing this variety.

Stoke

This variety appears to be the result of a natural Chinese-Japanese cross. The original tree was grown by H. F. Stoke, Roanoke, Va., whose attention was attracted to it because of its habit of maturing early. He reports that in southwestern Virginia, burs often begin opening during the third week of August. In appearance, the nuts greatly resemble pure Japanese. The parent tree bears well but the nuts are lacking in good quality. Insofar as known propagation has been discontinued.

Yankee (Syn. Connecticut Yankee)

The Yankee originated as a chance seedling on property of E. E. Hunt, Riverside, Conn. It was first propagated by Dr. J. Russell Smith, Swarthmore, Penna., in northern Virginia by whom it was first catalogued in 1935. The writer has seen no specimens but according to Dr. Smith, the size and other features are very good. The parent tree is said to bear well and to be hardy where it is located, which is not far from Long Island Sound in the extreme southwestern corner of Connecticut.

Zimmerman

This originated as a 1930 selection made by the late Dr. G. A. Zimmerman, Linglestown, Penna. Very few sound nuts of Zimmerman have ever been produced, for soon after the first crop the identity of the tree became lost and eventually it was destroyed together with others in an overgrown nursery row where it stood. In one known case where there are grafted trees of bearing age, the nuts are regularly destroyed by weevils. Such nuts as have been seen by the writer have been of a dull brown color and have had surface down only about the apex.

The Zimmerman was first catalogued in 1938-39 by Dr. Smith. It is probable that as many trees of this variety have been sold and planted as of any one variety but performance records are difficult to obtain.

Potential Varieties

Other varietal selection are being made, mainly by the Bureau of Plant Industry, Soils, and Agricultural Engineering from trees at its various field stations. Some of these are already under test as grafted stock in various parts of the country. The most promising will be released to commercial nurserymen as soon as their superiority over existing varieties is established.

Pollination

There is much evidence that chestnut pollen is largely carried by insects although this has not been fully established. The Chinese chestnut is largely, although apparently not wholly, self sterile; more than a single seedling or grafted variety should be included in any planting. Several seedlings or several varieties would be better. In seedling plantings, all trees that produce inferior nuts should be removed in order to avoid danger of undesirable pollen influence, either on nut characters, or on the genetic makeup of the embryos if the nuts are to be used as seed.

Harvesting and Curing

Chestnuts should be harvested daily as soon as some begin to ripen and drop to the ground. They should be placed at once on shelves or in curing containers with wooden or metal bottoms through which the larvae of any weevils with which the nuts may be infested cannot penetrate and reach the ground. In areas of infestation, these grubs soon begin to bore their way out of the nuts and leave conspicuous holes in the shells. All infested nuts should be promptly burned.

In order to cure chestnuts to best advantage, they should be spread thinly on floors, or on shelves, or in shallow containers as just described, and held in a well-ventilated room. They should be stirred frequently and held for from 5 to 10 days depending both upon the condition of the nuts and the atmospheric conditions at the time of harvest: During the period of curing, the nuts will shrink rapidly in weight and the color will change materially.

Both luster and brightness will largely disappear and, although still attractive, the nuts will quickly become dull brown. Three weeks is about as long as Chinese chestnuts usually remain edible without special treatment.

Chestnuts should be marketed as promptly as possible both to minimize deterioration and to take advantage of good prices which are usually highest early in the season.

Storing

Chestnuts in sound condition when stored may be kept fit for eating or planting for several months by any one of several methods. When available, cold storage with temperatures somewhat above freezing is the simplest and generally the most satisfactory method. Stratifying method. Stratifying in a wire-mesh container buried deeply in moist but well-drained sand is very satisfactory and successful. Another method is to hold the nuts in a tightly closed tin container either in a refrigerator or in cold storage at 32 deg. F. Burying under a porch or in the shade of a house or even in a bin of grain, preferably wheat or rye, is also a good method. Regardless, however, of temperature or other conditions, germination is likely to begin in early March and nuts intended for planting should be hastened into the ground as promptly as possible after that time.

Insect Pests

The two chestnut weevils are the principal insects attacking the nuts. These are exceedingly well-known in certain large areas where the chestnut is grown and in these areas both are often extremely abundant. Unless checked in some way they often render whole crops unfit for use. One of most effective means of control is to plant trees only in well populated poultry yards; however, in large developments, this is impracticable and other methods must be employed. In preliminary work carried on by the Bureau of Entomology and Plant Quarantine at Beltsville, DDT has given very encouraging results in the control of the weevil. The weevils have sometimes been called curculios, under which name they were well discussed by Brooks and Cotton.[3] The Japanese Beetle is also a serious pest as chestnut leaves are among its favorite foods. Control methods have been given by Hadley.[4] Another insect pest which feeds on the leaves is the June bug or May beetle. It works mainly at night and feeds on the newest leaves. It is seldom seen and usually disappears about the time when the operator becomes aware of its presence.

[Footnote 3: Fred B. Brooks and Richard T. Cotton, "Chestnut Curculios."
U. S. Dept. Agr. Tech. Bul. 180. 1929.]

[Footnote 4: C. H. Hadley, "The Japanese Beetle and its Control." U. S. Dept. Agr. Farmers' Bul. 1856. 1940.]

Diseases

Blight is the disease attacking the chestnut tree with which the public is most familiar. The Chinese chestnut is strongly resistant although not immune as few old trees entirely escape attack in areas where blight is prevalent. In most cases healthy vigorous trees of this species overcome the disease within a few years after being attacked. The ones that die are usually those that have been devitalized in some way. The nuts are subject to attack by any of several diseases either before or after the harvest. A preliminary report on these has been made by Gravatt and Fowler.[5]

[Footnote 5: G. F. Gravatt and Marvin Fowler. Nor. Nut Growers Ass'n. Proc. 31: 110-113. 1940.]

Present Extent of Planting

With few exceptions the known plantings consist of small numbers of trees about residences. Occasionally there are one or two hundred trees in orchard arrangement. Production is not large and in most cases all sound nuts are either consumed locally or used by nurserymen and others for planting. The quantity that has reached the wholesale market is known to be small although a beginning in that field has been made.

Future Outlook

Extensive expansion has not appeared possible in the near future until the 1946 crop was harvested. This was unexpectedly large and a number of tons are known either to have been planted immediately or set aside for planting in the spring of 1947. It is conceivable that annual production of nuts available for seed purposes will increase rapidly. In this case, the extent of planting within the next few years will be entirely a matter of guesswork.

Extensive planting in the early future cannot be considered economically safe for in addition to the usual number of problems that must be solved in establishing any new horticultural enterprise, chestnut growers must expect keen competition with imports from both Europe and Asia. At the outbreak of World War II, an average of more than 16 million pounds of chestnuts were yearly being imported into this country.[6] These imports will doubtless again appear with the return of normal international relations.

[Footnote 6: Computed from Table 541, p. 413, Agricultural Statistics 1938. U. S. Dept. Agr. 19]

Furthermore almost an exact half-century ago, the chestnut outlook was regarded as being so bright that it could hardly go wrong. During the middle and late 'nineties extensive chestnut developments were established in certain eastern districts mainly by use of Paragon and other varieties of European parentage. Thousands of small plantings were developed about home grounds and occasionally there were large orchards. The greatest developments were conducted by top working suckers that sprung up from stumps of native chestnut trees on cutover mountain land. Hundreds of acres were handled in this manner. Without exception, all ended in financial disaster.

Summary

The nut of the Chinese chestnut is an excellent product. It is unexcelled in sweetness and general palatability by any other known chestnut. The tree bears well and is about equally as hardy as the peach. It appears to require much the same conditions of cultural environment as does that fruit. It is practically the only species of chestnut now being planted by informed growers in the eastern part of the United States.

It is thus far grown in this country almost entirely as seedling trees. Variation is about what was to be expected, with the majority of bearing trees proving to be poor producers and, in most cases, with nuts too small to sell well.

Varietal selections of much promise are being made; the first appeared in 1930 and were first catalogued in 1935. Some of the earliest have already been dropped as their defects came to be known, and others of greater apparent promise have originated. The process of selection is constantly going on and further introductions should shortly appear.

By taking certain simple steps chestnuts in sound condition may be kept in usable condition for many weeks.

The Chinese chestnut is subject to attack by certain serious natural enemies. These include both insects and diseases and the tree as well as the nuts are affected. However, all that are now known appear controllable.

Past planting has been largely limited to small numbers of trees mainly about residence grounds. The total number of trees available for planting has never been large, due chiefly to the scarcity of seed nuts needed for nursery use. Production, however, rose sharply with the harvest of the 1946 crop which was unexpectedly large. Annual production may continue to increase since the number of trees of bearing age is likely to become appreciably greater each year. Nursery planting is likely to be proportionately greater. The extent of future planting will doubtless be correspondingly influenced.

Present enthusiasm over the Chinese chestnut is very great and it is possible extensive planting may soon take place. It is believed, however, that this would be unwise from an economic point of view. There are many uncertainties in connection with the industry in its present state of development, and, not improbably there will be keen competition in the market with imported chestnuts from both Europe and Asia as soon as international relations become normal.

Bearing Record of the Hemming Chinese Chestnut Orchard

By E. Sam Hemming, Easton, Maryland

Our Chinese chestnut trees have aroused such interest that we are sure the readers of the Proceedings will wish to hear of the large crop harvested in 1946. A year ago an unseasonal spring brought a frost that killed back the six inches of soft new growth. As a result, the 1945 crop amounted to less than 250 pounds. This year the 18 trees produced 1138 pounds, 938 by actual weighing and 200 estimated. This is an average of 63 pounds per tree, with the largest crop of 124 pounds on No. 19, and the smallest on No. 14 of 22 pounds.

These trees are now 18 years old and were unfortunately planted too close. But using a spacing of 30 feet x 30 feet, they would have borne 3000 pounds per acre and if planted 40 feet x 40 feet would have borne 1600 pounds per acre. Figure this crop at 25c a pound and you would get a really high return. This year the price was much better than that, but we planted the crop.

The tree record was as follows: Number 1—38; Number 2—25; Number 3—30; Number 4—52; Number 5—44; Number 6—30; Number 7—42; Number 8—40; Number 9—45; Number 10—58; Number 11—56; Number 12—48; Number 13—58; Number 14—22; Number 15—50; Number 16—80; Number 18—86; Number 19—124; Total of 938 + 200 (estimated) = 1138.

It is also worthy of note, that No. 19 is spaced 30 feet from No. 18 and No. 16 is the same distance from No. 18, while all the other trees are spaced 16 feet apart. An acre of trees like No's. 16, 18 and 19, spaced 30 feet apart, would average 96 pounds per tree or 4200 pounds per acre, a really tremendous crop.

We had one disappointment this year, in that our method of controlling the weevil was not completely effective. To our chagrin we found that, while we were diligently picking the nuts up each day, some of the larvae were escaping through the cotton bags to reinfest the ground. Next year, we will use metal containers and we are sure that will stop them. We will fumigate if necessary. We do not particularly fear the weevil as we are sure that spraying, and fumigation will clean them up; after that proper harvesting should control them. We have heard that the U. S. D. A. has found the use of DDT to be effective. In another county a raiser of hybrid corn seed dusted his corn with DDT by plane, to kill the Japanese beetle, for $3.00 per acre. Surely that method would be adaptable to chestnut orchards to control the weevil.

At the present time we are using our entire crop for seed purposes and this year we sowed 40 to 50 thousand nuts. We carefully grade the seed, not only discarding any infested nuts, but all moldy, split or undersized nuts, so that we get trees grown only from the choicest. By doing this we feel that although the trees are seedling raised, they come from parent trees that are bearing well, and from which all extraneous pollen is excluded so that the customer has a good chance of getting a tree that will bear well.

The seed is sown in the fall, because it keeps better that way and germinates better too, although we have some trouble from a mole-mice combination. The seeds are sown in shallow trenches 6 inches apart and 2 inches deep and back—filled either with sawdust or light soil. On top is mounded a further 4 to 6 inches of soil which is removed in the spring. This reduces damage from freezing and thawing.

We do not doubt for a moment that the Chinese chestnut is here to stay as an important food crop for the United States.

Walnut Notes

G. H. Corsan, Islington, Ontario

I find the Ohio, Ten Eyck, Stabler, Allen and Wiard black walnuts inferior and unsuitable. The Stabler has only a small crop every five years. Very excellent varieties, I find, come from Thomas seedlings.

The black walnut makes an excellent stock for the Persian walnut in low and slightly damp ground. I bud the Persian on the black during August.

The Japanese heartnut and the butternut x heartnut hybrid can be grafted on black walnut. The Persian walnut when grafted on the black decidedly outgrows the latter. The reverse is the case when Japanese heartnut, Japanese butternut, or hybrids of either are grafted on the black.

So far I have not found one good butternut worthy of naming, but there is one Japanese butternut that grows in clusters of 17 or even more that has a very thin shell; it is the Helmick. I have, however, very many named as well as unnamed black walnut seedlings that are very excellent nuts.

This has been a very cold summer and I cannot state yet as to the maturing of the larger black walnuts, as they require a long season to mature properly. Pecan and hican trees grow well at Echo Valley and the small twigs harden up so that there is never any winter killing but the nuts do not fill well; in consequence I am using the trees as stocks for grafting with good shagbarks. The Weiker hickory ripens nicely with me and I consider it one of the best varieties in every way.

Self-fruitfulness in the Winkler Hazel

By Dr. A. S. Colby University of Illinois, Urbana, Illinois

To insure fruitfulness in nut plants it is generally recommended that more than one variety of each kind be planted in reasonably close proximity to help in bringing about cross-pollination. Then, with other conditions being favorable, the grower would be more certain of good yields of well-filled nuts.

With specific reference to the filbert, the literature contains references to the effect that provision for cross-pollination is essential. However, one exception is listed. In the report of the proceedings of the 26th (1935) annual meeting of the Northern Nut Growers' Association, D. C. Snyder of Iowa says on page 47, "The catkins of Winkler always come through the winter bright and the variety can be depended upon to bear without other varieties near for cross-pollination."

The writer has been interested in this subject for several years. The question arises; how near were Mr. Snyder's Winklers to other varieties and in what direction with reference to the prevailing winds? It is not known just how far filbert pollen may be carried and still function. A planting of Winkler filberts consisting of about 30 bushes was set on the University Farm at Urbana in 1940. Crops have been borne annually since that time. The planting was isolated from other filberts to the southwest by about one-fourth of a mile.

In an effort to determine whether the variety was self-fruitful, plants were dug in the early winter of 1943 after the rest period was over and reset in the greenhouse. The plants leaved out in January, 1944, and both male and female flowers appeared soon after. The pollen was applied to the pistils both by shaking the branches and by means of a camels hair brush. Nearly all the blossoms set and the nuts carried through to maturity. The experiment was repeated in 1944-45 with the same results.

It is therefore concluded that the Winkler filbert is self-fruitful and may safely be planted alone where climatic conditions are favorable for filbert production.

Hickories and Other Nuts in Northwestern Illinois

By A. B. Anthony, Sterling, Illinois

I have something like 25 grafted hickories of my No. 1 (Anthony) variety. The largest tree now has a trunk of 5-1/2 inches in diameter; has 20 nuts on it this year; and while it has had but few nuts each year, has missed bearing but one season in the past seven years. Other No. 1 trees run from 3-1/2 inches, in diameter down to about 1 inch. One 3-1/2 inch tree is offering its second bearing with five nuts this season. All these trees were grafted in cutover woodland tracts and moved here except the largest one which was moved in 1930 and grafted in 1933, 30 inches high and never trimmed for a higher head. Heavy annual catkin bloomer, few pistillates so far.

Of my No. 2 variety, one tree transplanted in 1927 now has something like 25 nuts on it. The No. 3 hickories, five of them, have never borne either pistillate or staminate blooms. No. 4 is a hican from the parent tree of which I have had but three good nuts. The weevil moth works so well in dense woods that rarely are the nuts good there. The nuts are attractive and should not discolor like the lighter hickories, should their opening husks get rained upon when maturing.

Men of the future must decide on the merits of these trees. Of the two Hagen trees grafted in 1931, one now has its first nuts, eight in number. I have been told that some one will cut these trees down some day. One of our county or state officials said a short time back that "if hog troubles keep coming on as of late, in 50 years we will not be able to raise hogs." With corn being the main hog food and the corn borer coming, this may come to be quite true, and then perhaps more men will get new vision as to where their meat is coming from.

The past three years have offered almost no hickories at all. Hickories do not like shade, but they have to grow where the squirrels have planted them. Carrying a nut 100 yards to bury it would doubtless be about a squirrel's limit. I have noticed in timber of sizeable growth a north slope showed no young hickories, while a south slope showed a scattering few. Oak trees in this section predominate when it comes to groves of one species. Cottonwood trees come up here and there, probably because their seed is wind-carried. Willow sticks get carried down stream and get lodged, and grow. I have known a few young oaks to come up on my place all of a mile and a half of such woods. How come? It is probably the combination of the blue jay and squirrel, this time. No trouble for the blue jay to travel some distance and put his acorn in a bark crevice of cottonwood or willow tree. Along comes a wandering squirrel, finds the acorn, and if not hungry enough puts in the ground where it has a chance to grow. I have seen blue jays start off with chestnuts and the nearest trees they could reach were willows one-fourth mile or further away.

For some reason there seems to be a tendency for the hickories to bear in seasons when the black walnut does not and the walnut to bear when the hickory fails. Last year, except for filling, walnuts did reasonably well and this year, at least with my Rohwer variety, the yield is still better except that the nuts are unusually small, doubtless because all of July and up to the 9th of August it was very dry.

Throughout my years there have always been walnut trees on the place, first started by a pioneer land owner, then squirrels took it up, so I have a choice of stocks I did not have in hickories.

Two of my Rohwer trees have trunks 12 inches in diameter; one is 11 inches and the other 14 inches in diameter. For years these trees, grafted in 1931, have been very profuse with catkins, but with few nuts. I have heard other complaints of it not bearing.

My complaint with all walnuts grown in Northwest Illinois is that so many kernels turn out black and immature. I am inclined to blame it, in part, to the walnut shuck, which takes in so much moisture. The hickory shuck is much dryer and never has so many immature kernels. Late summer is generally the dryer part of our growing season, which can well be the cause. In the year 1940, we had an excess of moisture in that it rained day after day all through August, and that is the only season I can say we had good walnuts with practically all good, light-colored kernels.

I have a few Thomas walnuts planted on the edges of the lowest flat ground I possess, hoping that they may there get more moisture and produce completely matured nuts.

We had on August 9th about one inch of rain and since that 2-1/2 inches more. So far, throughout this month, I have been carrying about 15 gallons of water daily to two Rohwer trees and hope for some better filled walnuts, though they are unusually small. I am writing this August 24th.

Nut Trees for Ohio Pastures

By Dr. Oliver D. Diller, Wooster, Ohio

Today I would like to discuss for a few minutes the possibilities of nut trees for shade and nut production in permanent pastures on Ohio farms.

One of the most important developments in Ohio agriculture during the past decade has been improvement of pasture land through fertilization, new varieties, and combinations of grasses and clovers, and better methods of management. As one drives over the State it is evident that many farmers practice "clean" agriculture which means clean fence rows and treeless fields. Shade on a hot summer day is an important item to contented cows, so today I am going to plead the case for a cow out on pasture on a sweltering day. I believe that nut trees, particularly black walnuts, can be of real service in the fence rows and the interior of hundreds of permanent pastures in Ohio.

In 1939, L. R. Neel,[7] of the Tennessee Agricultural Experiment Station, published an interesting article on the effect of shade on pasture. The results indicated distinct improvement in the carrying capacity of the pastures which had black locust and black walnuts spaced regularly throughout the fields. Improvement was evident both in the amount of Kentucky bluegrass and the pounds of beef produced. So far as I know, no evaluation has ever been made of the direct effect of shade on the contentment and consequent increase in efficiency of cattle for either beef or milk production. I believe this is an important factor and is frequently used as an excuse for woodland grazing.

[Footnote 7: Neel, L. R., 1939. The effect of shade on pasture. Tenn. Agr. Exp. Sta. Cir. 65.]

Another study similar to the one in Tennessee was conducted by R. M. Smith in southeastern Ohio during the period 1939 to 1941.[8] Dr. Smith made an intensive study of the effects of black locust and black walnuts upon ground covers and he found that in poor pastures black walnut trees improved both the species composition and chemical content of the plants growing under the trees. He rated walnut high as an ideal pasture tree because of its period of leaf activity; its light crown canopy; its small, fragile leaves which decompose rapidly, and are high in mineral matter and nitrogen; its deep tap root which competes very little with the surface rooted grasses for moisture and nutrients; its hardiness; and finally its high commercial value.

[Footnote 8: Smith, R. M., 1942. Some effects of black locust and black walnut on southeastern Ohio pastures. Soil Science, Vol. 53, No. 5.]

It seems apparent, therefore, that the introduction of improved black walnut trees into permanent pastures would be practical from the agronomic angle to say nothing about the beneficial effect of shade to livestock and possible income from occasional crops of high quality nuts.

One stumbling block to the adoption of this idea is the protection of the trees during the period of their establishment. The conventional cattle guard with three or four long posts supporting a wire fence is expensive in both labor and materials.

During the spring of 1946 in connection with my forestry instruction at Ohio State University, I had as one class project the planting of 50 black walnut seedlings of selected parentage in the cattle and poultry ranges on the University farm. Thirty of these trees were planted along a fence row at 32 foot intervals and were protected by a single electric wire connected to a battery charger.

The set-up is illustrated in figure 1 which shows the charger at one end of the line and the wire supported by the line posts and a short single post opposite each tree. The one year old seedlings were planted 4 feet from the fence at alternate posts and the wire zig-zagged along the line to create the guards around the trees. Within a few days after planting and completion of the electric guards the trees were mulched to control weeds and conserve soil moisture.

While this experiment has been in effect for only one growing season, the results, to date, indicate that this method is effective in providing protection from livestock. Growth and survival of the trees has been very satisfactory thus far.

The advantages of this method appear to be the rather low cost of labor and materials and ease of installation.

Within the next decade, we should be able to determine how the nuts from these seedling trees compare with the parent tree and there should be adequate shade for all classes of livestock on either side of the fence.

How Hardy Are Oriental Chestnuts and Hybrids?

By Russell B. Clapper and G. F. Gravatt Plant Industry Station, Beltsville, Maryland

One of the questions most frequently asked in regard to the Oriental chestnuts is, will they thrive in a given locality? Broadly speaking, with respect to temperature requirements these chestnuts have been found about equally hardy with the peach. Some strains of the Chinese chestnut appear to be superior to the Japanese chestnut in hardiness.

The Chinese chestnut is more widely planted in this country than the Japanese chestnut and more information has been collected on the hardiness of the former species than of the latter. The Chinese chestnut is growing satisfactorily in certain plantings as far south as Orlando, Fla. and the other Gulf States, northward to the southern tip of Maine, and westward as far as Iowa. But many areas within this large zone are unsuitable for growing Chinese chestnuts because of more severe climatic conditions.

Specific data have been obtained relating to several types of winter injury of Oriental chestnuts and hybrids. This information is limited to the performance of mostly young trees and to a comparatively small number of locations.

The fall freeze that occurred in mid-November, 1940, was studied in detail by Bowen S. Crandall,[9] formerly of this Division. Widespread damage occurred to Oriental chestnuts in the central parts of South Carolina, Georgia, and Alabama. Temperatures before the freeze had been mild, and heavy rains in early November had broken a drought. On the nights of November 15 and 16, temperatures of 12 deg. and 14 deg. F. were reported by various farmers, and a drop to 20 deg. F. was general on the night of the 16th. The damage to chestnuts by this freeze was increased because of the mild temperatures and heavy rains that preceded the freeze. The chestnut trees were not able to attain complete dormancy. Those trees, however, that were growing on uplands or on sites that were well air-drained suffered much less damage. Apparently equal damage was inflicted to Chinese and Japanese chestnuts.

[Footnote 9: Crandall, Bowen S. Freezing injury to Asiatic chestnut trees in the South in November, 1940. Plant Disease Reporter 27:392-394. October 1, 1943.]

On one farm near Columbus, Ga., four plantings were located at different elevations. The planting at the lowest elevation, maintained as a well cultivated orchard, suffered almost 100 per cent loss from this fall freeze. The trees at the highest elevation, in a forest planting, were practically uninjured. The damage from this freeze varied from killing of buds and shoots to killing of complete trees. Many owners of chestnut plantings did not notice the damage until the following spring. Fortunately fall freezes of this magnitude occur only infrequently.

In the winter of 1944, this Division lost 23 per cent of its hybrids at Glenn Dale, Md., from freezing following abnormally high temperatures. The hybrids had been fertilized in October of the preceding year, but the effect on the extent of freezing damage is not known. The months of November, 1943, through March, 1944, were characterized by extremely variable temperatures. For example, in November a minimum of 15 deg. F. occurred on the 17th, a maximum of 72 deg. on the 19th; in December a maximum of 66 deg. on the 3rd, a minimum of 2 deg. on the 16th; in January a minimum of 8 deg. on the 17th, a maximum of 74 deg. on the 27th; in February a minimum of 11 deg. on the 2nd, a maximum of 72 deg. on the 25th; in March a minimum of 8 deg. on the 10th, a maximum of 81 deg. on the 16th.

The extremes of temperatures in any one of these months may have been sufficient to cause damage to chestnut, although the extent of damage is influenced by the physiological conditions within the tree. The usual type of injury to the hybrids was a killing of the cambial cells extending from the base of the trunk up to varying heights. The cambial region was grayish-black and the inner bark was sappy and greenish-brown. More trees were injured and killed on the lower portions of the plot than on the higher portions.

This catastrophe afforded opportunity to study resistance of the hybrids to freezing. In the lower part of the plot there were several 3-year-old American chestnut seedlings that were not damaged. Sixteen per cent of first generation hybrids of Chinese and American chestnut were killed.

Chinese by American backcrossed with Chinese were killed to the extent of 36 per cent. Chinese by Japanese chestnut of the second generation were killed to the extent of 35 per cent.

Despite this extensive killing of hybrids by extreme variations of winter temperatures, older Chinese and Japanese chestnuts on slightly higher ground in the same plot suffered no visible injury. These old trees have rough bark, which may serve as an effective insulator against extremes of temperature. In 1944, there was no damaging late spring frost, and these old trees produced the largest nut crop in their history.

Winter temperatures of -25 deg. F. or lower are usually injurious to Oriental chestnuts. A few reports of chestnuts surviving temperatures of -25 deg. F. have been recorded, but usually Oriental chestnuts do not thrive in those northern States or regions where such temperatures occur.

Many of our cooperators report that late spring frosts frequently cause failure of chestnut crops. Damaging frosts in late spring occur more frequently and over greater areas than early fall frosts or extreme winter temperature variations. A late spring frost in 1945 reduced the chestnut crop at Glenn Dale, Md., from 50 bushels expected to 3 bushels actual. A freeze of 24 deg.F. on the nights of April 4 and 5 was sufficient to inflict this damage after two weeks of abnormally warm weather. Many of the trees were visibly injured, with wilted or dried unfolding buds. Other trees on higher ground were not visibly affected, yet they produced no crops.

Again it was noted that the American chestnut, followed by American chestnut hybrids, sustained none to little damage. The American chestnut, besides its inherent resistance to freezing, leafs late in the spring. Most of the crop of nuts obtained in 1945 was produced by the American chestnut hybrids.

Late spring frosts in 1945 were very extensive, reaching throughout the eastern and northeastern States, and there were practically no chestnut crops. There were also numerous reports of late spring frost injury to chestnut in the Central States.

In order to reduce freezing injury to Oriental chestnuts, it is essential that they be grown on sites that have excellent cold air drainage. As an approximate rule, these chestnuts should be planted on sites similar to those that are best for peaches. The orchard planting is not the only type that is subject to winter injury; forest plantings, ornamental plantings, and plantings for wildlife are also subject to winter injury especially if they are not on the most favorable sites.

Growing Chestnuts for Timber

By Jesse D. Diller Plant Industry Station, Beltsville, Maryland

Before the turn of the century, and even before chestnut blight had swept through our eastern forests, destroying one of our most valuable commercial timber trees, European and Asiatic chestnuts had been introduced. They made variable growth in the Gulf States, along the eastern seaboard from Florida to southern Maine, the southern half of Pennsylvania, southwestern Michigan, southeastern Iowa, down the Mississippi River Valley and on the Pacific Coast. These trees were grown for horticultural purposes, and for the most part, represented large-fruited varieties of Japanese chestnuts. They were not regarded as having forest-tree possibilities for in the open situations in which they were usually planted to insure early fruiting, the trees developed low-spreading crowns, resembling orchard trees. However, after the blight became fully established and it became apparent that our American chestnut was doomed, and that these scattered Asiatic chestnut trees had a natural resistance to this disease, a new interest developed in the Asiatic chestnuts as a possible substitute for the American chestnut.

The interest in and need for resistant, forest-type chestnuts became so great that the U. S. Department of Agriculture imported from the Orient seed of strains that might be suitable for the production of timber, poles and posts, with tannin and nuts as valuable by-products—qualities inherent in our native chestnut. The Division of Forest Pathology, Bureau of Plant Industry, Soils, and Agricultural Engineering has been carrying on the project of testing Asiatic chestnuts as timber trees. Professor R. Kent Beattie of this Division was in China, Korea, and Japan from 1927 to 1930, and collected over 250 bushels of seed for shipment to this Division. The seeds represented four species: Castanea mollissima—the Chinese chestnut; C. henryi—the Henry chinkapin; C. seguinii—the Seguin chestnut; and C. crenata—the Japanese chestnut.

Direct Seeding Studies

At the very beginning of these investigations in growing Asiatic chestnuts as timber trees, it was believed that greater success in establishment could be obtained by planting seedlings, rather than by direct seeding. In direct seeding trials during the early thirties the planted nuts were promptly devoured by rodents. Sixteen years of field experience has proven the soundness of this belief. The imported nuts were planted in the Division's nursery at Glenn Dale, Md., and the resulting seedlings distributed as 1- and 2-year-old trees to cooperators throughout the eastern United States.

In order to thoroughly test the possibilities of direct seeding as an economical method of establishment, this Division during seven years (1939 to 1942, and 1944 to 1946) planted over 7,000 nuts by direct seeding in 200 locations in 18 eastern States. It was suspected that the greatest hazard to direct seeding in or near forests would be rodents. Accordingly, in the spring of 1939 and 1940, 400 nuts and 600 nuts, respectively, were coated with a strychnine-alkaloid rodent repellent, and a comparable number of seeds, for both years, were left untreated to serve as checks. The checks were held in sphagnum moss at Beltsville, Md., and the nuts to be treated were packed in sphagnum moss and expressed to Denver for treatment by the Division of Wildlife Research, the Fish and Wildlife Service, Department of the Interior. Only 5.9 and 2.5 per cent of the treated seeds developed into seedlings, whereas 22.6 and 13.5 per cent of the untreated seeds produced seedlings. Not only did more of the treated seeds fail to germinate than of the untreated seeds, but the seedlings from the treated nuts were less vigorous. Because of the results obtained, the rodent-repellent study was discontinued at the end of the second year.

In 1941 and 1942, over 4,000 untreated chestnut seeds, representing 22 strains, were planted in 12 locations in eight eastern States. The seed source was entirely from American-grown, Asiatic chestnut trees growing in 28 locations in 16 eastern States. They represented Chinese, Japanese, hybrids, and also a limited quantity of American chestnut seed. Seed of the American species was included primarily to determine whether or not it differs from the Asiatic species with reference to establishment by direct seeding. The results for the two years confirmed our earlier beliefs: Only 2.2 per cent in 1941, and 4.0 per cent in 1942, developed into seedlings, of which only a remnant have survived. No species or strain differences were apparent.

"Tin Can" Method

In 1944, the tin-can method was employed in planting 400 nuts in four eastern States. By this method 15.5 per cent of the planted nuts developed into seedlings, representing a fourfold increase over results obtained for the three previous years. One end of a No. 2 can is removed, and a cross is cut in the other end with a heavy-bladed knife. The open end of the can is then forced into the ground, over the planted nut, so that the top lies flush with the ground level. The four corners at the center of the cut top then are turned slightly upward, to allow a small opening through which the hypocotyl of the developing seedling can emerge. The can completely disintegrates by rusting within two or three years, and does not interfere with the seedling's development.

An examination made of the various burrows about the tin cans, and also of the teeth marks on fragments of chestnut seedcoats lying about, indicated that not only squirrels, but other rodents, such as chipmunks, field mice, moles, and even woodchucks were probably involved in the direct seeding failures.

In 1945 and 1946, the tin-can method was tested widely on farms, to determine its possibilities in securing establishment of blight-resistant chestnuts without a great outlay of cash to farmers. In 1945, five seeds were distributed to each of 90 cooperators residing in the Piedmont and southern Appalachian regions, and in the lower Mississippi and Ohio River valleys; and in 1946, to 38 cooperators residing in the Middle Atlantic States. Preliminary results indicate that 40.0 and 37.2 per cent of the nuts planted by the farmers developed into seedlings. It should be pointed out that these results are not strictly comparable with those of previous years, because most of the farmers preferred to plant the chestnuts in their gardens, and under these conditions the nuts were not exposed to the severe competition and the extreme rodent hazards that occur in the forests.

Further proof of the superiority of planting seedling stock over direct seeding as a method of establishment is indicated in the results of an experiment initiated in 1939. One hundred and fifty 1-year-old seedlings and 150 nuts, all of the same Chinese strain, were planted on cleared forest lands in the Coastal Plains, the Piedmont, and the southern Appalachian regions, and in the Middle West. At the end of the eighth year, at each location, establishment and development of those originating from the 1-year-old transplants were better than those originating from seed, and their average survival was six times greater.

Distribution of Planting Stock

During the period 1930 to 1946, the Division of Forest Pathology distributed thousands of Asiatic chestnut seedlings to Federal, State, and private agencies for experimental forest plantings in 32 eastern States. The ten States receiving the most planting stock, in the order named were: North Carolina, Tennessee, New York, Pennsylvania, West Virginia, Virginia, Ohio, Georgia, South Carolina, and Maryland. The purpose of this seedling distribution was to obtain information concerning the little-known characteristics of the Asiatic chestnuts—their soil and climatic requirements, and their range adaptability.

Selection of Planting Sites

At first the selection of the planting sites was left entirely to the judgment of the cooperators, and most of them assumed that the Asiatic chestnuts have site requirements similar to those of the native American chestnut.

Because the American chestnut often occurs on dry ridges and upper slopes, especially where soil is thin and rock outcrops are frequent, the cooperators proceeded to plant the Asiatic chestnuts on similar "tough" sites. They believed that the planting of forest-tree species is justified only on defrosted areas that have reverted to grassland, or worn-out, unproductive agricultural land, or on wastelands—sites that we now know are better suited to the growing of conifers rather than hardwoods. As a result of this unfortunate choice of site selection, together with the several severe drought periods recurring in the early thirties, the cooperators lost most of their trees during the first and second years after planting.

Inspections of some of these planted areas after a lapse of from 10 to 15 years indicated that the sites still support only a scant herbaceous cover, with broomsedge and povertygrass predominating, and with no evidence of native woody species encroaching on the areas. The few surviving Asiatic chestnut seedlings were sickly looking, multi-stemmed, misshapen trees, heavily infected with twig blight and chestnut blight, and severely damaged by winter injury. But despite these heavy losses, a few plantations succeeded at least in part, and from these limited areas, together with an appraisal of the situations where some of the earlier planted chestnuts grew well, valuable information as to the site requirements of the Asiatic chestnut species was obtained.

Site Requirements

These field studies clearly showed that the site requirements of the Asiatic chestnuts, particularly with reference to soil moisture, are more nearly like these of yellow poplar, northern red oak, and white ash, than like the American chestnut or the native chinkapin species. On fertile, fresh soils that support the more mesophytic native species, Asiatic chestnuts remained relatively disease-free, developed straight boles, made satisfactory growth, and were able to maintain themselves in the stands in competition with the other rapid-growing associated hardwood species.

The indicator plants that suggest good sites for Asiatic chestnuts are: (a) Tree species—yellowpoplar, northern red oak, white ash, sugar maple, and yellow birch; (b) shrub species—spicebush; (c) herbaceous species—maiden hair fern, bloodroot, jack-in-the-pulpit, squirrelcorn and/or Dutchman's breeches. Plants that indicate sites too dry for forest-tree growth of Asiatic chestnuts are: (a) Tree species—the "hard" pines, black oak and scrub oak; (b) shrub species—dwarf sumac, and low blueberry; and (c) herbaceous species—broomsedge, wild strawberry, and povertygrass. Plants that indicate sites too wet are: (a) Tree species—black ash, red maple, and willows; (b) shrub species—alder; (c) herbaceous species—sedges and skunkcabbage.

Climatic Test Plots

On the basis of the experience gained from the earlier, extensive distribution of Asiatic chestnut planting stock, the Division of Forest Pathology, during the years of 1936, 1938, and 1939, established 21 Asiatic chestnut climatic test plots on cleared forest lands in eight eastern States on the most favorable sites obtainable. These plots, with their isolation borders, aggregating slightly less than 32 acres, and accommodating nearly 22,000 trees spaced 8 by 8 feet, occur from northern Massachusetts, along the Alleghenies southward to the southern Appalachians in southwestern North Carolina, and from the Atlantic seaboard, in southeastern South Carolina through the Middle West to southeastern Iowa. More than 20 strains are being tested at each place, including Chinese, Japanese, Seguin, and Henry species, as well as hybrids, and progeny of some of the oldest introduced chestnuts. Most of the plots are fenced against livestock and deer.

Although the results from these plots are as yet entirely preliminary, during the 8- to 11-year period of testing, valuable information has already been obtained: (1) The range of the Asiatic chestnuts tested does not coincide entirely with the range of the American chestnut or the native chinkapins. All Asiatic chestnut species that have been tested have failed at Orange, Massachusetts, where the American chestnut grew in abundance. In southeastern South Carolina, where the several species of native chinkapin thrive, some of them attaining a height of 20 feet, the Asiatic species have largely failed. On the other hand in northern Indiana and southeastern Iowa, entirely outside the botanical range of the American chestnut, a few Chinese strains have done remarkably well. (2) The Chinese chestnuts have a much wider range adaptability to site than the Japanese chestnuts; the latter are more restricted to mild climate and appear to require somewhat better site conditions. Of ten Chinese strains tested, only four can thus far be recommended for future planting in the Middle West. One Chinese strain that has thus far proven far superior to the others, in all the climatic plots, was introduced by the Department of Agriculture as seed from Nanking, China in 1924. (3) Poorly aerated soil is an important limiting factor in all regions where the chestnuts were tested.

Establishment by Underplanting and Girdling

On the basis of the field experience gained from the wide distribution of Asiatic chestnut planting stock and the information thus far obtained from the climatic test plots, a new method of establishing Asiatic chestnut under forest conditions was initiated in the spring of 1946, and is now being tried on a limited scale. It consists of underplanting, with chestnut seedlings, a fully stocked stand of hardwoods ranging from 4 to 8 inches in diameter breast height in which the predominant species are yellow poplar, northern red oak, white ash, and sugar maples. All overstory growth 5 feet and over in height is then girdled. As the girdled overstory trees die, they gradually yield the site to the planted chestnuts in transition that does not greatly disturb the ecological conditions, particularly of the forest floor. Rapid disintegration of the mantle of leafmold is prevented by the partial shading, which the dead or dying overstory, girdled trees cast. At the same time, the partial shading hinders the encroachment of the sprout hardwoods and the other plant invaders (which would normally become established if the planted area had been clear cut) until the chestnuts have become fully established. Not only does this system provide the best site conditions conducive to the development of forest-tree form in the Asiatic chestnuts, in limited areas, but also under establishment conditions that require a minimum amount of maintenance.

Summary

In general, Asiatic chestnuts, when grown for timber purposes, are best adapted to northern slopes, above frost pockets on cool protected sites, on deep, fertile soils having a covering of leaf litter and humus in the top soil, a soil that is permeable to both roots and water, and that has a good water-holding capacity. The plant association, above mentioned as indicating ideal sites for Asiatic chestnuts for best timber development, occur in rich soils of slight hollows in moist hilly woods and on the mountains in cove sites.

Improved Methods of Storing Chestnuts

By H. L. Crane and J. W. McKay Plant Industry Station, Beltsville, Maryland

Trees of the Chinese chestnut, Castanea mollissima, are quite resistant to the chestnut bark or blight disease. The heavy bearing of the trees together with the good quality of the nuts produced has stimulated planting of trees to replace those of the American species largely killed by that disease. Although a few horticultural varieties of Chinese chestnuts have been introduced and propagated, the great majority of the bearing trees are seedlings. In seedling plantings seldom do two trees produce nuts of the same size, color, and shape. All of these nuts when properly harvested, treated, and stored are sweet and edible and nourishing as food either raw, boiled, roasted, or combined with other foods in poultry dressing, salads, or pancakes. Then too, there is a big demand for Chinese chestnuts as seed for the purpose of growing seedling trees to be planted in orchards or to be used as rootstocks in propagating horticultural varieties. In either case, it is often desirable to store the nuts for several months before using them.

Chestnuts are not like the oily nuts, such as pecans, walnuts, almonds, filberts, or peanuts, that must be dried to a moisture content of 5 to 8 per cent to store well. Chestnuts are starchy nuts, containing about 50% moisture when first harvested, and on drying they become very hard. In experiments conducted at the U. S. Horticultural Field Station, Meridian, Miss., it was found that the loss in weight of chestnuts ranged from 16.2 to 30.5% when stored for 4 months in open containers at 32 deg.F., and 80% relative humidity. In an experiment in which chestnuts were stored 4-1/2 months at 32 deg.F., they lost 18.8% in weight when stored in burlap sacks, 3.7% when stored in waxed paper cartons with tight-fitting lids, and 2.0% when stored in friction-top cans. Furthermore, chestnuts on drying lose their viability and become worthless. Chestnuts lose moisture rapidly and become subject to spoilage due to molds and other fungi and therefore must be considered as highly perishable and handled accordingly.

There is a great difference in the keeping quality of the nuts produced by different trees in that some are very susceptible to infection by molds and bacteria and spoil quickly while others keep quite well. At Meridian, Miss., nuts from 5 different seedling trees ranged from 2 to 34% mold infection at harvest. Studies made by John R. Large at U. S. Pecan Field Station, Albany, Ga., showed that much of the infection of the nuts by molds occurred after they had fallen from the burs and while the nuts were in contact with the soil. It is, therefore, essential that the nuts be harvested promptly after they are mature.

As a general practice the nuts should be gathered every other day during the ripening season. Burs that have split open and exposed the brown nuts should be knocked from the trees, and all of the nuts on the ground should be gathered up cleanly. It would be difficult to emphasize too strongly the importance of harvesting the nuts promptly as soon as they are mature. Prompt and careful attention must then be given to the conditions under which they are stored if they are to remain for long in an edible and viable condition.

After the nuts have been gathered[10] they should be held in a layer not exceeding 1 or 2 inches deep for 3 or 4 days. It is important that they be kept in a well-ventilated building and that the sun does not strike the nuts during curing. After the preliminary curing, the nuts should be placed in friction-top metal cans (slip-top cans) and the lids should not be tight for the first month of storage. The nuts contain enough moisture after the short curing process that the lids will "sweat", or surplus moisture will accumulate on the under side. This will disappear slowly by evaporation during the first month or 6 weeks of storage and the lids may then be pushed firmly into place, making the can nearly airtight. The containers of nuts should be held in cold storage at temperatures of 32 deg. to 36 deg.F. While some nuts have kept quite well at temperatures as high as 45 deg.F., the tests indicate that the nearer the storage temperature is to 32 deg.F., the less is the mold development. Placing the cans in an ordinary home refrigerator should prove fairly satisfactory with nuts that have good keeping quality.

[Footnote 10: If the nuts are infected with weevils, they should immediately be treated after harvesting with the hot water or methyl bromide treatment as recommended by the United States Department of Agriculture, Bureau of Entomology and Plant Quarantine.]

It is essential that the nuts be placed in storage immediately after they have had the preliminary curing. Any delay may increase the possibility of mold development.

In the winter of 1945-46, nuts from 6 seedling Chinese chestnut trees were stored separately in five-gallon friction-top cans at the Plant Industry Station, Beltsville, Md., at 32 deg.F. for approximately 6 months. The results are given in Table 1. It will be noted that there was some variation in the percentage of spoiled nuts in the different lots, but the loss was small when compared with results obtained by other methods. All of the sound nuts in these lots were planted in a rodent-proof coldframe immediately after they were removed from storage, and from 90 to 95% germination of the seed was obtained throughout.

It is almost impossible to keep some varieties satisfactorily with even the best of care. Because of the great difference in keeping quality of the nuts of different varieties and from different seedling trees, each chestnut grower should study the keeping performance of the nuts from the different trees in his own orchard. He should save for permanent trees those producing nuts that keep well.

The method of storing chestnuts that perhaps has been more widely used than any other is to pack the nuts in slightly moist sphagnum moss or fresh hardwood sawdust in boxes and place them in cold storage at 32 deg.F. to 34 deg.F. A little less volume of packing material than of nuts is customarily used. The correct amount of moisture may be attained by adding 4 fluid ounces of water to 1 pound of dry sphagnum moss. There is great danger of getting too much moisture, which will tend to cause spoilage. If the cold storage compartment is one that has a tendency to dry the stored material, it may be necessary at some time during the year to open up the boxes and add a little moisture to the sphagnum, but in most storage houses this is not necessary.

Based upon results obtained during the last 2 or 3 years, it seems probable that the method of storing chestnuts in friction-top cans will prove to be more efficient than other methods now in use. Tests are under way to determine the most desirable moisture content of nuts at the time of storage. If this can be determined the present period of preliminary curing will become a matter of reducing the moisture content of the nuts to a known amount before they are stored. It is likely that other refinements of the method will be made in the near future, but the procedure here described has given results that merit further trial by those concerned with chestnut storage problems.

TABLE I—Record of Keeping Quality of Nuts from 6 Seedling Chinese Chestnut Trees Stored In Friction-Top Cans At 32 deg.F. for Approximately 6 Months At Beltsville, Winter—1945-46[11]

Tree Number	Total Weight of Nuts 4-24-46—Lbs.	Weight of Sound Nuts Lbs.	Weight of Spoiled Nuts Lbs.	Percent Spoiled
7861	23.69	23.08	.61	2.57
7881	25.20	24.63	.57	2.26
7930	26.85	26.48	.37	1.37
7932	24.29	23.80	.49	2.02
7938	29.00	27.48	1.52	5.24
8174	15.82	14.80	1.02	6.45
ALL LOTS	144.85	140.27	4.58	3.16

[Footnote 11: Weighed and examined 4/24/46.]

Essential Elements in Tree Nutrition

(Paper presented before the Northern Nut Growers Association Convention, September 3-5, 1946, Wooster, Ohio.)

By J. F. Wischhusen Manganese Research & Development Foundation, Cleveland 10, Ohio

Mankind has harbored an age-old grudge against insects and fungi, so that under the heading of crop protection from these pests there has developed a large insecticide and fungicide industry.

Relatively little attention has been paid to the effects of a nutritional character that can be obtained from simultaneous applications of essential elements. Insects will probably always constitute a problem of destruction, either of them or by them. But fungi, bacteriae, viruses, can be made to combat, control and balance each other; depending on the conditions under which their propagation is either facilitated or inhibited.

There is evidence that so-called essential nutrients, also variously referred to as "minor", "trace", "rare", or "micro" elements play a direct as well as indirect role of considerable importance in this matter, and that trees can be fertilized, sprayed, injected or treated with them in other ways to insure their growth, health, crop bearing ability, longevity, disease—frost—and drought—resistance. There still exists a paucity of scientific explanations on these subjects, but there is already a good deal of scattered information, which it is my purpose to draw to your attention. People do not care about scientific facts if they can obtain results without them, and then scientific concepts too may undergo changes. The manner in which trees obtain their nutrients from soil, air and water, however, will forever remain unchanged, whether we understand it or not, and it behooves every grower to observe effects from causes, and to reflect upon them, and report his observations to his association for the benefit of all.

Physical Soil Characteristics

That the primary requisites for tree growing are the physical characteristics of all soils favorable for that purpose requires no discussion. The successful nut tree planting starts with the soil, whether it be on the scale of an orchard, grove, or just a few trees around the farm or garden.

The better soils for general crop production are on limestone, basalt, dolemite, dolerite, diorite and gabbro formations, whereas sandstones, aplites, granites, pierre shale, cretacious rocks and volcanic formations weather into inferior soils. Gneiss can be sometimes good, sometimes unfavorable for building of fertile soil.

It is well to bear in mind that geology and botany are our two fundamental sciences, and that all our other sciences are in reality departments of these. Chemistry can be either a branch of botany if it deals with organic chemistry, or else a branch of geology, if it deals with inorganic chemistry, and it would appear that the modern scientific grower of nut trees or any other crops is wittingly or unwittingly concerned with both. Biology and zoology both are branches of botany.

The Essential Elements

In the past, economics have governed any crop production, whether of trees, grains, fruits or vegetables; not nutrition and health. The future in all likelihood will demand improved crops from the standpoint of nutritional purposes as foods. It is gradually being realized that the production of better crops can be brought about by greater application of essential nutrients to soils or as nutritional sprays direct to trees, and that such practices also reflect true economics. The same principle should govern wood production.

According to our today's knowledge, there are at least nineteen elements invariably essential to life, viz:

Primary: Hydrogen, carbon, nitrogen, oxygen, phosphorus.

Secondary: Calcium, magnesium, sodium, potassium, iron, sulphur, chlorine.

Micro: Manganese, copper, boron, silicon, aluminum, fluorine, iodine.

Then there are another eighteen elements at least variably necessary to life, viz:

(1) Variable Secondary Elements: Zinc, titanium, vanadium and bromine.

(2) Variable Micro-Elements: Lithium, rubidium, caesium, silver, beryllium, strontium, cadmium, germanium, tin, lead, arsenic, chromium, cobalt and nickel.

Elements in Soils Essential for Plant Growth

It is furthermore safe to state at the present time that fertile soils should contain at least the following twenty elements: Nitrogen, phosphorus, potassium, calcium, magnesium, sulphur, hydrogen, carbon, oxygen, iron, sodium, chlorine, aluminum, silicon, manganese, copper, zinc, boron, iodine, and fluorine.

Until quite recently many scientists believed that only the first ten elements were necessary for growth and maturing of crops; that only the first three should be considered as fertilizer ingredients, and that the others were supplied by soil, air and water, or were present as natural fillers in manures and fertilizer raw materials.

The modern agronomist, however, takes all these twenty essential elements into consideration, and many so-called "complete" fertilizers contain at least sixteen to eighteen, if not all of the elements mentioned above. Cobalt, essential to animal nutrition, can also most economically be supplied through the soil, even though crops grow without it.

As long as we have sufficient experimental research data that at least nineteen elements are invariably essential to all life, it stands to reason, that they at least must also be present in one way or another for the normal, or better the optimum growth of nut trees, and a crop of more nutritious nuts. Therefore, every time one of them is considered, all the others must also be borne in mind. It will neither prove difficult nor costly to experiment with them. It is a matter of finding the proper balance of everything essential for optimum nut tree growing.

Indeed, to ascertain the true balance of all elements that are invariably essential to life, and their relationship to the elements which are variably essential, would quite naturally appear to constitute the quintessence of research still to be performed. We cannot control such essential factors as climate, weather, sunshine, but man can control the supply and adjustment of nutrients to trees, and it rests entirely with him to do so.

There is one advantage a nut crop has over some other crops; it does not have to be harvested before fully mature. Nut crops obtain the benefit from elements that may be slowly assimilated during the season.

The following experimental and historical evidence and opinions have come to my attention, and I record them for what interest they may have. Past experience is often discarded as too old, but many a time an experimenter was ahead of his time, and his work remained unrecognized, so that now some old references can be revived and presented as novelties.

What the past ignored may indeed be due to the ignorance of those who did the ignoring.

1) The Chestnut Blight

The chestnut blight, for instance, of a generation ago, may be re-examined in the light of the proceedings before a chestnut blight conference, held at Harrisburg, Penna., February 20-21, 1912. A chestnut extract manufacturer, a Mr. W. M. Benson[3], stated at the time that in his experience the best extracts were made from trees high in lime. "A blighted tree," he stated, "is simply a tree in the process of starving to death for lack of lime." Maps showed that the blight was worst where there was least lime, and that the chestnut trees died last in Tennessee, where soils are high in lime. Analysis showed that chestnuts contained 40% lime, an unheard of amount. That this high test may reflect a faulty condition is pointed out later.

All I can add to this is that there is an English Walnut Tree, Alpine variety, on the farm of Mr. Deknatel, on Route 202, Chalfont, Penna., which is remarkable for its virility and crops of large nuts. This tree grows in a place protected by house and barn near a well, in limestone soil. It resisted the severe winters of 1935 and 1936, when many other English Walnuts in the vicinity died. My opinion is that any tree in that location would be an outstanding tree; and vice versa, had that particular tree been planted in another location, it would have done no better than any trees there located. Nuts from that tree might well be tested and compared with nuts from other trees.

2) The Banana Blight

The banana blight in Central America threatened for a while to be as destructive as the chestnut blight in this country. It was due admittedly to an attack by soil fungi, but no fungicide to foliage or to the soil served its purpose. However, the proper restoration of bacterial life in soils to keep the soil fungi in check proved effective. This was a matter not of the presence or absence of any one inorganic nutrient, but of restoring to soils the balance of fertility, an abundance of organic matter as food for bacteriae. Dr. George D. Scarseth, West Lafayette, Ind.[4], is one of those largely responsible for correcting this epidemic. His experience may prove useful to nut growers, so that they may not live in constant fear of another blight epidemic such as the one that exterminated our chestnuts only a generation ago.

3) Tree Nutrition, Microbial

From England comes interesting information about "Tree Nutrition"[5]. Evidence shows that the healthy growth of trees such as pines and spruces is intimately bound up with an association between their roots and fungi present in woodland soil. Poverty in mineral nutrients is no longer regarded as a necessarily critical factor in the failure of growth of trees of this kind, since the associated fungi have at their disposal sources of supply inaccessible to the roots of higher plants.

Experiments carried out during the past ten years at Wareham in England fully confirm the opinion expressed long ago by Professor Elias Melin, Upsala, Sweden, that the growth of trees and other plants on poor soils of the raw humus type is greatly influenced by the root-fungus association. By fostering the appropriate combination it has been possible to carry out successful afforestation of heathland so poor that ordinary cultural methods prove inadequate for the least exacting tree species. Satisfying the mineral requirements of the trees by direct application of fertilizers is not in itself sufficient treatment to ensure continued healthy growth; biological factors also play an essential role in promoting soil fertility. The experiments have shown that failure of the trees to establish a satisfactory biological equilibrium with the necessary fungi is due in this case, not to the absence of these fungi in the soil, but to their inactivation by toxic products of biological origin. The factors inhibiting the activity of the fungi can be removed by the application of comparatively small amounts of organic composts which produce dramatic and lasting effects on the growth of roots and shoots.

The special composts used are prepared from organic materials such as straw, hop waste and sawdust. The mechanism by which they stimulate growth is still obscure. All of them contain small amounts of directly available plant foods such as phosphates and potash, but careful investigation both in laboratory pot cultures and in the field, has shown that these can account for only a relatively temporary effect on growth. It is suggested that the composts act mainly by modifying the course of humus decomposition, thus bringing about drastic changes in the biological activities of the organic substrate of the soil.

This demonstration of the profound influence of biological factors on the nutrition of trees challenges the attention of foresters and has important practical applications. By making use of suitable composts, it will be possible to carry out the successful afforestation of land formerly regarded as wholly unproductive.

For further information see "Problems of Tree Nutrition"[5].

From the two foregoing examples it is seen that in the case of banana blight, fungi had to be suppressed by bacteriae, but that for pine trees on poor English soils fungi had to be activated for proper tree nutrition.

4) Inorganic Tree Nutrients

Other information also from England concerns the use of so-called "minerals" which I prefer to call "essential inorganic nutrients," and name by the element or the compound in which the element is contained. "Minerals", strictly speaking, refers to compounds formed by nature as rocks, ores, brines, salt deposits, etc.

Professor Wallace, Director of Britain's Long Ashton Research Station[6], has laid the foundation for diagnosing mineral deficiencies by leaf symptoms. These are reliable indicators of what nutrients to furnish plants when they are distinct and easily recognized. But for subacute deficiencies, plant analysis and injections are resorted to. Injections of manganese sulphate as pellets into holes drilled in trunks of cherry trees caused orchards that had been barren, to bear heavy crops a few months later.

Manganese, boron, zinc, copper, iron, magnesium also lend themselves quite readily for applications as nutritional sprays, when applied as suitable compounds such as the sulphates. Both spray applications and tree injections have great diagnostic values, because a response to them, if needed is relatively quick. When trees are deficient their foliage will show marked improvement from a spray application within a few days, so that a test can be made on a few trees before an entire orchard is treated. Trunk injections should of course be made during the dormant season for results to show the following summer.

5) Nutritional Sprays

Florida and California lead in the application of nutritional sprays on citrus and other fruit[7]. Vegetables, too, respond remarkably thereto[8]. I see no reason why nut trees likewise should not benefit from them, especially when other spray materials are used. Copper sulphate, zinc sulphate, manganese sulphate, magnesium sulphate, iron sulphate, cobalt sulphate and borax are all compatible with each other and with most other spray materials. Combination sprays seem to perform better, anyway, than single sprays, and the only objection would seem to be that some element is applied that is not deficient. It can be taken for granted, however, that nothing is wasted, even though the benefits may be invisible. Soils benefit in the long run from sprays. One element, even though not noticeably needed, may make another available or it may antidote toxicity of some element present to excess. Indirect results in all likelihood are always obtained.

In Florida, recommendations for spray applications to citrus are made annually[9]. They can be obtained from the Florida Citrus Commission, Lakeland, Fla. A typical formulae is as follows:

3-5 lbs. zinc sulphate |
3-5 lbs. manganese sulphate | per 100 gallons of water or
2-5 lbs. copper sulphate with | other spray material
equal amounts of lime. |

1 gallon of lime sulphur or 1-1/2 lbs. of lime is used for every 3 lbs. of sulphate of manganese or zinc.

Cherries, apples, plums are quite responsive to such applications, and I have seen the defoliation of prune trees in New York State corrected with a mixture containing:

Manganese 10% | All as metallic, in the form of hydrated oxides,

Copper 10% | and applied at the rate of 4 lbs, for the combination
Zinc 5% | material per 100 gallons.
Boron 1% | The addition of 2 lbs. lime is optional.

In California a manganese deficiency has been observed on English Walnuts[10], and 5-15 lbs. commercial manganese sulphate was used per 100 gallons of water during late May, through June, to correct this.

Sprays should be applied at ten day intervals until the deficiency symptoms no longer persist.

Plausible reasons for the somewhat quicker action of sprays than fertilizers may be furnished by two prominent authorities:

McCollum[11], one of our foremost nutritionists, first noted the discovery that the leaf of the plant is a complete food, and that none of the storage organs of plants, seeds, tubers, roots, fruits enjoy that distinction. In the leaf, biological processes are most active. It is the site of synthesis of proteins, carbohydrates and fats. The leaf is rich in actively functioning cells which contain everything necessary for the metabolic processes, and they supply all the nutrients which an animal requires. ("All flesh is grass").

Hoagland[12], another authority, writes on this subject thus:

> "It is now certain that soils are not invariably capable of supplying enough boron, zinc, copper and manganese to maintain healthy growth of plants. This knowledge has come mainly during the past ten years. Within this period thousands of cases from many parts of the world have been reported of crop failure, of plant disease, resulting from deficiencies of micro nutrient elements…. The statements do not imply that most soils are deficient in any of these elements, but the areas involved are large

> and important enough to warrant the view that the recognition of micro nutrient deficiencies constitutes a development in applied plant nutrition of major significance.
>
> "When I refer to deficiencies of boron, copper, manganese, or zinc, it is not a question of absolute deficiency in total quantity of the element present in the soil, but rather a physiological deficiency arising from the insufficient availability of the element in the plant; in other words, not enough of the element can be absorbed and distributed in the plant for its physiological needs at each successive phase of growth."

Nutritional sprays under such circumstances may prove the remedy, and we have experimental evidence to support this. Nut trees as is shown by the above mentioned experiment, may respond to spray applications equally as well as citrus, other fruit and vegetables, and effects, too may possess special diagnostic values, showing the need of trees, and therefore also the need of soils on which they are grown.

Investigators are constantly confronted with determining whether foliage shows symptoms of disease or starvation, and whether this is due to a deficiency or an excess of any particular nutrient; whether fungicides inhibit the generation of fungi from the spore state, or whether the plant is fortified from sprays or dusts to become disease resistant, or repellent.

Fungicides are valueless where plant disease is caused by bacteriae which invade the water conducting tubes, (roughly corresponding to the blood vessels of mammals), of plants, tree trunks, etc. and prevent the flow of water and nutrient solutions from roots to leaves. Deprived of water and nourishment, the plants or trees will wilt and die. Where, however, soils furnish these plants with protective inorganic nutrients, such as manganese, copper, iron, zinc, borax, etc. these bacterial diseases are prevented. Similar actions may take place in leaves.

Deficiency Symptoms. Kodachrome Slides.

Many acute deficiency symptoms have been identified by authorities and photographed, and I am able to show Kodachrome slides of the following:

> Manganese starvation on Swiss chard, spinach (five illustrations), courtesy of Dr. Robert E. Young, Waltham, Massachusetts.
>
> Apricot, sweet cherry, lemon, onions, peanut, soybean (two illustrations), tobacco (4 illustrations), sugarbeets, walnuts, wheat, all by different authors.
>
> Manganese deficiencies in Indiana on soyabeans, hemp, corn, by courtesy of George H. Enfield, Purdue University.
>
> Manganese on beets (mangels), (4 illustrations), and Romaine lettuce, Nassau County, Long Island. Courtesy of Dr. H. C. Thompson, Cornell University.

Many more are published in "Hunger Signs of Crops," an illustrated reference book popular with scientific farmers and growers[13].

Other deficiencies that have been observed on nut trees are the so-called "little leaf" or "rosette" of pecans and black walnuts[14], which is due to a lack of zinc. Strangely enough, healthy orchards in this case contained a preponderance of fungi, whereas in affected orchards the soil microflora was predominantly bacterial[15].

We now have definite experimental evidence that lime, manganese and zinc are required in appreciable quantities for the growth, health and bearing quality of nut trees. It is well to make sure of these elements in the soils devoted to nut tree planting, but it cannot be emphasized too often that all essential elements and factors should be taken care of; anyone of them may be the limiting factor in crop failure; the one that is absent is always the most important.

In regard to inorganic nutrients, more attention has probably been devoted to citrus trees than to any other tree species, largely because the soils of Florida and California require additions thereof. It would be unfair to say that such main fruit crops as apples, cherries, peaches, plums have been neglected; we merely possess more information on the nutrients of citrus trees than on other tree crops, as far as the micro essential nutrients are concerned. Most orchards and groves are fertilized only with nitrogen, phosphorus and potash, and limed when necessary. Nitrogen can stimulate size of fruit at the expense of quality.

A paper by P. W. Rohrbaugh[16], Plant Physiologist of the California Fruit Growers Exchange, Ontario, California, deals with eleven mineral nutrient deficiencies and their causes, viz: calcium, magnesium, potash, phosphorus, sulphur, nitrogen, iron, boron, zinc, manganese, copper, and this might well be used as a guide for nut trees.

6) Miscellaneous

A few oddities may also be mentioned for anyone inclined to experiment:

From Holland it is reported that an avenue of large handsome shade trees close to a century old, all died in one year, except where a junk dealer had stacked a pile of old metals. The trees had exhausted the inorganic nutrients within reach of their roots in the soil, but the junkpile had replenished them sufficiently, so that those within reach of it kept alive to this day, twenty years later.

A rock mulch is reported to have improved the growth of lime and lemon trees considerably[17], and it would seem that similar experiments should be made on young nut trees, just before bearing age in a comparative test with a check planting. Stones can be selected for the nutrients they contain, and a geologist can easily point out those containing the greatest number of elements. No one could go wrong in placing a few rocks of limestone or dolomite near the base of a tree, and let rain and sunshine, heat and frost attend to the fertilizing in a slow but perpetual manner.

Maple sugar contains manganese[18], showing this as a distinct quality over cane sugar. Manganese and other essential nutrients are known to facilitate the production of proteins[19], and the question of better quality nut production may well be examined from the viewpoint of the indirect effect from activities of soil microbiology by manganese, copper, cobalt and zinc. Some of these elements have also been classed as inorganic plant hormones[20]. "Chlorosis," the yellowing of leaves, may not only be a deficiency symptom of manganese, but also one of iron, copper and magnesium. Lack of manganese can cause a decrease in photosynthesis[21], so much so that in manganese deficient leaves the CO_2 assimilation may be reduced to half of normal. Herein, too, may lie the cause of low yields, smaller roots and lowered resistance of those roots to invading detrimental organism.

Contemporary work on soil microbiology may show that manganese and other essential nutrients are perhaps most important in their functions for the preservation and balancing of microbial life and actions in soils. There is where tree nutrition must begin; whatever is neglected in soils can at best only temporarily be adjusted afterwards. After all, deficiency symptoms on foliage show lack of soil fertility, and while we should welcome them for their diagnostic value, our corrective measures to be most economical must be taken on soils.

Transmission of Inorganic Nutrients from Soils to Plants to Animals

Soil analysis and plant tissue tests both have their value, but also their limitations. Many laboratories and experiment stations are equipped to make rapid soil tests, and some engage in leaf analysis. It is important that they be correctly interpreted. For instance, at the Citrus Experiment Station, Riverside, California[22], bark and leaves were collected from healthy and diseased Persian Walnuts. They were analyzed for calcium, magnesium, inorganic phosphate, manganese and iron. A higher percentage of ash was found in the diseased than in the healthy bark, and calcium, magnesium, manganese and inorganic phosphates were also generally higher.

It would be a fallacy I think to conclude therefrom that these elements were not necessary, or were present to excess. They were probably present because they had failed to function properly, due to changes in weather, excessive rains or droughts, and could not eliminate themselves.

We must consider the results from the functions of the essential elements, and discard the popular belief that inorganic nutrients in soils are transmitted from soils to plants, and therein contained for the express purpose of satisfying the need of animals and humans[23].

The plant has only one purpose to perform which is to grow and to reproduce itself, and such is the case with all other forms of life.

Plants contain very often inorganic elements in a form in which they cannot be utilized. It is therefore quite easy to mistake their presence either as a toxicity symptom or as a high requirement, when as a matter of fact these elements are present due to conditions unfavorable to metabolism, and they remained in bark and leaves as end products, in an inert form. Rather than being transmitted from soils to plant, their functions may consist of the formation of enzymes, proteins, hormones, chlorophyll, antibodies, vitamins, in carbon assimilation. When they have served such purposes they are not likely to be present in plants in anything like the amounts or forms as present in soils. They may come into question as catalysts or bio-catalysts, as sources of energy for microorganism, from which their optimum effects have been secured when they are not transmitted at all, causing changes, but remaining themselves unchanged. They are essential in the sense that the elements composing soils, sea, atmosphere are constantly energized, changed and used over and over again to create plant, animal and human life. In this cycle nothing is lost, only changed from old to new generations.

Summary

Soil factors for tree growth are physical, chemical and biological. To control the organisms of soils and plants is probably the most difficult problem in microbiology. It is not wise to alternate neglect with feverish attention when blights or other pests become epidemic or threatening. They may be of a nutritional, preventable rather than curable nature. Pathology and tree nutrition may as well become a constant part of your activities.

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Nut Tree Propagation As a Hobby for a Chemist

By Dr. E. M. Shelton, Cleveland, Ohio

Not so long ago we saw a movie by the title of "Cluny Brown." The heroine was possessed with a passion for repairing plumbing, but was continually inhibited by well-meaning relatives who told her that she "didn't know her place." A scene early in the story shows Cluny on the floor under a stopped-up kitchen sink explaining her problem to a sympathetic professor who states a philosophy something like this. "To be happy, one should not have to be bound by what is appropriate. If it is customary to throw nuts to the squirrels and you prefer to throw squirrels to the nuts, it should be all right to throw squirrels to the nuts."

It is obviously not always advisable to be so unconventional, but it seems to me that in matters pertaining to one's hobby it should be permissable to throw "squirrels to the nuts."

A hobby, like a shadow, is necessarily a very personal thing. Without the person with which it is associated it could not exist. Therefore, I feel that it is appropriate to present throughout this paper a liberal use of the pronoun in the first person.

Years ago, as a boy on an Ohio farm, I tried repeatedly, without success, to graft on small hickory trees along the river bank scions from one especially good tree that stood out in a cultivated field. Time that followed was too crowded for further attempts at nut tree propagation until about fifteen years ago, when, living in Connecticut, I bought a grafted walnut, a Thomas, and set out to produce more like it. Before we left Connecticut, I had been able to present grafted walnut trees to many of my neighbors who had persisted, hitherto, in calling hickory-nuts "walnuts." They would listen with some show of interest while I expounded on my enthusiasm for black walnuts, but sooner or later would inevitably ask, "Do you mean the shagbark kind?"

Last summer we drove back to Connecticut for a brief visit, and, on calling at the home of one of these friends, we found that the first nut borne on their Thomas tree had been carefully saved. Forthwith there was a solemn nut-cracking ceremony, and all present tasted the meat and pronounced it good. We hope that that tree and many others will thrive for years to come to add to the bonds of friendship with these neighbors we have known.

Lately I have arranged my work so that we may once again live in Ohio not too far from my boyhood home. Last year I tried once again to graft along the hillside scions from that prized hickory, and this time six out of seven grafts have grown.

My field of work has been that of a chemist, engaged in industrial problems related to animal and plant products. Hence, my hobby and my day's work are productive of mutually helpful ideas. The literature which I review frequently contains suggestions applicable to the various phases of tree propagation. Though a few references are quoted in the bibliography at the end of this paper, these are for illustration only and comprise a very small number of those which have appeared.

My experiments in nut tree propagation have been reported from time to time in the yearbooks of the N.N.G.A. and I intend in the remainder of this paper only to outline problems under a number of general headings in which I am particularly interested, and give some indication of procedures which seem worth while investigating.

An important phase of nut growing to which I have given little attention is the search for new varieties. I find my interest in this aspect growing as I associate with the group of nut growers in Ohio, who through prize contests and active personal work are trying to discover superior nut trees in nature, yet I do not find in this the opportunity I seek for experimentation unless it may be in the matter of hybridization.

Rootstock Propagation

Rootstocks for walnuts and hickories are very easily grown from seed. Chestnuts are grown with variable success, and it would seem that particular care in drainage of the seed bed, and possibly the use of one of the seed fungicides, should improve chestnut germination.

The present trend in the propagation of fruit trees is toward selection of particularly suitable rootstocks. Do some nut tree seedlings accept grafts more readily than others? We do not know. Numerous writers have discussed the idea of varying degrees of compatibility of rootstocks with scions and Jones[1] has brought together considerable evidence to relate incompatibility among plants with something parallel to allergy in animals. Initial growth of the scion leads to a flow of foreign bodies into the stock. The theory is advanced that the stock develops antitoxins to these foreign bodies which succeed in killing the scion a few weeks later.

If a particular strain of nut tree stock is some day found to be of particular value for grafting, or for propagation of a disease resistant type, as in the chestnut, the propagation of such stock vegetatively would be essential. A present illustration is the series of Malling apple rootstocks which are grown from cuttings.

I have tried many times to grow chestnuts from cuttings with no success. A few experiments now in progress are limited to Malling IX apple stocks which I assume are not especially difficult to root. I am trying several modifications of a principle of making the cuttings at some time after girdling the stem. The hope is that in this way there will be accumulated at the base of the cutting more than the usual reserve of nutritive elements together with whatever plant wound hormones and plant growth substances the twig is capable of synthesizing.

Scion Storage

In earlier papers I described the use of sodium sulfate crystals (Glauber's salt) for controlling the humidity in scion storage. This season I have adapted the practice to the shipping of fresh walnut bud sticks. A sack of Glauber's salt in the bottom of the mailing tube keeps the cuttings moist, and if, in addition, the container is kept in a refrigerator when not actually in transit, the buds have been kept in condition for use up to twenty-five days.

A low temperature is essential in storage of any scions. Variations in this factor may have been the cause of some of the objections which have been raised to the practice of coating scions with wax when they go into storage. If wax is to be applied over a scion, it can be done more uniformly and in a thinner coating by immersion of the scion in melted wax. The scion so coated seems to be in better condition than an uncoated scion when it comes out of storage provided the storage temperature has been low. However, if the wood has not been kept dormant by low temperature, gases are evolved which form blisters under the wax and injure the scion. It is quite probable that a wax coating then aggravates this damage.

Grafting and Budding

Until this year I had not tried budding, and have gotten into it first of all to learn whether an ordinary laboratory cork borer is not a usable substitute for a patch bud cutter. It seems to do very well. The patches are small, but as an aid in tieing them in I prepared short strips of painter's masking tape with a thin coat of a plastic grafting wax on one side. In the center of each piece of tape is a hole just large enough for the bud to show through. The tape is pressed on over the bud patch, after which the usual binding with rubber strips is applied.

The whole technic of budding is fascinating and I plan to experiment as extensively next season as time and stock permit.

Wax and Tape

In 1937, Shear[2] published a report on a number of wound dressings for trees in which he observed that lanolin exerts a marked action in stimulating cambial growth. This led me to try various wax combinations in which lanolin was incorporated, and a mixture of equal parts of lanolin and beeswax has become the base for most of my experimental grafting wax mixtures. I have commented already on the importance of incorporating an opaque ingredient to exclude light. Experiments in progress this season have had to do with introduction of green vs. red dye and with the incorporation of a wax soluble pyrridyl mercuric stearate[3] as a fungicide.

I have recommended painter's masking tape for tying in scions in all cases in which moderate tension is sufficient. A winding of such a tape of course excludes the grafting wax from contact with the line of cambial contact, so any favorable action which any ingredient in the wax might have must be largely interfered with. If a tape is prepared with a thin coating of plastic grafting wax on one side to serve as the adhesive, it should be possible to bring the wax into contact with the cut cambial surface without, however, introducing such a mass of wax as would make its way between stock and scion and interfere with contact.

Nutrition

My own field of work has recently changed to nutrition, infant feeding, and I shall undoubtedly come to have more of an understanding of plant nutrition as well as of babies as I study longer on this subject.

Our recollections of the "good old days" are often mistaken, but I think there is no doubt that the nut trees bore more and better nuts when I was a boy than we can find now. Can it be a matter of nutritional failure?

The first consideration in plant nutrition seems to be the water supply, and perhaps in many localities the water table has fallen sufficiently to threaten our trees with malnutrition.

The supply of the common mineral elements may or may not be adequate. These elements should not be difficult to supply. The matter of the trace elements and their significance catches our fancy at present and many of us will undoubtedly begin to explore the effect of this or that panacea for restoring a favorite old tree to a second youth.

Medication

It is only a step from the consideration of nutrition of a plant or animal to that of medication. Remedial agents are readily introduced into plants, either through the roots, or by spray on the foliage, or by direct injection into the trees. Going a little further, such methods become means of killing trees.

A few years ago, I became interested in killing trees in a way which would prevent sprouting and also protect the wood to some extent from insect attack and decay organisms. More recently my interest has turned toward the use of hygroscopic chemicals injected in the living tree for the purpose, not only of killing the tree, but of preventing the wood from cracking radially or drying. A number of government publications[4-10] have contributed information along this line.

To inject enough chemical to accomplish this purpose it seems necessary to introduce the chemical solution through a cut the depth of the sap wood and extending entirely around the tree. A collar of water-proof paper cemented to the tree provides a means of supplying the chemical solution to the cut. All this is described in the literature cited. The only contribution I have made is the use of urea in the solutions.

Many salts are more soluble in a water solution of urea than in water alone, and many such mixtures are very hygroscopic. Moreover, it seems that in the presence of urea higher concentrations of salt may be introduced into the sap stream of trees, though I do not as yet have experimental data to confirm this statement quantitatively.

An example of a solution injected into a small ash tree is as follows:

90 grams urea

120 grams copper sulfate crystals

300 cubic centimeters water

I hope in another year to cure a number of varieties of woods on the stump and later to compare their qualities in the shop with lumber cured in the usual way.

By-Products

Any object as juicy and colorful as a black walnut hull may well become a subject for search in recovery of by-products. The thermally active carbon made from the shells has actuated laboratory thermostats for me for several years.

But more real and immediate by-products have been the personal associations which have arisen from this hobby. Physicians, engineers, teachers, farmers, persons from every calling are among those whom I have met through a common interest in nut tree propagation. I can recommend this hobby to anyone mature enough to take an interest in the future, and to chemists in particular.

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Notes on Propagation and Transplanting in Western Tennessee

By Joseph C. McDaniel, State Horticulturist

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These observations are presented as a preliminary report of the results obtained by three enterprising amateurs of nut growing in the western counties of Tennessee, whose work points the way toward overcoming some of the weaknesses previously encountered in nut culture in the northern part of the cotton belt states. These growers are the "three R's" of our Association in west Tennessee: Dr. Aubrey Richards of Whiteville, Mr. George Rhodes of Covington, and Mr. W. F. Roark of Malesus. I am giving this brief account of some of their experiences, with the hope that it will stimulate others to try their methods under various conditions, and to report their results at later N.N.G.A. meetings. We do not expect these methods to work equally well in all parts of the United States and Canada represented here today, but they are giving promising results in the mid-South territory, and perhaps will have value in a wider area. As Mr. Davidson has so ably done at this meeting in the case of his Ohio plantings, we expect to give you a follow-up report on this work in west Tennessee at the Toronto meeting or later.

"Twin-T" Budding in Chestnut Propagation

Of the nut trees grown in this area, the chestnut has been the most difficult to propagate by budding. Nurseries in the upper South have propagated their pecan and walnut trees mostly by patch-budding or the similar ring-budding method, with very good success. When applied to chestnuts, patch-buds have seldom grown. The common T-bud, likewise, has been a general failure on chestnuts in America, though reported successful in Japan. Chip-buds have not been much-better.

Several years ago, Dr. Max B. Hardy told me that the inlay bark-graft had been used successfully with Chinese chestnuts at the U.S.D.A, laboratory in Albany, Ga., following Dr. B. G. Sitton's use of this method with pecans in Louisiana. (It is described in a bulletin from Michigan State College, East Lansing, Mich.) I tried it in a small way, and had some success using it on chestnuts in July and August. This spring I suggested it to Mr. Roark and Dr. Richards, both of whom tried it out, using Castanea mollissima stocks and various scion varieties.

Mr. Roark used the inlay bark-graft in the spring, topworking a C. mollissima seedling with scions of the Colossal, a hybrid variety from California. About 50 per cent of these have grown this year. Dr. Richards tried it during July, on C. mollissima seedlings from a different source. None of the Colossal would grow on his trees, but he was partially successful with scions of the C. mollissima varieties, Hobson, Carr and Zimmerman. He then devised a variation in the method which was highly successful with C. mollissima varieties. This I shall call the Richards "Twin-T" bud.

In "Twin-T" budding, a vertical slit is made in the bark of the stock. Then horizontal cuts are made through the bark at both top and bottom of the vertical cut. The bud piece is cut from the well matured part of a current season's twig, leaving a rather thick slice of wood beneath the bud. (It may be as thick as half the diameter of the twig.) The bud is inserted in the stock as in ordinary T-budding, then wrapped with a large sized rubber budding strip. (Westinghouse electrician's tape and Curity adhesive tape have also been used. Some other brands poisoned the buds.) The "take" of Chinese chestnut buds by this method has run from 60 to 90 per cent on Dr. Richards' trees of various sizes this year. In a short nursery row, buds were placed under first or second year bark, while larger trees were topworked by placing the buds mostly under the bark of second year limbs.

The Colossal failed again on Dr. Richards' trees when budded by the "Twin-T" method, but Carr and other Chinese varieties were budded successfully. The graft-compatibility problem in chestnuts is one of considerable complexity. Thus Carr, which has presented incompatibility with certain stocks of C. mollissima at other places, grew on these trees, and Colossal, compatible on another C. mollissima tree, failed on trees which are apparently compatible with Carr. The Chinese chestnut species varies in its graft-compatibilities possibly as much as in other characteristics (growth, productivity, size and quality of nuts, etc.) so that nut nurserymen should begin to select their seed for chestnut understocks with a view toward getting strains with a greater degree of compatibility to the leading scion varieties.

Mr. Roark has been able to propagate the Colossal upon its own roots by layering a small tree in his orchard. Two limbs pegged into the ground in the spring of 1945 had produced roots a year later, and were then detached from the parent tree. This is a slow but sure method of propagating nut tree varieties that are not congenial with the stocks available for grafting or budding. He has also layered sweet cherries and prune trees by this method which is described in U.S.D.A. Farmers Bulletin 1501 with reference to filberts.

A Heartnut Variety Compatible with Black Walnut Stocks

Seedling black walnuts are common on farms of west Tennessee. Dr. Richards and Mr. Rhodes have been most active in showing that these can be topworked readily to improved black walnut varieties under the conditions prevailing there. Mr. Rhodes has also fruited such older Persian walnut varieties as Lancaster, Mayette, and Franquette on black walnut stocks, but finds them generally unproductive in his climate. Newer varieties, including some selections of the Carpathian strains are now being tried and should be of fruiting age soon. Mr. Rhodes has also found, at Covington, a heartnut that is vigorous and productive under west Tennessee conditions. He finds that it buds readily on the native black walnut. Some budded trees of it are over a dozen years old. They have medium sized nuts, smooth shelled (with fairly thick shells for a heartnut) and kernels of good flavor, coming out whole when the nuts are cracked carefully. I am giving this variety the name Rhodes, and suggesting it for use in west Tennessee because of its adaptability and the fact that it can be budded upon black walnut. Others have reported Japanese walnut (including heartnut) varieties incompatible with black walnut at other locations. Dr. Richards has propagated some other heartnut varieties on black walnut, but finds them more variable than the Rhodes, in obtaining a good union.

Paper Wrap Gives Summer-Long Protection to Transplanted Trees

Too commonly, transplanted nut trees suffer from sunscald injury on their southwest sides during the first summer in the orchard. This injury is particularly common on pecans, which suffer a severe shock from transplanting and are slow in re-establishing vigorous growth. In west Tennessee, as one grower puts it, "A pecan is doing well if it holds one green leaf its first year." Pecans have been known to remain dormant in their tops until the second spring after planting, and then start growth. During this initial period of establishment in the orchard, it is beneficial to give some kind of shade to the tree trunk, to keep the bark from "cooking" and dying on part of the most exposed side. Waxing of the trunks before planting helps reduce drying out of the tops before the roots are partially regenerated and top growth begins, but waxing alone, under our conditions, is not sufficient to prevent the frequent occurrence of a dead area starting on the southwest side of the trunk during the summer following tree setting.

Dr. Richards has found that a heavy wallpaper of a cheap grade, cut in strips and wrapped spirally to cover the tree trunk from the ground up, lasts through the season and eliminates nearly all of the sunscald injury on pecans which he has moved from his farm nursery row to the orchard. With trees that are shipped long distances, and allowed to dry out too much before resetting, the results are not so uniform. We are still in favor of the use of wax coatings on trees that must be shipped, but would recommend that they be given additional protection by some means, to shade the trunks throughout the first growing season. This paper wrap of Dr. Richards seems as efficient as any method, and is the most economical I have observed. It should be beneficial on most species of nut trees under summer conditions in the mid-south region.

Propagating Nut Trees Under Glass

By Stephen Bernath, Poughkeepsie, New York

About ten years ago I decided to try a few nut grafts in my small propagating house. The results were so satisfactory that since that time I have grafted from a few hundred to several thousand each year.

I found by experiment that I could not graft nut trees exactly as I did ornamental trees and shrubs, due to their extra sap content. Nut trees bleed excessively and I had to overcome this or my losses were heavy. I use no wax on grafts. My method is as follows: I take a strong light string and wax it with beeswax and parafin mixed fifty-fifty. I use a modified side graft, tying with this waxed string.

Late in December or early in January, I pot the understock, using black walnut seedlings for four varieties (Persian walnut, butternut, black walnut and heartnut). I make sure the understock has had its rest period by not digging and storing them until they have been really hit by frost and left for a period, to be sure the wood has matured for the season. The mature understock is then stored in moist sand in a cool cellar.

In late-December, as I have stated, I place the understock in benches using 3-1/2 to 4 inch pots, wetting them thoroughly after imbedding them in peat moss. Keep the moss damp and at a temperature of 55 degrees at night. After two or three weeks examine the roots by knocking several loose from the pots. If root action has started, the roots will show white thread-like fibers and are ready for grafting. This is important, because if grafting is done too soon the loss is heavy. If delayed too long the top starts growing. So I caution, do grafting when the understock is ready.

Place newly made grafts on their side, imbedded in moss, and refrain from watering until the union has formed. Open grafting case after third day and daily thereafter, until union is complete. Each day wipe glass off with cloth to prevent moisture from dripping on grafts. Increase bottom heat after grafts are laid in benches from 68 to 75 degrees. In about three to four weeks, if union has formed, place grafts in up right position, then watering is resumed and heat is reduced to around 60 degrees at night. When graft shows two inches or more growth, cut understock off close above the union, and then give house plenty of ventilation to avoid soft growth.

I find nut trees very tender subjects and delay planting these under-glass grown grafts out in nursery rows until every vestige of frost has passed. Also be sure to sever the waxed string as this is tougher than the green graft.

If this method sounds like a great deal of work and trouble generally, remember the reward will be heavy rooted, easy to transplant, healthy, named varieties of nut trees. Who can say that, at the present, there is an abundance of such trees in this country.

The Economic, Ecological and Horticultural Aspects of Intercropping Nut Plantings

By F. L. O'Rourke Michigan State College, East Lansing, Michigan

Mature nut trees are usually large trees, and large trees demand space. Young nut trees, therefore, must be planted relatively far apart from each other and for the first few years, at least, there is an abundance of unused land between the trees, which may be used for intercropping. The choice of just what crop or plants to use is often perplexing and should be considered for several aspects.

The economic factors are of prime importance. The cost of growing the crop, the specialized farm machinery and equipment needed, the availability of labor, the distribution of the seasonal labor demand, the time of the critical cultural practices or of harvesting, the potential market, and the expected price of the saleable product must all be considered.

The staple farm crops of the region are often preferable to specialty crops, particularly from the labor standpoint. Corn, wheat, oats, potatoes, and legumes can all be grown with a minimum of labor and the use of power machinery. There is less risk involved with farm crops than with specialities, both in securing an adequate crop and in the price received for the product. Fruit, vegetables and ornamentals often have very critical requirements. They must be sprayed, harvested, and shipped at exactly the right time or all the proceeds will be lost. Staple crops are not so demanding in either culture or harvesting.

The labor distribution throughout the season or even throughout the year must be considered and well planned in advance. No two crops should require exact and demanding attention at the same time. They should be chosen and planted so that a regular, even distribution of labor can be maintained with as little of a rush period as possible and yet with a minimum of idle time.

The general agricultural pattern of the region must be considered. In a sparsely settled grain and livestock region it would be quite inadvisable to grow strawberries or other crops which require a maximum of hand labor during a very brief period. Berries, however, may be perfectly well suited to sections where either transient workers or city children can be secured with little effort.

The crop should suit from the ecological viewpoint. It must not compete with the young, growing trees for mineral food and water, particularly during spring and early summer when the trees make most of their annual growth. On the other hand, if planted too close to the trees, some intercrops may be shaded too severely to produce a normal yield.

Success in intercropping is usually found between plants which are quite dissimilar in form and habit. Black walnuts and pasture grasses furnish a typical example. The long taproots of the walnuts penetrate deeply into the soil, while the grass roots are shallow and fibrous and feed in the soil surface layer. The aerial portions of these plants are likewise quite different, the walnuts tower high in the air, while the grasses form their crowns on the very surface of the ground. The light shade cast by the walnuts does not interfere with the photosynthetic activity of the grasses, but it is sufficient to discourage growth of broad-leaved weeds which have a higher light requirement than that of grass. This light shade also tends to provide a greater supply of available moisture for the grass, in that it reduces temperature and, consequently, water loss from the grass and soil by keeping down both transpiration and evaporation.

Experiments in both Tennessee and Ohio have shown that the quantity of grass produced from beneath walnut trees is greater than on equal areas in the open and that the quality, as represented by a larger protein content, is also higher. For this reason, one may well consider livestock as the income-producing portion of a walnut-pasture planting. Over one fourth of the agricultural land of the United States is devoted to pasture and much of the land is suitable for interplanting to walnuts, butternuts, and other pasture trees, as honey locusts and black locusts, all of which are known to improve the pasture grasses to some extent. The potential income which may be derived from such plantings over this vast acreage is enormous and is the more striking in that these pasture trees occupy a plane that is now idle and unproductive, that is, the area lying above the grass tops. The nuts produced on this "upper story" will represent almost all "clear profit" in that very little care need be given these walnut trees after they have been properly planted. Livestock guards will need to be placed about the trees at planting time and kept there until the trees have grown to the point where they may no longer be harmed by straddling and browsing.

Pastures are excellent sites from another angle. The closely grazed sod furnishes an ideal place to rake the nuts together at harvest time. Anyone who has hunted for nuts in a dense ground cover will appreciate this factor.

While the walnut responds best to the deep, fertile soil of the river bottoms and flood plains, it will grow well on the lower portions of slopes if water is available and the site is not too exposed to the force of drying winds. Contour strips should be prepared by plowing several furrows downhill, each a little less in depth than the preceding, and the walnuts planted thereon. The walnut is a spreading tree and plenty of space should be allowed. Perhaps it may be wise to plant the walnuts at extended intervals and fill up the contour row with black locusts, for post wood, and honey locusts to produce succulent pods for cattle feed. In any event, it is better to allow too much, rather than too little space, as walnuts are long-lived trees and will thrive best where there is least competition. In Iowa, black walnuts are responding well to "basin culture" in sites which were prepared by "scalping" the sod from the upper portion of a slope and depositing it on a lower portion in order to catch and retain more water.

Nut trees are like all other trees in that they react favorably to good horticultural practice. Fertilizer, particularly nitrogen, is usually always helpful. The addition of lime when the soil is acid and of organic matter when humus becomes depleted will aid in better soil aeration and an increased moisture supply. This, in turn, will be reflected in more vigorous tree growth and greater nut production. Occasional spraying may be necessary to control the Datana caterpillar in the summer.

Chinese chestnuts seem to be admirably adapted for interplanting with mulberries, cherries, pears, and the like in poultry runs and hog lots where the pigs and chickens will control the weevils by gleaning the prematurely dropped and overlooked chestnuts which contain the grubs of the weevil. The fruit portion of the integrated planting will maintain a high carbohydrate ration during the season for the use of the livestock. Here, again, plenty of space should be allowed between trees to allow each its full measure of water, food, air and sunlight.

Careful and thorough research is needed to determine the full requirements of nut trees and to work out the interplanting relationships. In view of the vast potentialities for their use, investigational programs may soon be under way and much more definite information be made available to the farmer and landowner.

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Nut Work At the Mahoning County Experiment Farm, Canfield, Ohio

By L. Walter Sherman, Superintendent

My interest in nuts dates back to the turn of the century when, as a boy in high school, I delighted in gathering wild nuts for my own use. I knew of several black walnut trees bearing very fine nuts and also one excellent hickory. These were near my home in northern Ohio.

After my school days were over, I married and went to Oklahoma, where I found the most miserable wild nuts imaginable. However, I stayed but a short time and returned to my native state where the wild nuts were reasonably good. In 1935, I made a trip to California and visited the Persian walnut orchards at harvest time. As if that were not enough to convince me that it would be worth my while to do what I could in behalf of the nut industry, the Agricultural press of the time published several intriguing accounts of Persian walnuts growing in and near Toronto, Ontario which had been brought there by Rev. Paul C. Crath from the Carpathian Mountains of Poland.

My constant talk about hardy strains of Persian walnut prompted friends to tell me of several plantings already growing in northern Ohio with more or less success. I promptly obtained scions and undertook to graft a number of these, but I had the usual ill-success of a beginner. I failed in attempts to top work trees and had no better results with bench grafting although I began early in the season and continued my efforts till the time arrived for planting the trees. I stored the grafted material in a cool apple storage house from the time they were grafted until they were planted. Then somehow I learned that walnut wounds would not callous over except at relatively high temperatures. Accordingly, I placed my next bench-grafted trees in a warm greenhouse, where growth started at once. This marked my first successful grafting of black walnut. Later, Mr. W. R. Fickes of Wooster, explained to me his technique of "boxing off" or "bleeding." By following his instructions, I was able successfully to top work some of the seedlings I had grown for the purpose. My next steps were to procure some of the nuts from Rev. Crath which he had brought from Poland and to make a personal importation of seed from an experiment station in Russia. With these two lots I started out to raise Persian walnut seedlings.

The first grafted trees set out at the Farm were obtained from Homer C. Jacobs of Kent, Ohio, in 1937. That year we began planting a three-acre tract. The trees were grown with scions cut from prize winning seedlings brought out as a result of the Ohio nut contest held in 1934. The trees were set 25 feet each way in order to conserve room. This distance allowed for but 69 trees to the acre and available space was quickly occupied. By 1944, it became necessary to add two more acres. The new land was from an abandoned berry ground. It was plowed, limed heavily and fertilized. The alternate rows were used for peach trees as fillers. The main rows were mostly filled with new varieties of Persian walnut from northern Ohio which had been grafted on black walnut stocks. Some of the room was used for growing black walnut seedlings for use in grafting with scions of prize winners in the next Ohio contest, plans for which were already under way.

In 1944, four plantings of Persian walnut trees located some distance from each other in northern Ohio, all had good crops and all produced superior nuts. A half bushel of the nuts were planted at the Farm during the following spring. All lots grew remarkably well. The resulting seedlings, together with grafted trees, which by then were growing in the Farm nursery, made it necessary to further add to the orchard room. The increase this time was eight acres, of which five were planted to trees during the spring of 1946. In all plantings, the distance between trees has remained the same as at first, not that 25 feet is enough for bearing trees but because it is expected to do a large amount of thinning out as bearing begins and many trees prove their inferiority.

The problem of propagating desirable varieties has been our greatest difficulty. The kinds we wanted were not to be had from nursery sources as they were entirely new. Commercial nurserymen would not even undertake the task of grafting. We were forced to rely upon our own ingenuity. Not only did we have to master the art of grafting but we had to drive hundreds of miles in order to obtain scions of the various kinds. We still know too little about grafting. We often raise the question as to how it happens that surgeons can do almost anything they wish in the way of cutting and splicing parts of the human body, yet with nut trees, 75 per cent of success is rarely attained.

Last spring I began a rather elaborate comparison of paraffin with beeswax—lanolin for use in grafting. Dr. Shelton had demonstrated that the latter was a good dressing for wounds and I assumed that in grafting, it would promote callousing. My experiment was partially frustrated by the loss of my melting pot which burned at about the time the work was half done. The grafting had to be finished without wax of any kind. Out of 60 grafts so set, only five grew. The five survivors had been merely "boxed off" or "bled," none grew which had been treated with hot wax of any kind.

Research with nuts has but barely begun at the Farm. We feel, however, much encouraged and that the worst is over. We have a total of 725 trees in the planting, many of which have already borne a few nuts. Production should increase rapidly and we will soon have considerable quantities of nuts and other material with which to work. We have the following genera, species, varieties, and hybrid forms: Butternut—Craxezy and Vincamp; Chestnut—Carr, Hobson, Yankee (Syn., Connecticut Yankee), and Zimmerman; Hickory, including hybrids—Bixby, Bogne, Boor Nos. 1 and

2, Bowen, Cranz Nos. 1 and 2, Fairbanks, Frank, Haskell, Leach, Lozsdon, McConkey, Nething, Reynolds, Ridiker, Russell, Stratford, Weschcke, and Wright; Pecan—Busseron, Greenriver, and Posey; Black Walnut—Barnhart, Brown, Cowle, Fulton (Syn. Miller of Ohio), Hare, Havice, Horton, Jansen, Krause, Lisbon, Mintle, Mohican, Murphey, Ohio, Rohwer, Snyder, Sparrow, Stabler, Stambaugh, Thomas, Tritten, Twin Lakes, and Wanda; Persian Walnut—Alliance, Baxter, Blosser, Broadview, Diller, Elmore, Gligor Nos. 1 and 2, Graber, Hall, Lieber, Lopeman, Oehn, and Schafer; Heartnut—Bellevue, Canoka, Fish, and Keck. In addition there are 55 black walnut seedlings of Brown and Lisbon varieties; 65 seedling black walnuts of unknown parentage; 280 Persian walnut seedlings of known percentage; 37 heartnut seedlings; 30 Chinese chestnut seedlings; and 22 seedling filberts.

The Ohio Black Walnut Contest of 1946

The contest was sponsored by the Ohio chapter of the N.N.G.A., Inc., and was publicised through the cooperation of the Ohio Forestry Association and the Ohio Farmer magazine. There were 692 separate black walnut entries, showing the great interest aroused.

The nuts that won first place were grown by Mr. Duke Hughes, of Coal Run, Noble County, O. He states the tree is about 50 years old and stands in well-limed permanent pasture near the crest of a ridge, in Muskingum silt loam.

The system of judging was that set up by the TVA at Norris, Tenn. The judges were Oliver D. Diller, Secretary of the Ohio Forestry Association; L. Walter Sherman, Superintendent of the Mahoning County Experiment

Farm; and C. W. Ellenwood, Associate Horticulturist at the Wooster Experiment Station. They were assisted by William H. Cummings, Spencer B. Chase and Thomas G. Zarger, all of T.V.A., and several members of the Ohio chapter of NNGA. The prize winners are listed in order of awards.

[Illustration: Mr. Duke Hughes, Coal Run, Washington County, Ohio, and the tree producing the first prize—Duke black walnut.]

	Name Applied	Weight, Grams	First Pick, Grams	Final Pick, Grams	Percent of Kernel
1. Duke Hughes, Coal Run, Washington County, Ohio	Duke	27.2	6.8	6.9	25.3
2. J. C. Burson, Rt. 5, Athens, Athens County, Ohio	Burson	21.5	4.9	6.2	28.8
3. Mrs. C. E. Campbell, Lowellsville, Mahoning County Ohio	Kuhn	19.0	5.5	5.8	30.5
4. Ed. Smith, Rt. 3, Athens, Athens County, Ohio	Athens	23.5	4.9	6.5	27.6
5. Mrs. O. Shaffer, Lucasville, Scioto County, Ohio	Oliver	22.6	5.3	5.8	25.5
6. Wm. J. Davidson, Xenia, Green County, Ohio	Davidson	13.5	4.6	4.8	35.5
7. A. C. Orth, Rt. 5, Dayton, Montgomery County, Ohio	Orth				

8. H. C. Williamson, Southside, Williamson
Mason County, West Virginia

9. Herbert Penn, Otway, Penn
Scioto County, Ohio

10. Mrs. A. L. Jackson, Little Jackson
Hocking, Washington County, Ohio

[Illustration: The Judges At Work]

1946 Iowa Black Walnut Contest

By C. C. Lounsberry, Secretary I.N.G.A.

The 1946 black walnut contest sponsored by the Iowa Nut Growers' Association was held at the Hoyt Sherman Place, Des Moines, Iowa, on November 14 and 15, 1946. The judges were Prof. H. E. Nichols, Dr. H. H. Plagge, and Dr. J. M. Aikman.

Following the policy set in the 1942 contest, the Iowa State Horticultural Society put up cash and ribbons with special reference to standard and previously shown varieties, while the Iowa Nut Growers' Association was interested in new varieties. The following are the premiums awarded:

Standard Varieties:

Prize Name Variety

1 Schlagenbusch Bros., Ft. Madison Thomas
2 Russell Krouse, Toddville Krouse
3 Schlagenbusch Bros., Ft. Madison Stambaugh
4 E. F. Huen, Eldora Thomas
5 Seward Berhow, Huxley Ohio
6 Seward Berhow, Huxley Myers
7 R. S. Herrick, Prole Thomas
8 Schlagenbusch Bros., Ft. Madison Hepler
9 E. F. Huen, Eldora Ohio
10 E. F. Huen, Eldora Rohwer

New Varieties:

Prize Name Variety

1 Schlagenbusch Bros., Ft. Madison Schlagenbusch
2 F. J. Wagner, Danville Wagner
3 Tom Bandfield, Shell Rock Shepard
4 Roy A. Wood, Castana Wood
5 Mrs. Minnie Waldo, Grand Junction Waldo
6 E. F. Huen, Eldora Huen
7 Ira M. Kyhl, Sabula Tinker
8 Schlagenbusch Bros., Ft. Madison Kramer
9 Sam Moncrief, Center Junction Acme
10 C. E. Brockway, Grundy Center Birchwood

There were only 22 entries in standard varieties and 22 entries in new varieties so we did not make much of a showing as compared with the 1946 Ohio contest. However, very good walnuts came in. They were all sampled with a mechanical cracker. An interesting development to me was the fact that machine cracking left the center of several of the best varieties of walnuts looking much like the core of an apple, instead of being broken in two as in hand cracking.

Grafting Methods Adapted to Nut Trees

By H. F. Stoke, Virginia

(The notes I contributed to the 1945 Report under the title "Experiences With Nut Grafting" were so fragmentary as to be of little value. In an effort to correct the error I am offering the following supplementary notes in the hope that amateurs like myself may find them of some practical use.)

My best success with the propagation of nut trees has been with the following methods. For budding, I use the plate bud exclusively. For grafting on stocks up to one inch I use either the splice graft or the modified cleft graft. For larger stocks I use either the simple bark graft or the slot bark graft. Each will be discussed in order.

In making the plate bud, it is cut from the scion or bud stick the same as for the familiar T bud. Usually a bit of wood is cut away with the bud, which should not be removed. A bud, or a bit of bark, should similarly be cut from the stock at the desired point, and discarded. The area of exposed cambium on the stock should correspond as closely as possible with the cambium area exposed on the bud. The bud is then laid on the exposed cambium of the stock, and bound in place, preferably with rubber budding strips. The point of the bud should be left exposed.

[Illustration: SIMPLE BARK GRAFT Useful with thin-barked species.]

Choice of time when conditions are right is quite as necessary for success as the proper procedure. There are two separate periods when the plate bud may be used on walnut with the greatest success. The first period, in Virginia, is the latter half of May, when the black walnut stock is in almost full leaf. If done earlier the bud is likely to be drowned by the excessive bleeding of this species. Dormant buds cut the previous winter are used.

The follow-up care is vitally important. The stock should be cut off above the bud within five to seven days after budding. If successful, the bud will start into growth within another week or ten days, and may be a foot long within 30 days.

[Illustration: 1. Slot bark graft; useful in top-working.

2. Splice graft; unexcelled when scion and stock are of equal diameter.

3. Modified cleft graft; for all general purposes.

4. Plate bud; for small and medium stocks.]

The tying material should be cut and removed within a few days after the bud starts, to prevent strangulation of the tender shoot. Be sure to keep native growth of the stock trimmed off until midsummer to force growth of the bud.

The second period for successful plate budding of the walnut centers around August first, varying somewhat with the weather conditions. Buds of the current season's growth are used. The time must be late enough for these buds to be well matured, and early enough so that the stock is still growing and the bark slipping. If the buds are immature, or the bark tight, the operation will be a failure.

The buds remain dormant during the following winter, and are forced into growth by cutting off the stock above the bud early in the spring. The tying material, if durable, should be removed about 30 days after budding.

If conditions are right and the work is properly done, a high percentage of "takes" may be expected. In summer I preferably place the bud on the shady side of the stock, or shade it with a little skirt of white paper tied just above the bud.

Chestnuts can be budded by the same method, but the spring budding should be done earlier, while the stocks are in bud, and the summer budding should be done two or three weeks later than with the walnut.

I have not tried the plate bud on hickory or pecan, but it is the only budding method I use on walnut and chestnut, and I have tried them all.

When it comes to grafting, the simple splice graft, as illustrated, is very successful, but it should only be used when scion and stock are of the same size. It works splendidly on chestnut, filbert and hickory, and can also be used on walnut; however, I prefer the modified cleft graft for the latter, because of the bleeding problem.

In making the splice graft, the diagonal cut should be about four times as long as the diameter of the scion, to prevent slippage in tying.

For the modified cleft graft I cut the stock off at the selected point at an angle of from 45 to 60 degrees. This greatly facilitates the healing of the entire wound.

The cleft is made not by splitting, but by making a cut with a sharp knife, beginning at the apex of the stock and cutting diagonally downward and inward toward the center of the stock.

Before making the cut, the scion should be selected, and the wedge cut, with one face slightly longer than the other. This enables one to properly judge the depth and angle of the cleft, thus securing a fit on all four cambial lines. The longer face goes toward the main body of the stock, and is left slightly above the top of the stock. The apex of the stock is squared off slightly before the cleft is cut, and the knife is set very slightly on the wood at the starting point, rather than between the bark and the wood. Care at this point guarantees very rapid healing, with no dead tissues or "heel" on the stock, sometimes called "dieback."

Remember to watch all ties in grafting to prevent strangulation of the tender new growth. This, with removal of sprouts or suckers from the stock below the graft are two very important features of after-care, and neglect can nullify the most expert work in the grafting operation.

In grafting the black walnut I prefer to use the side graft because of the bleeding problem. This is precisely the same as the modified cleft graft except that the cleft is made about three-fourths of an inch below the apex of the stock. By making the graft a little below the top of the stock one can tie and wax it, without waxing the top of the stock, which is permitted to bleed at will. This freedom to bleed relieves the pressure of the sap at the graft, where healing takes place without flooding.

For stocks under an inch in diameter, I use the splice and modified cleft grafts exclusively. For larger stocks, such as are encountered in top working, other methods are preferred.

One can cut the main stock off just above a small limb, and graft one or more of the limbs. Again, one may cut the large stock off a year in advance, and bud or graft one or more of the suckers that are thrown out.

If neither of the above methods are applicable, one can use either the simple bark graft, or the slot bark graft.

In making the simple bark graft, I cut the stock off at a 45 degree angle as for the modified cleft graft. The scion is prepared by making one long wedge face, and on the other side make two short faces so that the point is triangular.

To insert the scion make a cut through the bark downward from the apex of the stock. Insert the scion between the bark and the stock, with the long face next to the wood, and force gently down until just a little of the face of the wedge shows above the top of the scion. It is well, in case the stock is large, to place three or four scions around the stock, removing all but the strongest after a year of two.

This graft is satisfactory for thin-barked species, but for the hickory, the slot bark graft is preferable.

For this graft, the scion should be trimmed as a wedge, with one face about twice as long as the other. Two parallel cuts are made through the bark at the top of the stock a distance apart equal to the width of the scion wedge. This strip of bark, or "tongue" is loosened at the top, and the wedge is forced between it and the wood, with the long face next to the stock, as in the simple bark graft.

Secure tying and waxing should be practiced in all grafting. Small nails or tacks driven into the top of the stock will help in anchoring the tying material to the sloping surface.

Inexperienced propagators should get it clearly in mind that union takes place only in the new growth. This new growth builds up from the cambium layer, which is the outside layer of wood cells that lies just beneath and in contact with the bark.

This is why it is so vitally necessary that the lines between the bark and cambium be placed in parallel contact as closely as possible, in the splice and cleft grafts. Never mind if the outside of the bark of scion and do not match perfectly, due to differences in the thickness of the bark. It is the inside line of the bark that must match.

Actual union takes place along this cambial line. The old wood of the wedge and cleft cannot, and never does, unite.

A word about scions. I seldom use a scion with more than two buds. The best scion wood is of the previous season's growth, if it is of good diameter and well ripened. Thin, slender twigs give poor results. On old, slow-growing, bearing trees it is sometimes not possible to get good scion wood one year old. In this case it is best to take some of the older wood in cutting the scion. When used, the wedge should be cut from the two-year wood, just below the one-year wood, with the top of the scion carrying two or three buds on the new wood. The tip of the scion should be waxed, if cut.

Scions should be cut when perfectly dormant and kept in cold storage until used. If kept too warm and wet the buds may swell, making the scions worthless.

It is quite possible to cut the scions about three weeks before the buds begin to swell and get good results by grafting immediately. The chief danger from this practice is that late frosts may nip the buds after starting, which is fatal to the new scion.

Waxing all cut surfaces, including the tip of the scion, should be practiced except as explained when the side graft is used for walnuts. Some advocate waxing the entire scion, also. If this is done I think it better to leave the buds unwaxed.

Have your knife very sharp. A broad blade is desirable in a grafting knife, as it helps in making smooth, flat surfaces in wedges and clefts. For budding, use a knife with a narrow blade, but also very sharp. Develop skill in making the scion wedge, and in cutting the cleft just the right depth and width for the scion selected. Experiment on worthless material until you get the knack. If you are a good, natural-born whittler you will find it a greater asset than a college degree.

Beginnings in Walnut Grafting

By C. C. Lounsberry, Iowa

Anyone who has studied propagation manuals from ancient to modern times cannot help but see how methods are carried down from older books to modern ones. However, in walnut grafting one suspects there were trade secrets not permitted publication. How different this was from friendly and helpful cultural and propagation directions given by Mr. J. F. Jones, Dr. W. C. Deming, Dr. Robert T. Morris, and others of the Northern Nut Growers' Association.

Beginning with Ancient Times

Greeks: Theophrastus mentions hazel nuts but nothing about walnuts.

Romans: Pliny, Cato, etc. have little to say about walnuts. Pliny refers to planting seeds of walnuts but no other method of propagation. However, he states oaks and walnuts are poisonous to soil, and walnuts are only used in a few cases for human remedies.

English: Loudon, Evelyn, Knight, etc. Loudon sticks to propagation of walnuts by seed. Knight[8] followed the French practice of grafting walnuts by approach up to the time of his discoveries in 1832, which were similar to Dr. Morris's "immediate" grafting.

French: The French used grafting by approach (inarching) early in the 19th century. Mortillet[11], 1863, states only one-third to one-half of walnut grafts are successful. These were probably Persian walnuts. We are not sure what other methods the French used. Mr. C. E. Parsons of the Felix Gillet Co. in 1940, sent us a picture showing Felix Gillet in his greenhouse at Barren Hill Nursery, Nevada City, California. This picture he states was taken in 1900-1902. It shows one year grafted walnut trees, and bench grafted walnut trees covered by tumblers six inches high, grafted by the "Treyve" process.

Beginnings in the United States

The first grafting of black walnuts thus comes down to the beginning of the 20th century.

William P. Corsa[3] with the USDA gave much information from replies to a questionnaire sent out in 1890, on nut culture and grafting, including bench grafting, in 1896. Mr. G. W. Oliver[13] in 1901, describes a method followed by Corsa in bench grafting walnuts and hickories. He used an incubator. Mr. Jackson Dawson[15] previously, working with hickories, had success in the greenhouse.

Andrew S. Fuller[4] in his Nut Culturist, published in 1896, advises that the South had not yet perfected pecan grafting. This seems to have been a challenge to Mr. J. F. Jones[1 & 7], for we find he moved from Missouri to Monticello, Fla., about 1899, and specialized in pecan grafting. He developed the slanting cut he later advocated in walnut grafting. However, again showing "there is nothing new under the sun" the author's uncle, Owen Albright, is credited by Corsa[3] with suggesting it in 1894, and it is also suggested by Mortillet[11] in 1863.

Grafting Wax

The necessity to protect graft unions by excluding air and moisture from cut plant tissue led to the use of balls of mud in ancient times. Later, various kinds of waxes were used.

In 1879, Prof. J. L. Budd[2], head of the Horticultural Department at Iowa State College, using resin and linseed oil, side grafted 150 varieties of Russian apples received from the interior of Russia in the winter of 1878. A boy swabbed hot wax on the grafts, using a lantern heater not too different from those used nowadays.

Mr. F. O. Harrington and Mr. S. W. Snyder, Iowa nurserymen were teaching grafting to members of the Iowa Horticultural Society in 1900, 1901 and 1902, at their annual meetings. Mr. J. B. McLaughlin[9], College Springs, Iowa, speaks of successfully grafting walnuts in 1900 in a discussion of the horticultural society led by Van Houton, Edwards, etc.

In 1909, Mr. E. A. Riehl[14] gave a talk before the Iowa State Horticultural Society in which he advocated covering the whole walnut scion, buds and all, with liquid wax. His first Thomas grafted tree is in a ravine back at his barn at Godfrey, Illinois. It was planted about 1902[12].

In 1910, the Northern Nut Growers' Association was organized by Prof. John Craig of Cornell University, Dr. Robert T. Morris, Dr. W. C. Deming, Mr. T. P. Littlepage and others. Craig had previously been at Iowa State College where he and Budd had shown much interest in nut trees.

In 1912, Mr. J. F. Jones [1][7], came up from the South where he had been successful in pecan grafting and started a black walnut nursery at Lancaster, Penna. He had been in Florida up to 1907. While in Florida he became acquainted with Mr. John G. Rush, of Willow Run, Penna., and did some walnut grafting for him. It was Mr. Rush who advised him to go to Lancaster and start a nursery for northern black walnuts. Jones patented his patch budder in 1912, and using the hot wax method developed by Mr. E. A. Riehl was very successful in walnut grafting.

In 1914, Dr. W. C. Deming and President T. P. Littlepage of the N.N.G.A. and Messrs. C. A. Reed and C. P. Close of the USDA had a conference in Washington which resulted in the publication of the American Nut Journal.

Paraffin In Grafting

Dr. Robert T. Morris[10], writing in the American Nut Journal in 1929, advocates the use of paraffin to cover walnut grafts instead of wax. Both he and Dr. J. Russell Smith[15] credit Mr. J. Ford Wilkinson with first using paraffin instead of wax on walnut grafts. Mr. Wilkinson wrote that he got the idea from seeing a careless workman splash paraffin on the buds as well as on the union in fruit tree grafting at the McCoy Nursery about 1914. The author bought apple and plum grafts about 1922 from the Gurney Nursery which were all covered with paraffin. It was at conventions of the Northern

Nut Growers' Association that new methods like this were passed along to members.

Bench Grafting

In 1932, on account of the difficulties in outdoor grafting of the walnut, the author became interested in bench grafting of walnuts in the greenhouse as a means of supplementing outdoor grafting. However, like many other so-called new methods, it was discovered when we looked up the literature in 1937 that William P. Corsa[3] had used methods that were similar about 1896. He cut off the seedling above the crown instead of below the crown as we did. The completed graft was packed in layers of sphagnum and placed in an incubator instead of using a greenhouse.

Notwithstanding all that has been done in black walnut grafting, the straight grained and brittle wood, the heavy sap flow, the almost instant oxidation of cut tissues, the liability to frost injury in the North in short seasons lowering vitality of scions, all combine to make walnut grafting with best methods available, a seasonal gamble.

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Forest Background

By John Davidson, Xenia, Ohio

(Read at the Ohio Nut Growers Annual Meeting, Ohio Agricultural Experiment Station, August 16, 1946.)

Where did the Persian, or so-called "English" walnut come from? Why is it a good commercial nut? The Pecan? How far can it be carried north beyond its natural, or original, environment? The Pawpaw? Why is it not a good commercial fruit? Why don't most people like it? What is the matter with the mulberry in America? In China and Japan it has a score of uses and great popularity.

These questions need an answer, and the answer almost invariably is that the poorer varieties and species have had but little attention and development by human beings while the better ones, Persian walnuts, grapes, melons, apples, dates, figs—all have had much attention and painstaking selection—in some cases for centuries. Upon the other hand, to cite a contrasting case the black walnut has no such history. It is the baby among nuts—a pure American baby—waiting for some nursemaid—for many nursemaids—to tend and develop it as a prince among trees should be developed.

Let us look back into the story behind a few—a very few—of our better known fruits and nuts and see, if we can, how they happened.

In America once lived a man nicknamed "Johnny Appleseed." His neighbors called him a "crackpate." He had a mania for planting tree seeds wherever he went. As a rule they were haphazardly selected seeds, but usually appleseeds.

What started him upon this crazy journey through the wilderness? Whatever it was, it would be worth while to isolate the germ and with it inoculate our present-day soil wasters.

But he was not the first one of his kind. Hundreds of pre-historic planters had gone before him. For years, now, explorers have been searching out and sending back to America certain valuable discoveries. Tremendously interesting, all of them. As one reads, it becomes increasingly evident that a considerable amount of scientific plant and animal breeding, selection, perhaps even grafting and artificial cross fertilization, budding and slip propagating may have been practiced by pre-historic, intelligent, forgotten men long before our modern times.

We usually find, today, that the best plants and animals have had their start in some center of old civilization. China, Manchuria, Japan, Indo-China, India, Persia, Asia Minor, Central America, Oceania—these places, the nurseries of all existing races of men are today the bonanza spots for these explorers. Such a coincidence could hardly have been due to chance. It must surely occur to the mind of anyone who cares to put two and two together that, in each of these centers, other ancient gatherers and planters had been busy in their day, just as our own explorers and experiment station scientists are carrying on today—our modern, scientific Johnny Appleseeds.

It is hardly possible, here, to follow to the ends of the earth all of the trails of the tribe of Johnny Appleseed. One little section will do well enough for purposes of illustration. Let us consider Iran, or, as our fathers knew it, Persia. Here is a field that, possibly because of previous plunderings, is not now the most fruitful of our sources of plant and animal discovery, yet it is an eye-opener, and will do very well as a type of similar test-plots throughout the world.

Here is a short list of only a few of the plants which have been developed for centuries, and were reported in the last century as growing in Persia—many, no doubt, descended from stocks which once grew in the famous hanging gardens of Babylon: apples, pears, filberts muskmelons, watermelons, grapes, peaches, plums, nectarines. And of flowers, these: marigold, chrysanthemum, hollyhock, narcissus, tulip, tuberose, aster, wallflower, dalia, white lily, hyacinth, violet, larkspur, pink and finally, the famous rose of Persia, from whence comes the attar of roses for which Persia is still famous. It would seem that someone must have possessed a knowledge of plant propagation in Persia centuries ago.

Several of these products have had their influence upon the history and poetry of the world. It will be remembered by most high school students that when the Caesars and big shots of Rome and Greece wished to create a big splash in the social ponds of their day, they sent, at enormous expense, for melons and dates from Persia. Melons, in particular, seemed to be the high spot in those Lucullan feasts, and, in this connection it is well to remember that Lucullus, himself, as commanding general of a Roman legion, had long lived in Persia and had, no doubt, acquired a taste for Persian delicacies. His princely estates near Rome, no doubt, grew rare plants from Asia Minor and were very likely tended by the skilled Aryan, early Accadian or Semitic gardeners of Persia. These slaves were probably descended from and were heir to the trade secrets of some of the very builders of that seventh wonder of the world, the hanging gardens of Babylon. Except for those forgotten workers from Persia, one may well wonder whether, today, our Rocky Ford, Ohio Sugar, or Hearts-of-Gold muskmelon delicacies would exist at all.

An interesting side-light may be found in the history of the peach. Originally this fruit was in all probability a poisonous variety of almond. What wizard, or succession of wizards, was it who created a peach from a pest—an asset from a liability? Persian, probably. Whoever did it, it constitutes one of the outstanding miracles of plant breeding, whether natural or artificial. The poison was sealed within the seed (where it remains to this day) and the nectar of the gods was bred into the pulp around it.

Consider also the Persian walnut, now, for some strange reason, popularly called "English" walnut. This delicacy, too, was unlikely to have happened merely by chance. It was, no doubt, bred by a race of men trained in observation and experiment such as the Persians preeminently were. Having first been nomads, domesticators and breeders of animals; they eventually became husbandmen, breeders of trees and plants, and they undoubtedly found that the principles which were so usefully employed in producing animal variations could also be used in producing and fixing plant varieties. The pollen or germ of an outstanding good male individual, when brought into contact with the pistil or ovum of an outstanding female individual of the same species will produce a scion that is more likely than any other to have good qualities. Here was the secret of most of the progress which has been made in both animal and plant breeding, a secret of immense value—so valuable, in fact, that it was guarded for generation after generation by a close-mouthed priesthood.

Just as, in the middle ages, the monasteries of Europe and Asia kept alive the tiny flame of Greek and Roman culture throughout the foggy ignorance of the Dark Ages, so did the priests of Baal, of Ashtoreth, of Marduk and of Ormuzd pass on the torch of their day to their successors who were Greeks and Romans. The Eleusinian mysteries, which at a later time were associated with a considerable amount of sensual, closely guarded ritual, were, in the Greek period, celebrated in the temple of Ceres in Eleusis. The origin of these sacred mysteries is lost in the shadow of profound antiquity. We know, only, that they were in the safekeeping of many generations of priests who jealously guarded them from thieving and ignorant conquerors. These mysteries were probably, at bottom, a body of scientific truths. They undoubtedly had to do with a store of information, painfully gleaned for generations, about those facts of reproduction, selection and beneficient fertility which are so close to the Holy of Holies of creation itself. Probably these precious mysteries could be simmered down to a few fundamentals and such as are now generally practiced by all plant and animal breeders. And they are not fully understood today, any more than they were fully understood three thousand years ago.

By the practice of these simple arts, hedged in with taboos and religious inhibitions, Persia, Assyria, and all Mesopotamia became the garden spot of the world where things seemed to grow as they grew no place else. Here, in fact, was said to have been located the only genuine and original Garden of Eden, pointed out to this day by the faithful as the veritable spot where the father and mother of the race lived in a laborless, exhaustless Paradise.

Mention has been made of the probability that the Persians, who originally were nomadic and therefore were chiefly interested in the domestication of animals—which means, really, selective breeding—used this knowledge in plant breeding when they finally settled down. The big leap from nomadic to settled life must have caused the old timers of that day plenty of headaches. It was a new deal to top all New Deals. Was it, perhaps, some Johnny Appleseed who engineered the New Deal of that day?

Let us guess at the method he used. As the nomad tribe passed from place to place with its goats, its sheep, its camels, Johnny with his sons and grandsons would take to prettying up the camp sites a bit. He particularly like the dates from one palm that grew upon an oasis far down the desert. He carried the seeds from this tree and planted them at various stopping places. He did the same thing with some especially sweet nuts from a walnut tree which he had found, let us say, in the Caucasus Mountains. He set out many bright-blossomed desert weeds in order to attract the wild honey bees. Bees! Wherever there were bees, he had found flowers that reproduced themselves, trees that bore fruit. Some of these bees he found to be good workers and others he found lazy, quarrelsome and inefficient. He killed out the quarrelsome colonies and built hiding places for the better ones. In short, he did so much to make the camping places cozy, comfortable and in every way desirable that finally it became more and more difficult for the tribe to tear itself away on moving day. By reason of the small irrigation arrangements which Johnny had found desirable for his plantings and his bees, grass became more abundant and the flocks did not need to be moved so often. In time, the whole tribe wakened to the fact that a revolution had taken place. They did not need to move at all, ever! There was plenty of grumbling from the die-hards, but here the tribe stuck. It refused to budge.

In time, a certain phrase, current throughout that part of the world, was used to describe this pleasant country: "A land flowing with milk and honey!"

Unfortunately, it was a land, also, which could not fail, in the flower of its wealth and luxury, to attract the attention of those savage northerners who lived beyond this favored land. They came, they saw, and eventually they conquered. When Rome had definitely destroyed the flower of Asia Minor's civilization, the Roman proconsuls and merchants "rescued" and carried back to Italy many of the rarest of Mesopotamia's possessions. Among these, perhaps, were those indispensable wonder-workers among the flowers, the better bees of Persia. And this may be the reason why, these many centuries later, our bee experts still recommend that, if we wish to increase the strength and productivity of a backward hive of bees, we buy and introduce into the hive an Italian queen. Her ancient and still prepotent virility can almost invariable be relied upon to transfuse the colony with new and fruitful vigor. An "Italian" queen, is it? We wonder, as we think of that venerable land of Eden which once flowed with milk and honey, whether this so-called Italian queen might not more correctly be named Persian.

You see, in this story we are traveling backwards into history like Ally Oop in his time machine. But beyond Persia one can go only in imagination. For the Persians, too, were a conquering nation and, no doubt, gathered their booty of gold and sheep and camels, of flowers and bees, from all the then known world which was subject to them. So perhaps Persia, too, has no more right to label her treasures Persian than has Italy with her presumably mislabeled Italian bees, nor England with her undoubtedly mislabeled English walnuts. However, the work of Johnny Appleseed has always belonged, not to his tribe nor to his locality, but to the world. These same Persian walnuts take rank among the better clues by which migrations of the Aryans may be traced over the face of the earth. For instance, not only do they take root easily in the mild, friendly climate of California, but much hardier strains are found to have climbed the Carpathians and the steppes of Russia almost to the very doors of Moscow. Scions of these hardier strains have very recently been made to grow and yield their nuts in America as far north as Toronto and are being set out in numbers in the northern part of the United States. How well they will prosper in this new, more variable and chilly climate remains to be seen, but the start is made. No doubt it will be by Johnny's old method of patient and repeated selection, first for hardiness then for quality, that the planned result will be accomplished.

The contributions of Persia and the plantings of its forgotten scientists have here merely been touched on. Nothing, for instance, has been said about her great groves of mulberry trees, which led to silk-worms, which led to silk, which led to the production of jewel-bright vegetable dyes, which led to the development of a decorative art in fabrics that is rivaled by China, alone, in all the world. And of course, Aryan Persia is only one of the many treasure centers of ancient civilization. In scores of racial settlements elsewhere our lives today are being changed and enriched in innumerable ways by the hands of those old miracle-workers whose names were writ in water and whose works are immortal. The accomplishments of China are of such magnitude that even now we are only beginning to discover our debt to her. India, Indo-China, Mongolia, Manchuria, Japan—all have similar backgrounds. Even in the United States, young as it is, the migrations of pre-historic races have left their trails in the gardens and forests around us. Pecans from the South, for example, have been carried

North and are gradually developing hardy strains that survived in Indiana and Illinois groves.

Enough has been said to blaze the way to the end at which I have been driving. It may begin to look as though modern plant explorers have now followed the plant-spoor of human migrations to their final limits. It may look, too, as though the ends of these converging trails will find civilization at last firmly established. Or will they?

The future race, let us admit, may eventually be able, by means of an almost unthinkable development of food, clothing, building and medical supplies of a synthetic or semi-synthetic nature, to dispense with some of the agriculture we know. This is the prediction of some scientists. Let it stand. What then is to be done with the land upon which our food crops had formerly been raised? Manifestly, it must again be covered with hurricane-control, flood-control, and erosion-control vegetation, chiefly trees, perhaps. Trees for safety's sake, trees for beauty's sake, for recreation's sake, trees for food's—yes, food's sake, for flavor and health, trees and vegetation as sources for the very synthetic that are supposed to supplant them; and last but not least, trees and vegetation for the protection and perpetuation of animal life, of bird life, and insect life. All these are inseparably bound up with human life.

Come what may at the hands of a short-sighted human race, no matter what surface changes may come about in human eating habits, housing styles, farming or factory practice, still the winds will sweep the earth in hurricanes where there is nothing to impede them; the waters and ice of the heavens will still tear apart and level the hills, will gash the valleys and will carry off the earth and dump it into the sea. Following this, the sun will burn the unprotected earth into a cinder. Nothing can change these facts. From the beginning of life upon the earth, trees and vegetation have been the chief means by which a balance has been maintained between the antagonistically destructive and creative natures of the elements.

Do we realize fully, I wonder, how important is the work of this group and the parent NNGA? The interest of its members is chiefly in "wild" trees that produce food crops—mainly, but not exclusively, nut crops. And they are interested not merely in planting and testing names and known varieties, but in finding and testing the best individuals among the wild trees, planting selected seed, enjoying the exciting gamble which is always sealed up in the magic, unknown potentialities of a hybrid.

As, centuries ago, the Persian walnut was rescued from the forest and developed into the splendid nut we know today, so the American black walnut can be rescued; its nut can be improved and developed by selection and cross-breeding. It is a grand mahogany-like timber tree which is becoming far too scarce. Each war takes its toll for gun stocks. Its nuts are the only nuts within my knowledge, not even excepting our lost American chestnuts, that retain their full distinctive flavor through cooking. Nothing can replace its flavor in candy or cake making. The tree is indigenous to America and, in contrast to the Persian, has only decades, rather than centuries of selective breeding behind it. No one can tell what even one short century of intelligent selection may make of this great tree.

We Americans, in fact, have barely started on the Appleseed trail, a trail which tends toward the development of a permanent perennial, rather than annual, type of agriculture, with trees, shrubs, vines and perennial grasses its chief interest. For, no matter what chemistry has in store for us in the way of plastics for construction and of synthetics for foods and drugs, the good earth is still our sole source of supply. The chestnut, the mulberry, persimmon, pawpaw, pecan, hickory, wild cherry, the grape, the elderberry in fact the whole tribe of fruits and nuts with flavors found nowhere else on earth—all are growing along this ancient trail. They offer an infinite variety of opportunity for exploration and discovery. To work with them gives one a sense of sharing in the work of creation.

Graft the Persian Walnut High in Michigan

By Gilbert Becker, Climax, Michigan

The rule to plant the Persian walnut where peaches and sweet cherries do well is a good one; but not infallible and certainly can't be too closely relied upon here in southwestern Michigan. Since 1933, I have placed several hundred grafts of the Persian walnut upon black stocks. Many of these are top worked trees, but there were 68 grafted seedlings in nursery rows, grafted in 1936. These were planted out two years later. Some are now about ten feet tall with a well branched head. Of this lot I have only harvested one ripe nut and that was four years ago. Two of these same trees were planted near some buildings and shrubbery at a neighbor's home, and they are now bearing well.

Before going further I must say that Persian walnut trees and peach trees are quite different. First, the Persian walnut cannot stand having its female flowers frosted when they are out or nearly so. Second, the peach can stand frost at, or shortly after, full bloom, and they will set a bumper crop of peaches. We have had two years of late spring frosts at the time nut trees were in bloom, and we have had bumper crops of peaches each year. Apples were badly hit, so many have failed to bear. Lilac blossoms failed to come out and be showy because of these severe frosts. However, I know of a peach tree heavily loaded right now growing between two Persian walnuts that haven't had a single nut either year, though they have borne nuts previously. Thus, peaches will bear in frosty springs when Persian walnuts are damaged. Further, good-air drainage, such as a high hill, with a deep valley below will save the Persian nut crop in a frosty spring. I have a small Persian walnut grafted in such a location, and it is the heaviest loaded nut tree I have. It has so many large nuts on its limbs that its lower limbs are actually resting upon the ground. This was grafted upon an established black seedling four years ago.

What I have so far told would lead one to think that there is no nut crop on my Persian grafts this year. This is not so, for I have one of the largest crops in the 13 years I have had grafted Persian walnuts. These are on top-worked trees high above the ground! Most of the top-worked trees are over 12 feet at the graft, or higher, and it is best to have them this high, because almost all lower limbs are simply minus nuts, due to our unfavorable spring. As for proof, I noticed that the lower limbs had blackened leaves, while the entire tops were undamaged a few days after the frosty weather. The lower branches leaved out the second time in late May. It seems as if the Persian walnut produces two nuts to every one that a grafted black walnut will on a top of equal size. We are troubled with walnut curculio as well as considerably by squirrels, and by a leaf disorder that often blackens the leaves and causes them to fall in early September, followed by premature dropping of the nuts. Even then, there should be a good crop this year.

Now, comes the question, should we graft the Persian walnut high, here in Michigan? It certainly saves time, because a middle-aged walnut tree produces, in terms of pecks and bushels, in eight to 15 years. Being well established it saves patience and disappointment. And I know it is far more profitable.

This writing of my experience is not intended to hurt the established nut tree nurseryman in any way. Any of you who may live in Michigan are certainly devoted to your hobby and have doubtless learned the skills and pleasures of top-working a good sized seedling black walnut. You will surely find it profitable. First, purchase the grafted Persian tree from your nurseryman, and later, from this, work your established seedling blacks at your convenience. Graft them at least 12 feet up and see if what I say isn't quite true.

Pecan Growing in Western Illinois

By R. B. Best, Eldred, Illinois

We need a consistent philosophy in this troubled world of ours. Working with nature and especially with nut trees helps us to develop this philosophy and to realize that there are no panaceas for our present day problems except as we work them out ourselves. After all our wishful thinking with panaceas and doctrines, we come back to the same conclusion. Those people with the best foundations built on reason and truth are those who are nearest the soil and growing things. Those who work with trees and other living things in nature possess the philosophy which acts as a breastwork against the forces which would destroy our society.

We started our propagation of nut trees in 1930 under the guiding hand of Mr. Wilkinson, Mr. Sawyer and Professor Ray Marsh of the University of Illinois, and later have had help from Dr. Colby of the University. We have at present about 2500 grafted pecan trees, a few varieties of hickories, black walnuts, chestnuts, filberts, persimmons, butternuts, heartnuts, pawpaws, etc. When people ask me what we expect from our trees, I tell them that the trees have already paid me in satisfaction if not in filling my purse. I do expect our nut tree project to give us a good financial return. The pecan is our leader in Western Illinois as a popular nut. Much of our Illinois river bottom land, if deserted by man, would immediately pass back to nature and exist as pecan groves. I have been working with pecan trees since 1930 and today find myself with more questions than answers. We are growing at present about 37 varieties of pecans. We are reaching certain notions which we hope are right. The hybrids are fine and make wonderful trees but I doubt if they are the answer to our problem. With these remarks I dispose of further discussion of the Burlington, Rockville, McCallister and Gerardi varieties.

The Major and Greenriver are excellent performers but are a little late maturing for us. The Posey nut is slightly earlier and makes an excellent quality but is not to be compared with Major and Greenriver for bearing. Our Butterick trees are excellent growers but bear few nuts. This variety is the poorest bearer that we have. Our earliest pecans of the better known varieties are Indiana and Busseron, of the newer varieties, Stephens and Gildig No. 2.

The Giles pecan which Mr. Wilkinson discovered in Kansas is our outstanding nut for yield, size and early bearing but it should also be earlier maturing. Although the Giles has been late when grafted on some of our native trees, it has been early on others. In 1945, which will always be known by the Illinois weather man as the year without a summer, we found a great difference in our Major, Greenriver, and Giles nuts from tree to tree as to size and maturity. This question of compatibility between stock and scion is of the utmost importance and it impedes investigational work, complicating comparisons we are trying to make. Some of our new varieties which we are trying out might be checked immediately if we knew the effect of the under stocks of our trees.

Our farms are about 50 miles north of St. Louis, Mo. Our first problem with pecans is maturity. The old named varieties are a little late for us. I personally feel that we should get grafts from no farther north than New Haven, Ill., or Rockport, Ind. I am interested in Mr. Gerardi's varieties at O'Fallon, Ill., because they should be early. Dr. Colby has brought to light three new ones from Cass County, Ill. which should make excellent maturity in central Illinois.

We are blessed in our community with large numbers of native pecan seedlings. The behavior of different nuts on different stocks is not the same. Before any nut should be condemned we feel it should have an opportunity to perform on different stocks over a period of years. For this reason we always try to graft a number of trees to each variety.

Most things taken from nature are subject to improvement and can be better adapted to the use of man. I would like to see some new varieties of pecans developed for our northern zone. I would like to see large plantings of nuts from all our leading varieties of pecans. From these seedling studies, great good would come and possibly a good variety. I would like to see Major, Greenriver, Giles, Posey, Busseron, Indiana, the Gildigs crossed with some early prolific nuts. I would like to see every nut that had any good quality crossed with every other good nut in a mass planting so that genetics could operate and have these trees planted where they might be permitted to reach maturity and the "get" of each union studied. We might get an early heavy bearer which would revolutionize the pecan industry. I would like to see some of our good Southern varieties like Stuart crossed with early northern varieties. This search for new nuts should be accelerated.

Let us rededicate ourselves to the problem of getting the "super-nut." Let us explore these new fields of nut germ plasm which lie all about us, pull these old nuts apart genetically and recombine their good with the good of other nuts into new varieties. If we should fail 10,000 times and succeed once, success would be cheap.

Random Notes from Eastern New York

By Gilbert L. Smith, Wassaic, New York

During the past few years I have found it increasingly difficult to keep up my nut tree work. However, three years hence, I expect to retire from my job as Farm Manager at Wassaic State School and then to devote much of my time to nut work. Mr. Benton now has even less time than I do for the nut work. Our work of previous years is now beginning to show results, especially our variety tests which should become more significant each year as more varieties come into bearing and repeat crops bear out or disprove our earlier opinions. Following are some of our findings on such varieties as have borne enough for us to form an opinion.

Black Walnuts

THOMAS, no doubt, is still entitled to first place. We made a poor start with Thomas as our first graft was placed on a stock growing at the edge of low swampy ground and the nuts of this graft have never matured properly, while those from two younger grafts, on higher ground, have matured their nuts well. This shows that black walnuts should not be planted in low wet ground, that is, land that is actually swampy; low ground which is well-drained is all right.

We have found Thomas to be a fast growing and very good type tree. The nut is large, thin-shelled and cracks excellently, giving light-colored fine appearing kernels, largely in whole quarters. We do not consider the flavor of Thomas to be one of the best. I have tested this many times by cracking nuts of Benton, Snyder, Sparrow and Thomas, and then, without revealing which is which, have had various people try them and pick out the ones they like best; Benton and Sparrow in all cases were liked best, Snyder second and Thomas always least in favor. Thomas is a consistent bearer here.

SPARROW is a little known variety which has a good many good points in its favor. In my opinion, it surpasses Thomas in everything except size of nut and cracking quality. In cracking quality I consider them to be about equal. Sparrow originated near Lomax, Ill. Wood of it was sent to us by C. A. Reed in the Spring of 1938. It has never been entered in any contest so is little known. The tree may not be quite as fast growing as Thomas, but it retains its foliage in the fall until cut by hard frost, long after its nuts have ripened, while Thomas will be nearly bare of leaves for some time before frost or its nuts are ripe. Sparrow ripens its nuts a full two weeks ahead of Thomas.

The nuts of Sparrow are medium in size, being about 27 to the pound while Thomas will run about 19 or 20 to the pound. The nuts of Sparrow look small while on the tree because it has a thin husk. Yet it husks easily, coming out of the husk cleaner than any other black walnut I know of. Also I have never seen a husk maggot in this variety while some varieties with thick husks were badly infested. As the nut ripens, the husk turns yellow. The nut yields practically 30% kernel (29.94%) with 96% unbroken quarters. Color of kernel is bright and the flavor is excellent. Sparrow has borne consistently.

SNYDER is a fairly well-known variety, having won first prize in the New York and New England contest of 1934. The tree is a little slower in growing than most varieties, yet it bears young and consistently Like Sparrow, it retains its foliage well until cut by frost. The nut is large, being about 21 per pound, with a very thick husk, on which account it should be husked as soon as gathered, as the husk will turn dark and stain the kernel. It ripens at the same time as Sparrow, last of September here. The nut cracks well, yielding about 25% kernel of good quality, about 95% in unbroken quarters. The color of the kernel tends to be a little dark.

Certainly Snyder should prove to be a valuable variety for short season locations and possibly as a pollinizer for Sparrow. Also the retention of foliage in fall, until cut by frost, make this and Sparrow of considerable ornamental value. Early dropping of the foliage in the fall is a serious fault of some varieties as an ornamental.

BENTON originated with us, the original tree growing in Mr. Benton's dooryard. It won second prize in the New York and New England contest of 1934. The nut is rather small, running about 34 to the pound. However, it yields about 29% kernel of excellent quality, light in color and about 86% quarters. It ripens about a week later than Snyder and Sparrow. It is a consistent bearer, a fairly fast growing tree, but only fair as to retention of foliage in the fall.

STAMBAUGH is a well known variety, but we are a little too far north for it, 41 deg.45' N. Lat. It matures well here only in our most favorable seasons. It appears to be an excellent nut, large, good cracking quality and good flavor. It appears to be a little capricious as to bearing, two years ago our one graft was heavily loaded, but there was no crop last year and a light one only this year. In spite of the lateness in maturing the nut, the tree sheds its foliage early.

Hickories

WILCOX is the outstanding variety of hickory of those which have borne in our test orchard, so far. This originated near Geneva, Ohio. It. won second prize in the Ohio contest of 1934. It appears to be a consistent, alternate bearer. The nut is only medium in size for a shagbark, about 90 to the pound. It cracks almost perfectly, yielding about 38% kernel, mostly in whole halves. Color of kernel bright and of very good flavor.

MINNIE has also appeared very good. It is a trifle larger than Wilcox, being 85 to the pound. It cracks excellently and is of good quality. But so far it has not yielded as well as has Wilcox.

DAVIS has shown up quite well. Our oldest graft is on a bitternut stock; it has borne well but the nuts have not cracked as well as those from the original tree or the ones grown at Cornell. In size the nut is between Minnie and Wilcox, kernel bright, plump and of good quality.

FOX has been rather disappointing as produced on grafts so far. Not that it is a poor nut, in fact it is a good nut, but because it has fallen so far short of what was expected of it. Fox is the mystery variety of the hickories. How it could unanimously win first prize in the Northern Nut Growers Association contest of 1934, with a sample of nuts so excellent in every way and then for the grafts to bear only fair nuts, is a mystery. Some have advanced the idea of bud variation in the parent tree. To prove or disprove this, I made a trip to the original tree in the spring of 1943 and gathered grafting wood from various parts of the tree. This wood was grafted on various stocks in our test orchard, so that we now have living grafts from 13 different parts of the original tree. If there is a bud variation, we should certainly have some of the good ones and are anxiously waiting the time when these grafts begin to bear. To lend a little credence to the bud variation theory, I found that at some time in the past the Fox tree had been broken off in a storm and had since formed a new top, largely from a single leader. Mr. Fox stated that he had naturally taken wood from the lower portions of the tree as it was much easier to do so. (The late Dr. Zimmerman made a similar study of this tree and its nuts from different branches. He was firmly convinced that there were differences.—Ed.)

Heartnuts

We have really tested only two varieties so far, these are the Fodermaier and Wright. Both are very good, but we now consider Wright to be by far the better of the two. It is somewhat hardier than Fodermaier, nuts ripen earlier, and bears better with us. Fodermaier is also more severely affected by the butternut curculio than is the Wright, some years nearly all of the Fodermaier nuts have been destroyed by the curculio.

GELLATLY has borne only one year with us, so we cannot form much of an opinion on it. It appears to be a very good nut.

Crath Carpathian Persian Walnuts

Several of our seedling Crath trees have nuts this year. In all cases, there are only a few nuts on each as our trees are still quite small. I had to hand-pollinate the blossoms this spring; this resulted in a rather small percentage of sets; then the curculio took a rather severe toll, so we will have only a few of each variety.

In 1944 one of our seedlings bore 12 nuts. These were so good that we have named the variety "Littlepage" in honor of the late Thomas Littlepage, and are having it patented. We have published a little booklet on this variety, and upon request, we will be glad to mail a copy to anyone interested.

This is about all we have to offer at this time in regard to our variety tests.

We have a problem which I wish to bring before the members of the Association. It is that of controlling the butternut curculio. This insect is very bad on butternut, heartnut and Persian walnuts, with us it does not attack black walnuts or hickories. I fear that it is going to prove hard to control, as the larva is of the boring type, being found inside the green nuts, inside the new growth of the terminals and in the fleshy part of the leaf stems. In these places it cannot be reached by poisons. It appears that we will have to work entirely on the adult beetles. These eat very little and seem to make puncture-like holes, eating little outside tissue but mostly deeper tissues, thus poison will probably have to be applied heavily in order for it to get enough to kill it. D.D.T. is not effective against the apple and plum curculio so probably will not be so against the butternut curculio. It might be effective to apply a heavy coating of D.D.T. bearing dust under the trees so that as the larva drop to the ground to pupate, they will be killed while the adult beetle may be immune to D.D.T., it is not likely that the pupa could survive in heavily impregnated soil.

The adult beetles are present from the time the first leaves appear until late summer. A spray of 4 to 5 pounds of arsenate of lead and 12 to 15 pounds of hydrated lime to 100 gallons of water, applied once a week throughout the early part of the season might prove effective but it will certainly prove expensive.

Planting of the affected varieties at some distance from woodlands and wild butternut trees is helpful in avoiding this insect, but as the trees grow older the pests may build up a population of their own. Some sections of the county may not be affected; I hope so.

Maybe we can get some of our entomologists to work on this insect. Let's put a little pressure on our State Experiment Stations and the U. S. Department of Agriculture. Maybe Mr. Reed can help us.

Another subject I wish to mention is that of hardiness in nut trees. In reading the NNGA reports and in some of the letters I have received, I have found that many people confuse killing of the young leaves in the spring by late frosts, with winter hardiness. In my opinion there is no connection at all. I have seen many trees that were not hurt at all by -34 deg.F. in mid-winter yet had all of their leaves killed by a late frost in the spring. In fact all species and varieties of hickory and walnut will have their leaves killed by a hard frost if the leaves have opened out of the buds; this includes our native wild trees as well as the grafted varieties.

The only hardiness against late spring frosts is the characteristic of leafing out late, thus escaping most of such frosts. Of the different species, the black walnuts seem to be best protected in this way, with the hickories next and the heartnuts and Persian walnuts least protected. Of course there is a considerable varietal variation within each species.

Then the protection we can provide, is to plant nut trees on side hills or other high ground where there is good air drainage, thus avoiding the frost pockets. Of course many want to plant nut trees and have no place except in low frosty sites. To these I say that they can expect to lose an occasional nut crop by these late spring frosts, but that only in exceptional cases will the trees suffer permanent injury. In years when the crops are lost the trees will still be good ornamentals and shade trees. My door yard is quite a frost pocket, yet I have lost only one crop of heartnuts out of four or five crops, no permanent injury to the tree.

Yield and Nut Quality of the Common Black Walnut In the Tennessee Valley[12]

By Thomas G. Zarger, Tennessee Valley Authority

Black walnut occurs on open, non-crop land in the Tennessee Valley region. Trees grow around the farmstead, along fence rows, and in pastures on most farms. In recent years harvesting of walnuts for market from these trees has increased significantly. Looking forward to a fuller utilization of the wild black walnut crop, knowledge on the bearing habits of these open-grown black walnut trees was required. To supply this information a study of tree growth, nut yield, and nut quality was undertaken in 1940. Results on nut yield available from this study after six years are summarized in this report.

[Footnote 12: Contribution from TVA Forestry Relations Department, Forestry Investigations Division on a project conducted in the Forest Products Section.]

This study was initiated with the selection of representative open-grown walnut trees throughout the Tennessee Valley. In 1940, 96 sample trees were selected and 36 trees were added to the study in 1942. These 132 trees are located in 42 counties and afford a good representation of age, size, and growth quality of open-grown black walnut. Each sample tree has been visited annually. Entire crops were collected, carefully weighed and sampled: tree diameters and other measurements were taken for the tree growth phase of the study. When convenient, nuts were hulled in the field with a corn sheller, but more often they were brought to Norris and run through a hulling machine. After hulling, the nuts were dried until cured, then a sample for each tree was tested for percentage of filled nuts, nut weight, and cracking quality.

Yield

Results on nut yield and nut quality for the 132 sample trees have been condensed to the presentation in Table 1. For the six-year period the average tree in this study had a diameter of 13.3 inches and yielded 33 pounds of hulled, dry nuts a year. The yield of common black walnut trees in the Tennessee Valley is characterized by extreme variation. Tree size, of course, influences nut yield. One-half of the yields from a 6-inch diameter tree ranged from no crop to 4 pounds of hulled, dry nuts; whereas half the yields from a 22-inch tree ranged from 40 to 100 pounds. A yield of less than one-half pound of hulled, dry walnuts was considered "no crop". Some individual trees had unusually high or low yields. The outstanding bearer was tree 117. It had the highest average yield for the six-year period, and the heaviest crop of hulled, dry nuts for any single year. During the six years this tree yielded 953 pounds of dry, hulled nuts and 194 pounds of kernels—truly outstanding production for a common black walnut tree. Another notable bearer, tree 100, yielded 916 pounds of nuts and 189 pounds of kernels. However, this tree was almost 11 inches larger in diameter than tree 117. The exceptional bearers in each diameter class also had the highest single nut crops. The other extreme is characterized by low yields. Crops were lacking or insignificant for trees 60, 63, 211, and 221. Tree 37, with a 19.7-inch diameter, bore only one crop of 31 pounds during the entire six-year period. This tree has no value for nut production but would yield a good sawlog.

Variation of yield by seasons and locality was examined by grouping the 132 sample trees into six localities of 22 trees each. Greater variation in averages by crop years existed than averages by tree location groups. However, some variation was found between the eastern and western portions of the Tennessee Valley.

Indications on bearing habits were obtained for a six-year period on 96 trees, Nos. 1 through 140 (Table 1). Crop records for each of these trees were examined for relatively high and low yield by seasons. Convincing evidence on the alternation of bearing has accumulated during this six-year period with 46 percent of the trees having lighter crops every other year. Of these, 28 trees bore lighter crops in the odd years and 16 trees bore lighter crops in the even years. Tree 117, previously mentioned as outstanding in regard to yield, produced lighter crops in 1940, 1942, and 1944. This tree is located in west Tennessee.

Walnut trees bearing lighter crops in 1941, 1943, and 1945 are more abundant in the eastern than in the western portion of the Tennessee Valley. This occurrence undoubtedly accounts for much of the variation found between the eastern and western portions.

Four other yield patterns were recognized in 30 per cent of the trees. These indicate the existence of uniform annual crops and three-year cyclic bearing of black walnut. The bearing habits of the remaining 24 per cent of the trees is considered merely irregular, since definite patterns cannot be recognized until bearing records cover a longer period of years.

Nut Quality

The cracking quality of the nuts from the trees in this study was tested on a random sample of nuts from each crop that was collected and brought back to Norris. The nuts of each sample were weighed and the average nut weight computed. The nuts were then cracked in a hand-cracking machine, and kernels that could be extracted with the fingers were removed and weighed.[13] From this weight was computed the first-crack marketable kernel percentage. The nuts that still contained kernels were recracked and the remaining kernel removed. All kernels, including crumbs, were then weighed in order to compute the total kernel weight and kernel percentage. Finally, all of the quarters extracted were counted, and the average number of quarters was computed. Kernels recovered at first crack and the average number of quarters extracted indicate the relative ease of extraction of kernels.

Cracking quality of walnuts for individual sample trees averaged by crop years are presented in Table 1. Nuts of all crops collected from four trees, 57, 58, 60, and 139, were shriveled or abnormal, and afforded no test of nut quality during the six-year period. Thus, nut quality data, based on 440 nut crop samples, are complete for 128 of the 132 sample trees. From this study, the average common black walnut in the Tennessee Valley has a nut weight of 17 grams, a kernel weight of 3.3 grams, a total kernel content of 20 per cent, a marketable kernel recovery at first crack of 17 per cent, and a quarter recovery of unbroken quarters averaging 1.8.

[Footnote 13: The kernels were extracted over a 6-mesh wire screen. In commercial cracking, kernel pieces passing through this type of screen are not marketable as kernels.]

Table I—Yield of Nuts and Kernels, and Cracking Quality of Nuts from 132 Sample Trees of Common Black Walnut in the Tennessee Valley

Tree

Diameter

at 4-1/2

Sample ft.
number av. yr. 1940 1941 1942 1943 1944 1945 Av. yr.

inches pounds pounds pounds pounds pounds
pounds pounds 1 20.4 27 16 2 95 5 3 25 2 14.9 43
43 26 45 42 23 38 3 9.3 22 12 8 22 6 29 17 5 11.2
54 0 61 0 33 7 26 6 13.6 28 1 72 0 16 1 20 8 13.2
50 28 23 41 76 45 44

9 13.2 36 50 40 28 33 7 32 10 22.9 29 25 0 0 6 15 12 11 6.1
2 4 12 2 4 1 4 12 17.4 110 18 128 40 100 49 74 13 14.5 98 5 83
50 46 128 68 14 12.2 1 0 12 3 2 98 19

15 11.6 38 46 44 106 0 63 50 16 15.7 130 0 106 25 135 33 72
17 12.0 1 66 4 100 2 61 39 18 7.8 20 0 40 21 33 4 20 25 8.6 13 0
82 0 0 0 16 26 20.7 0 36 46 90 0 67 46

27 8.4 0 1 26 2 0 22 8 28 8.0 0 11 1 19 0 12 7 29 9.2 0 17 22 21
2 19 14 30 15.2 150 25 200 0 102 15 82 31 18.0 33 194 14 259 0
135 106 34 16.4 0 108 0 25 0 129 44

37 19.7 0 0 31 0 0 0 5 38 9.1 2 0 14 0 47 0 10 39 17.7 151 0 80
0 56 0 48 40 16.5 88 0 50 5 37 6 31 41 9.5 60 0 74 0 67 0 34 42
14.5 123 0 170 0 119 0 69

Av. Filled nuts Complete crack
Kernel

yield First-
bearing In terms crack Crops
yrs. of total Average marketable Kernel tested
number only weight weight kernel weight Kernel Quarters basis

---

pounds percent grams percent grams percent number
number 1 2.2 6 17 21 3.7 22 2.9 2 2 4.9 50 14 17 2.6 19 1.3
5 3 2.2 63 16 18 3.3 21 2.0 7 5 7.9 67 16 23 4.5 27 1.1 3 6
4.9 92 14 22 3.2 23 2.8 4 8 5.6 59 24 17 5.0 21 3.0 7

9 6.1 56 16 20 3.6 23 1.3 5 10 1.6 36 13 15 2.6 19 0.6 2 11 0.8 99
14 16 2.7 19 1.4 7 12 16.3 95 18 17 4.2 23 1.5 7 13 17.9 92 19 25 5.2
28 2.9 7 14 4.9 91 19 18 4.3 22 2.4 3

15 11.8 96 17 19 3.4 20 2.1 6 16 20.0 94 24 19 6.0 25 2.3 6 17 8.4 97
13 17 2.7 20 2.5 7 18 5.3 85 15 24 4.0 26 2.5 4 25 9.6 93 18 16 3.8 21
2.5 4 26 5.5 42 16 14 3.1 18 0.6 4

27 2.7 95 17 19 3.7 21 1.2 3 28 1.7 99 9 14 1.6 18 1.6 4 29 3.4 100 11
15 2.2 21 1.2 4 30 19.9 77 19 17 4.3 23 2.3 3 31 25.1 90 20 15 4.5 23 1.3
4 34 16.8 73 20 17 4.0 20 2.5 2

37 6.2 100 16 16 2.8 18 3.4 2 38 4.5 63 18 14 2.8 15 2.3 2 39 15.0 87
19 14 3.5 18 2.2 3 40 4.9 82 20 13 3.1 15 2.3 3 41 9.1 73 16 17 3.0 19
2.5 3 42 28.4 86 19 21 4.5 24 3.3 3 46 13.8 14 18 15 36 12 12 18 47 9.8
15 0 39 0 20 2 13 48 13.6 25 34 50 52 17 96 46 49 6.6 14 9 16 4 19 0 10
50 9.5 29 0 13 25 0 57 20 51 11.2 11 13 11 0 24 0 10

52 13.3 25 8 0 84 0 14 22 56 13.4 15 8 0 12 4 6 8 57 16.7 162 5 103
17 74 4 59 58 12.0 42 2 30 6 20 2 17 59 9.4 2 8 4 8 2 8 5 60 9.6 1 1 3 0 2

0 1

61 10.6 2 2 20 1 10 0 6 62 12.4 27 6 23 7 13 0 13 63 12.1 2 0 0 0 0 0 0 64 11.8 18 2 37 0 21 0 13 65 17.8 130 53 101 9 107 0 67 66 9.6 31 0 25 1 13 5 12 67 9.4 89 0 7 7 10 11 21 69 13.7 70 2 104 4 30 2 35 70 16.1 72 2 11 95 0 68 41 71 15.2 7 1 43 1 0 1 9 76 8.1 7 0 6 0 9 0 4 77 11.2 40 0 21 6 4 23 16

78 11.4 34 0 40 0 31 2 18 79 16.4 28 0 24 0 11 22 14 80 11.4 132 44 110 8 189 42 88 86 24.9 191 0 282 0 64 110 108 87 14.0 45 0 107 0 31 9 32 89 8.4 1 8 2 39 0 44 16

90 13.2 11 6 72 8 13 7 20 91 12.4 68 5 200 3 54 22 59 92 17.6 18 74 138 76 2 126 72 93 10.9 30 0 48 3 26 0 18 94 7.2 0 36 0 21 0 53 18

46 1.7 51 17 16 2.8 17 2.8 4 47 3.8 97 11 17 2.2 20 1.0 3 48 8.2 83 13 16 2.6 20 1.3 4 49 2.0 80 18 16 3.7 20 2.0 3 50 3.7 72 17 19 3.6 21 3.4 3 51 1.9 49 18 19 4.2 23 1.6 3 52 6.0 80 18 16 3.2 18 1.3 2 56 0.6 13 22 20 4.8 22 3.0 1 57 4 0 58 9 0 59 0.4 20 27 19 5.6 21 3.0 2 60 0.0 0 22 15 3.4 15 3.9 1

61 0.2 48 14 9 2.0 14 1.5 2 62 0.4 15 25 19 4.6 19 3.8 2 63 0.5 94 13 21 3.2 24 3.1 2 64 3.1 70 21 20 5.1 24 3.2 3 65 7.9 58 23 15 3.7 16 3.4 3 66 1.6 34 24 18 4.6 19 3.7 3

67 2.2 31 20 18 3.7 18 3.6 3 69 8.6 92 21 21 5.4 25 1.9 3 70 9.2 87 18 16 3.4 20 1.5 3 71 2.0 88 14 16 2.6 19 1.9 3 76 1.4 94 13 16 2.6 20 2.2 4 77 2.6 89 21 16 3.5 17 3.6 3

78 4.2 80 20 14 3.6 18 3.0 3 79 5.0 97 21 20 5.0 24 2.8 3 80 19.3 94 18 22 4.3 23 3.2 3 86 32.0 96 13 19 2.8 20 1.8 4 87 11.3 100 11 19 2.7 22 0.5 3 89 3.2 91 13 18 2.6 21 1.1 6

90 3.4 87 18 19 3.5 20 2.6 3 91 13.4 96 16 22 3.8 23 2.4 7 92 15.5 93 16 20 3.4 21 2.2 7 93 5.1 97 12 14 2.4 18 1.7 3 94 3.6 52 19 24 4.7 24 2.1 3

Table I——Yield of Nuts and Kernels, and Cracking Quality of Nuts from 132 Sample Trees of Common Black Walnut in the Tennessee Valley (continued)

____

Tree _______________________

Diameter at 4-1/2 ____ Sample ft. number av. yr. 1940 1941 1942 1943 1944 1945 Av. yr. ____

inches pounds pounds pounds pounds pounds pounds pounds
96 16.5 23 31 93 51 29 103 55
97 9.8 2 8 9 7 4 6 6
98 21.3 44 20 66 35 26 4 32
100 27.8 159 272 65 334 6 80 153
101 21.2 0 294 120 206 30 239 148
102 13.1 38 2 44 4 12 3 17

103 7.5 20 15 25 30 9 119 36 104 12.3 40 17 52 17 16 0 24 106 11.4 50 16 66 29 46 66 46 107 13.2 29 0 5 8 0 1 7 108 9.0 34 11 12 25 12 7 17 109 12.6 11 12 30 69 0 14 23

110 14.9 65 104 29 61 54 32 58 111 11.3 8 55 5 65 0 54 31 116 11.8 0 16 6 7 4 9 7 117 17.0 10 285 13 142 116 387 159 118 13.3 3 78 6 170 4 263 87 119 14.6 0 34 148 0 40 145 61

121 17.6 67 9 41 15 0 64 33 129 13.3 13 70 8 157 0 149 66 130 15.3 47 1 50 10 0 24 22 131 16.2 78 1 33 89 0 69 45 132 14.2 6 8 22 10 0 17 10 134 13.3 9 20 11 17 24 3 14

135 14.1 12 55 0 15 0 94 29 136 15.1 7 1 18 14 0 2 7 137 9.4 27 0 38 13 5 28 18 138 14.5 36 18 28 35 69 8 32 139 10.2 14 9 19 64 51 0 26 140 11.1 0 18 62 53 28 34 32

Av. Filled nuts Complete crack
Kernel ___________________ ___________________________
yield First-
bearing In terms crack Crops
yrs. of total Average marketable Kernel tested
number only weight weight kernel weight Kernel Quarters basis

---

pounds percent grams percent grams percent number number

96 9.6 100 12 15 2.1 17 0.7 3
97 0.6 51 11 13 1.7 15 2.0 3
98 2.7 44 18 10 2.4 13 0.4 7
100 31.4 97 22 17 4.6 21 2.8 7
101 31.7 94 25 15 4.8 19 1.5 3
102 3.4 49 18 19 3.6 20 2.5 3

103 5.9 94 19 16 3.7 20 0.9 3 104 4.4 84 15 15 2.5 17 0.8 3 106 6.2 81 18 15 3.1 17 1.4 3 107 2.1 78 13 14 2.1 16 0.7 4 108 3.4 99 16 17 3.4 15 0.9 3 109 5.0 98 14 14 2.6 18 0.6 3

110 9.1 77 23 14 4.8 20 1.1 3 111 7.7 100 15 17 3.0 21 0.6 3 116 1.4 100 15 16 2.7 18 0.7 3 117 32.2 86 29 15 5.7 20 1.5 7 118 15.2 96 19 12 3.3 18 0.2 3 119 12.2 72 20 18 4.0 20 1.6 3

121 7.1 92 16 17 3.0 19 0.7 3 129 14.2 98 16 15 3.0 18 0.8 3 130 4.4 97 13 14 2.2 16 1.3 4 131 10.2 95 19 17 3.8 20 1.7 3 132 2.7 98 17 16 3.5 21 1.0 3 134 1.7 75 16 14 2.6 16 0.7 3

135 3.4 58 20 15 3.3 16 1.5 3
136 1.8 41 10 13 1.7 17 0.1 3
137 2.7 66 15 17 3.0 20 0.7 3
138 2.7 49 16 15 2.6 19 0.6 4
139 8
140 7.6 92 13 19 2.9 22 0.8 3
199 13.3 15 4 2 2 6
200 10.4 18 17 1 1 9
201 13.1 30 28 23 117 50
202 15.1 2 4 14 0 5
203 13.7 13 30 8 21 18
205 22.6 56 34 33 77 50

206 9.3 46 26 39 4 29 207 5.8 1 0 9 1 3 208 10.4 2 8 4 19 8 210 6.6 35 0 15 0 12 211 12.6 2 4 3 1 2 214 13.1 32 11 19 24 22

215 6.9 3 5 6 0 4 216 10.8 0 6 2 5 3 217 19.1 111 12 62 25 48 218 7.1 18 0 1 0 5 219 12.0 5 13 26 14 14 220 10.7 13 0 8 6 7

221 6.4 0 0 0 3 1 222 15.3 29 6 6 7 12 223 19.2 22 3 6 0 8 224 13.9 53 11 16 29 27 225 16.8 16 57 27 48 37 226 15.6 119 26 101 13 65

227 6.6 9 12 0 33 14 228 7.3 4 9 0 2 4 231 18.4 74 41 0 184 75 232 21.1 47 0 0 180 57 236 22.3 8 204 0 120 83 237 20.3 121 29 86 95 83

240 6.6 5 7 3 13 7 241 13.0 50 24 44 2 30 242 6.4 11 8 10 1 8 243 22.0 82 0 13 11 26 246 21.1 93 220 52 216 145 247 19.1 2 57 17 1 21

199 1.8 49 19 22 4.8 25 3.0 4 200 1.7 24 17 16 3.3 19 2.0 3 201 10.9 100 17 19 3.8 22 2.2 3 202 0.4 46 21 14 4.1 20 1.0 3 203 4.3 97 18 23 4.4 25 3.1 3 205 6.4 19 11 22 2.6 23 4.0 1

206 4.2 98 19 12 2.7 15 2.2 3 207 0.7 21 17 15 2.8 16 2.0 2 208 1.1 66 16 22 3.8 24 2.4 3 210 4.8 98 17 14 3.2 19 0.7 3 211 0.3 83 11 11 1.6 15 0.2 3 214 3.0 87 18 16 3.1 17 2.7 3

215 0.9 100 13 18 2.7 20 0.7 3 216 0.7 97 11 13 1.7 16 0.6 4 217 12.1 93 18 21 4.4 25 0.8 3 218 1.7 100 18 17 3.0 17 2.6 2 219 2.5 94 11 17 2.0 18 0.8 3 220 0.8 61 20 13 3.2 16 1.9 3

221 0.4 53 16 20 3.4 21 3.2 2 222 1.3 72 16 16 3.0 18 1.9 3 223 0.8 55 12 19 2.6 20 0.5 3 224 4.9 93 17 16 3.5 20 1.1 3 225 7.8 94 15 18 3.4 22 1.0 3 226 9.8 96 12 12 1.9 16 0.2 3

227 4.2 99 16 19 3.6 23 0.7 3 228 1.1 99 15 21 3.3 22 2.3 3 231 7.3 52 10 17 2.3 19 1.1 3 232 26.6 98 19 17 4.5 24 2.0 3 236 10.5 26 15 17 3.0 19 0.5 2 237 16.1 100 18 14 3.5 19 1.5 3

240 1.2 100 14 15 2.5 17 0.9 3 241 4.6 96 14 15 2.3 16 1.2 3 242 1.4 98 13 17 2.5 20 0.7 3 243 4.9 98 14 12 2.0 14 1.3 3 246 29.6 98 14 16 2.9 21 0.7 3 247 1.2 23 15 12 1.7 15 0.2 3

Results of cracking tests show that, in general, cracking quality of nut samples from the trees in this study is poor. When cracked, the kernels crumble badly, making extraction difficult and quarter recovery low. Variation in cracking quality can be seen by studying the values in Table 1. Nuts from trees 28 and 136 were extremely small, averaging 9 and 10 grams, respectively. Nuts from trees 61 and 98 had generally poor characteristics. Trees bearing walnuts of better-than-average quality are trees 5 and 18 with high total kernel per cent, and trees 8, 16, and 59 with high nut weight and an unusually high kernel weight. Other trees, of interest as exceptional bearers, include tree 101 with large nut weight, and tree 117 with both exceptional nut and kernel weight. The outstanding tree in the study from the standpoint of cracking quality of the nuts is tree 13, which has exhibited those characteristics of thinness of shell and high kernel content sought for in improved varieties. This black walnut selection is being propagated at the Norris Nursery under the appropriate name of Norris.[14]

[Footnote 14: Kline, L. V. A method of evaluating the nuts of black walnut varieties. Proc. Amer. Soc. Hort. Sci. 41:136-144. 1942.]

Results from this study on the common black walnut have application in the evaluation of the relative yield and nut quality of improved selections suitable for use in the Tennessee Valley. This summary should also prove of value to other workers dealing with black walnut in other regions. It provides a basis for comparison, brings out the possibilities for making selections, and emphasizes the importance of nut production from improved varieties.

The 1946 Field Tour

By C. A. Reed

Attending the indoor sessions of the meeting for two days in Wooster, visiting the Station orchards and plantings near town and contacting personally some of the big men of the Staff together with the wives of some, called for intensive attention on the part of everybody. It was time exceedingly well spent and created a feeling in everybody that they would like soon to return for another convention of the same kind. But the good things that had been planned were not over when the delegates left on the morning of the third day in the general direction of their homes. No matter in what direction they went, hardly a route could be found which did not lead near or through the home town of some nut man.

A few took opportunity to visit the planting near Wooster of the late W. R. Fickes. A letter is before my eyes as these lines are being written which was directed to Dr. W. C. Deming by Mr. Fickes on January 9, 1924, in which he asked for information regarding certain kinds of nut trees which he did not have. He mentioned having Beaver, Fairbanks, and Siers hickory hybrids and asked about Weiker. He wanted to know about Barcelona and White Aveline filberts. He said he had procured seven varieties of filbert of European origin which were then being featured by Conrad Vollertsen of Rochester, N. Y. He was concerned over the chestnut weevil as he had about 125 trees of the Reihl varieties from Illinois and already weevils were troublesome.

Those who had the privilege of keeping in touch with Mr. Fickes during his later years know that he assembled together a good many varieties of other kinds of nuts. His was an excellent collection of black walnut varieties. Persons who knew him well still mourn his passing. He was the type of man who made others feel better to be in his presence.

It was 24 years ago last February that the American Nut Journal, then edited and published by R. T. Olcott of Rochester, N. Y., told of "x x the 57-acre farm of O. F. Witte near Amherst (in northern Ohio), on which Mr. Witte, who was then 72 years old, had been growing nuts for 52 years." The dispatch went on to say that the "x x farm was devoted exclusively" to nut trees. What a pity such men can't live on indefinitely! However, the spirits of Fickes and Witte live on. No one need go far in Ohio to see the evidence.

Going east from Wooster on the morning of the third day, a group of 50 or more persons stopped first at Kidron where they were shown the nut plantings of Mr. E. P. Gerber and his family of that small hamlet. A half mile north of town, Mr. Gerber led the party through his largest planting of nut trees mostly of bearing age. Of black walnuts he showed such varieties as Deming (purple foliage, especially in early spring), Lamb (the original tree had a figured grain), Ohio, Stabler, Ten Eyck, and Thomas. Of pecan, there were five varieties, Busseron, Butterick, Greenriver, Indiana and Posey. In the group of heartnuts, there were two named varieties, Bates and Faust, and one of which Mr. Gerber appeared not to have the name. He simply called it a "sport." There were filberts of various kinds, Barcelona, DuChilly and Jones Hybrids, being the ones bearing variety designations. Also there were Persian (English) walnut trees, principally Broadview and Crath. Mr. Gerber had more Chinese chestnut seedlings than trees of any other one kind. There was but one butternut and that appeared to have been unnamed. Altogether 40 black walnut trees, 20 pecan, 30 filbert, 20 Persian walnut, one butternut, and 140 Chinese chestnut trees were seen.

Upon finishing with the first block of trees, the party was taken into town where a large business house of Gerber and Sons was passed and a short visit paid to a second planting in the rear of various Gerber buildings, including the residence of Mr. Gerber. Here were some two or three dozen fine appearing trees of various species and hybrid forms.

Lastly at Kidron, the party, was piloted a half mile west to a small park which Mr. Gerber had developed as a public picnic ground and a source of water for the village. It was well planted with nut trees and it was here that the Gerber family had provided tables and various food delicacies, including fresh milk, peaches and ice cream for everybody. A great part of the work of preparation had been taken care of by Mrs. Gerber and her two youngest children.

The next stop on the tour was at the Mahoning County Experiment Farm, a half-mile south of Canfield, some 70 odd miles east and north of Wooster. Here transportation was provided and the entire group was taken in charge by L. Walter Sherman, Superintendent. The first impression one gained here was that of good buildings, excellent land, able management, and a lot of things under way. All is comparatively new. From a mimeographed list of species, varieties, hybrids, and strains which was prepared in June for another occasion, one gathered that there were perhaps more seedling nut trees here than grafted kinds. Mr. Sherman has reported fully elsewhere in these Proceedings regarding the nut work that is under way at this Station.

Report of the Resolutions Committee

The Northern Nut Growers Association in its annual meeting assembled at Wooster, Ohio, September 3rd to 5th, 1946, adopted the following resolutions:

> That our sincere thanks be extended to Dr. Edmund Secrest, Director of the Ohio Agricultural Experiment Station and other members of his staff for the courtesies extended, and for the

facilities provided in the use of the auditorium and exhibit room of the Station.

That we extend thanks to the speakers who unitedly made a successful meeting.

That we appreciate the fine work of our Secretary, Miss Mildred M. Jones, in formulating the program and that we are mindful of the valuable assistance rendered by Dr. Oliver Diller, Mr. Clarence A. Reed, and Mr. A. A. Bungart.

That we acknowledge appreciation to the estate of the late Zenas H. Ellis for providing in his will a gift of one thousand dollars to a special fund of the Association and that we thank Mr. Sargent H. Wellman for his legal efforts therewith.

That the members of the Northern Nut Growers Association fully appreciate and extend sincere thanks to our officers for their hard work and enthusiastic efforts in maintaining the Association during the past five years when war conditions precluded annual meetings.

RESOLUTIONS COMMITTEE
C. F. Walker, Chairman
J. L. Smith
Albert B. Ferguson

Obituaries

DR. J. H. GOURLEY

Members of this Association who attended the Wooster meeting in 1946 will not soon forget the cheery, witty and resourceful toastmaster who presided at their annual banquet, Dr. Joseph Gourley. Soon after this meeting, on October 19th, to be exact, Dr. Gourley was stricken with coronary thrombosis, and the field of horticulture lost a nationally known leader.

Dr. Gourley's passing came at a time of high tide in his work. "Less than an hour before he was stricken," said an associate, "he was engaged in planning a project that he knew would continue long after his active career must end. This is the spirit of the true research man."

He was a graduate of Ohio State University, had served as head of the Department of Horticulture in the University of New Hampshire and later in a like position with the University of West Virginia. In 1921, he was appointed chief of the Department of Horticulture at the Ohio Experiment Station and, from 1929, he concurrently held the position of Chairman of the Department of Horticulture at Ohio State University. He served both of these offices until the day of his death. He was the author of many bulletins and technical articles as well as of some better known text books which have had wide use in American Universities. He had acted as president of The American Society for Horticultural Science, President of The American Promological Society, and as president and member of numerous similar organizations to which he gave continued and enthusiastic service.

It is as a good teacher, companion and warm friend, however, that Dr. Gourley will best be remembered by those who knew him well. His life and fire have sparked many another teacher, research worker and common man to greater effort and better achievement. A close associate closed a press notice of Dr. Gourley's passing with these words:

"His consideration for his associates, both those equal and below in rank, marked his every contact through his long years of service. He was indeed, a truly great Chief.

His family and close associates in the two departments he headed for so many years will miss him most of all, but life for them and for countless others who called him friend has been made richer, fuller and deeper because he passed this way.

Teacher, scientist, Christian gentleman, friend and chief, we salute you."

* * * * *

MRS. I. E. BIXBY

Mrs. Ida Elise Bixby, wife of the late Willard G. Bixby, died at her home at Baldwin, New York, April 29, 1945.

Mrs. Bixby was a life member of the Northern Nut Growers Association, of which her late husband was a past president. Following Mr. Bixby's death in August, 1933, Mrs. Bixby interested the United States Department of Agriculture in taking over much of their large experimental planting of nut trees. Many specimens were moved to experiment stations under Government control, while other institutions as well as individuals benefitted by their collection.

Mrs. Bixby is survived by three children: Willard F., of Cleveland, Ohio; and Katherine Elise and Ida Tielke, of Baldwin.

Letters to the Secretary; Notes; Extracts

EXCERPT FROM LETTER TO SECRETARY FROM G. S. JONES

July 4, 1946. From G. S. Jones, R 1, Box 140, Phenix City, Lee County, Alabama.

My trees (Chinese chestnuts) appear to be healthy and grow vigorously. (They were given me by the Bureau of Plant Industry in 1934.) They began bearing in 5 or 6 years and have now been bearing quite large crops for 3 or 4 years. There are 22 trees in the orchard, and the approximate yields have been: 1943—550 pounds; 1944—450 pounds, and 1945—950 pounds. The enormous increase in 1945 was due partly, I am quite sure, to mineralized fertilizer (Es Min. El.) which I began using in 1944.

As my trees are seedlings they vary considerably in productivity and in size of nuts. Most of the nuts are of good size and quality when first gathered. This is where the trouble begins. The keeping quality is very poor, sometimes half of them spoil during the first month after being harvested. Since this is the case, you can see that germination may be very poor, unless they are handled in a special way. Refrigeration helps for a short while only. During the last two years, I have had good results in germination by stratifying the nuts under the trees, just as soon as they fall. In this way, the nuts are not allowed to become too dry as they are not exposed to the hot sun but are kept in the shade. Our falls are usually dry and our soil is sandy so there is little danger of the nuts becoming too wet during the winter. The danger of spoilage does not seem to be so great by the time winter rains set in. By this plan, I have had from 60 to 90 per cent germination during the last two years. I dig the nuts just as soon as they begin to sprout in late winter and line them out in nursery rows where they are to grow during the first year. Sometimes the sprouts become from 4 to 6 inches in length before I get to do the moving, but they transplant easily. I believe the micorrhiza from the soil of the old trees helps the young ones to grow better.

December 11, 1946—My chestnut trees this fall produced slightly over 1,722 pounds. The nuts seemed to keep better than usual which I attribute to the cool rainy weather which we had during the ripening period. Hot, dry weather causes the nuts to begin spoiling quickly. My records show August 7th as the beginning of the ripening period and October 3rd as the ending. So one can see that this is often a hot and dry period in our section.

* * * * *

EXCERPT FROM LETTER TO SECRETARY FROM MRS. W. D. POUNDEN

Dairy Department—Ohio Agric. Expt. Sta. Wooster, Ohio
October 14, 1946

I am glad to give you the method I used in canning pecan kernels.

Spread the shelled pecans in a shallow pan and place in a warm oven just long enough to heat the kernels through. Have clean jars—preferably pints so that the heat will penetrate more easily in processing—which have been warmed in the oven to be sure they are thoroughly dry inside before adding the pecans. Fill the jars with the pecans (do not add any liquid), place the lid on the jar (I prefer the Kerr self-sealing type), and process the nut-filled jars in a 250 deg. oven for 30 minutes.

I have kept pecan meats for over a year using this method and they are as crisp and good as when they came out of the shell.

HYBRIDS

At an informal meeting at Dr. Diller's cabin the evening before the Convention, Mr. Slate was asked to say something about hybrids.

Mr. G. L. Slate: Hybrids between black and Persian walnuts were made at Geneva about 1916 by Professor W. H. Alderman, now of the Minnesota Experiment Station. After these trees had fruited all but five were removed to permit the remaining trees to attain full size. The trees have produced very few nuts and have been absolutely no good. Various persons have attempted to raise second-generation seedlings from these trees, but from my observation no one has succeeded.

From what I know of these hybrids and what Reed has published about those with which he is familiar I am convinced it a waste of time and effort to attempt to produce hybrids between black and Persian walnuts with the hope of getting desirable nuts. The trees themselves are very rapid growing, handsome and well worth while as shade trees. But the walnut breeder will have more to show for his efforts if he confines his labor for the time being to improving the black and Persian walnuts by crossing among themselves the many clones within each species.

However, the unsatisfactory hybrids between black and Persian walnuts, of between butternuts and Persian walnuts should not blind us to the fact that there are many species-hybrids of great pomological value. The hybrids between the Rush variety of Corylus americana and various varieties of C. avellana produced by the late J. F. Jones are very much worth while. Some of our finest red raspberry varieties are hybrids of the European and American species.

The purple raspberry resulted from crossing the red and black raspberries. All our cultivated strawberries are descended from crosses between the native Virginia strawberry and the Chilean strawberry. The valuable new plums from the Minnesota Experiment Station resulted from crossing the native American plum, Prunus americana with the Japanese plum, P. salicina. Many of our best grapes, the Boysenberry, the Kieffer pear, and various citrus varieties are species hybrids.

We must not generalize too much as to the merit or lack of merit of species-hybrids. Some are very good and of great economic importance. Many others of which we never hear are without merit, often being discarded, leaving only a few lines in a notebook to record their characteristics.

* * * * *

Mr. Stoke: Would you consider chestnut hybrids worth while?

Mr. Slate: If you can get everything you need from the Chinese chestnut I see no reason for hybrids with any other.

Mr. Stoke: Dr. Arthur S. Colby has made a number of hybrids between Fuller and Chinese. I consider his hybrid No. 2 as promising; the nut is large, beautiful and of good quality. So far I have found no weevils in this hybrid. The bur is very thick and fleshy, with close-set spines. Possibly the curculio is not able to penetrate the thick husk in laying its eggs. Colby No. 2 is the most rapid grower of all my chestnuts.

PECANS WITH COMPANION EVERGREENS[15]

Twenty years of experimenting with pecan trees at the Iowa Park station have revealed that pecans in the Wichita irrigated valley of Texas do very poorly in buffalo grass or Bermuda sod, much better when given clean cultivation, but best of all when planted with or near evergreens, particularly conifers.

[Footnote 15: Forty-Eighth Annual Report, Texas Agricultural Experiment Station. P. 42. 1945.]

In 1926 some pecan trees were set along the west line of the farmstead. Most of these died soon after setting and the few that survived did not grow satisfactorily. Later, a general farmstead improvement program called for Arizona cypress along this line. In 1933, when these pecan trees were seven years old, they had made little growth and were in such poor condition that it was decided to ignore them and set the cypress on equal spacings. Some of the cypress trees were placed very near pecan trees while others were farther away. None of the pecans were removed, however.

As the cypress trees grew, the pecan trees near them began to take on new life, while the isolated pecan trees continued in their unthrifty state. As the years passed the pecans with companion cypress trees continued to increase in health and vigor until there was no doubt about the favorable influence of this companionship. At the time the cypress trees were set close to the older pecans, other pecan trees were being set in various locations on the farmstead; some in open sod and others with or near evergreens of various types. The behavior of these trees also confirms the value of companion evergreens for pecans in the Wichita irrigated valley.

At the age of seven years the pecan trees were about the same size and in equally poor condition. The treatment as far as cultivation and irrigation is concerned has been the same. Hence, the great contrast in size of the pecan trees is attributed to the favorable influence of the companion conifers.

[NOTE BY EDITOR—Heavy shade can reduce soil temperature, on summer afternoons, more than 20 deg.F six inches underground. This may largely explain the benefits of companion trees.]

* * * * *

SAWDUST MAKES GOOD FRUIT TREE MULCH

Many kinds of material ranging from paper to glass wool have been used as mulches for fruit trees, discloses J. H. Gourley, of the Department of Horticulture at The Ohio Agricultural Experiment Station. Straw, hay, and orchard mowings have been most commonly used.

In some areas, sawdust and shavings are available in quantity and have been used to some extent for mulches which raises the questions of whether they make the soil acid.

The Experiment Station has used both hardwood and pine sawdust and also shavings for a number of years in contrast with wheat straw, alfalfa, timothy, and others. No difference in appearance or behavior of the trees can be noted. Sawdust packs and gives poorer aeration than straw and it requires a large amount to mulch a tree. This mass also absorbs a large amount of rainfall before passing through to the soil but no injurious effects have been noted.

The chief question has been about soil acidity and it may be stated that after 12 years of treatment the soil is little or no more acid than it is under bluegrass sod. The soil under the latter has a pH of 5.22, the hardwood sawdust 5.07, the softwood sawdust 5.07, hardwood shavings 5.20, and wheat straw 5.35. Contrary to the common conception, no objection to sawdust from the standpoint of soil acidity is justified from Station experience.

* * * * *

TWO FAMOUS TREES

(Taken from "Bruce Every Month," December, 1938, page 17. Published by E. L. Bruce Company, Memphis, Tennessee.) Living monuments to a great governor of Texas are two nut bearing trees, a pecan and an old fashioned walnut. The last wish of Governor James S. Hogg was that "no monument of stone or marble" be placed at his grave, but instead there should be planted—"at my head a pecan tree and at my feet an old fashioned walnut; and when these trees shall bear, let the pecans and walnuts be given out among the plains people of Texas so that they may plant them and make Texas a land of trees." His wish has been fulfilled in its entirety, many trees from these two parent ones adorning the lawns of schools and court houses throughout the State of Texas.

* * * * *

OHIO TREES SERIES

No. 1.—Black walnut (Juglans nigra):—Black walnut is one of the most valuable of the forest trees native to the United States. It is regarded as the country's premier tree for high grade cabinet wood; it produces valuable nut crops; and under certain conditions is highly effective as an ornamental shade and pasture tree.

~Lumber~—As lumber, black walnut is used principally for furniture, radio cabinets, caskets, interior finish, sewing machines, and gun stocks. It is used either in the form of solid wood cut from lumber or in the form of plywood made by gluing sheets of plain or figured veneer to both sides of a core. Black walnut veneer is made by the slicing method and to a limited extent by the rotary-cut method.

~Nuts~—In recent years the black walnut has gained an important position in the kernel industry. There has never been a market surplus of black walnut kernels. The demand, mostly from confectioners and ice cream manufacturers, has steadily increased while the supply has been limited largely by the labor of cracking and extracting the kernels. The process of cracking the nuts and separating the kernels from the shells has been mechanized by a farmer in Adams County, Ohio, to the extent that he uses over 4,000 bushels of walnuts per year. He sends the kernels to markets in New York, Baltimore, Pittsburgh, Columbus, and Chicago. The facts all emphasize the economic importance of the black walnut in a market that is still far from saturated.

~Ornamental Value~—There are few trees whose utility is as great as the black walnut, that can rival it in beauty as a lawn tree. Its long graceful leaves provide a light dappled shade and grass will grow luxuriantly up to the very base of the tree. In its pleasing form and majestic size the black walnut can be a great addition to any landscape. Any tree yielding such fine timber and nuts, yet possessing beauty and utility for yard and pasture, can be nothing but a sound investment.

~Soil Requirements~—Black walnut grows best in valleys and bottom lands where there is a rich, moist soil but well drained. It does not generally grow on the higher elevations nor on wet bottom lands. It usually occurs as a scattered tree in hardwood stands and along roadsides, fence rows, and fence corners.

~Distribution and Growth~—The botanical range of this tree covers most of the eastern half of the United States. It is among the more rapid growing hardwoods. On good sites trees 10 years old will be about 20 feet high and in 40 years will reach 60 feet in height and 12 inches in diameter at breast height. According to Forest Survey figures, the estimated merchantable stand of walnut in Ohio in 1941 was 112,275,000 board feet while the cut during the same year was slightly over 3 million board feet.

~Pests~—The most serious pest is the walnut datana whose larvae eat the leaves. Other leaf-eating insects include the fall web worm and the hickory-horned devil. Several leaf spot diseases have attacked the leaves, also causing early defoliation. Leaf eating insects and leaf spot disease can be controlled by the application of one spray in June. This is composed of three pounds of arsenate of lead, ten pounds of powered Bordeaux mixture, and a good sticker in one hundred gallons of water.

~Selected Varieties~—Walnut trees vary greatly in the type of nut they produce. The most popular strains have been selected for propagation. The varieties which have been propagated by nurserymen are the Thomas, Ohio, Stabler, Ten Eyck, and Elmer Myers. Since the cost of grafted nut trees is rather high, many people are interested in planting the nuts of the better varieties for large scale planting. Seedling trees may be raised easily by anyone, whereas much skill and practice are required to produce grafted and budded trees. The degree to which the desirable characteristics of selected varieties are transmitted through seed is now being studied by the Ohio Agricultural Experiment Station.

A list of commercial nut nurseries may be obtained by writing to Miss Mildred Jones, Secretary, Northern Nut Growers Association, Lancaster, Pennsylvania.

~References~—A few of the most outstanding publications on black walnut are listed below.

1. Black walnut for timber and nuts. Farmers' Bulletin No. 1392, U. S. Dept. of Agriculture, Washington, D. C.

2. Nut Growing in New York. Bulletin 573, College of Agriculture, Ithaca, New York.

3. Top-working and Bench Grafting of Walnut Trees. Special Circular 69, Ohio Agricultural Experiment Station, Wooster, Ohio.

4. Growing walnut for profit. The American Walnut Manufacturers Association, 616 South Michigan Avenue, Chicago 5, Illinois.

Exhibits

Gilbert Becker, Climax, Michigan.

Crath strains of J. regia, hickory, black walnut kernels.

Hebden H. Corsan, Hillsdale, Michigan.

Cases of nuts, folders on nut planting for success.

H. F. Stoke, Roanoke, Virginia.

Chinese chestnuts, hybrid chestnuts, tree hazel hybrid, Jones hybrid filberts, hazelberts, black walnuts, E. Golden persimmons, J. regia, hickories, nut ornaments.

Edwin W. Lemke, Washington, Michigan.

Heartnuts, black walnuts, filberts, tree hazels, black walnut wood, a vacuum nut cracker.

Jay L. Smith, Chester, New York. Books, black walnuts, hickories, chestnuts, hacksaws, grafts.

E. J. Korn, Kalamazoo, Michigan.

J. nigra, hickories, filberts.

S. H. Burton, Indiana.

Petrified nuts, wild hazels.

Harry R. Weber, Cincinnati, Ohio.

Breslau Persian walnuts, filberts.

E. P. Gerber, Kidron, Ohio.

Photos, hickories, chestnuts, hicans, black walnuts.

U. S. D. A., Beltsville, Maryland.

Green hickory nuts of several varieties.

A. G. Hirschi, Oklahoma City, Oklahoma.

Heartnuts, J. regia, persimmons, chestnuts.

U. S. D. A., Beltsville, Maryland.

35 large pictures of famous nut and other trees fully described; many other smaller photos of famous trees remarkable for clearness.

John Davidson, Xenia, Ohio.

Cross-sections of seedling black walnut. A very remarkable exhibit of thin-shelled black walnuts.

Dr. A. S. Colby, University of Illinois, Urbana, Illinois.

A very desirable Crath (seedling I believe) Persian walnut.

Fayette Etter, Lemasters, Pennsylvania.

Large number of filbert varieties.

Attendance

Dr. and Mrs. Truman A. Jones, Farkesburg, Penna.
Geoffrey A. Gray, Cincinnati, O.
John W. Hershey, Downingtown, Penna.
Mr. and Mrs. C. A. Reed and Miss Betty Reed, Washington, D. C.
Mrs. G. A. Zimmerman, Linglestown, R. I, Penna.
S. B. Chase, Norris, Tenn.
Thomas G. Zarger, Norris, Tenn.
W. A. Cummings, Norris, Tenn.
Mr. and Mrs. John Davidson, Xenia, O.
H. C. Cook, Leetonia, O.
Mr. and Mrs. H. F. Stoke, Roanoke, Va.
Mr. and Mrs. Raymond E. Silvis, Massillon, O.
Victor Brook, Rochester, N. Y.
D. Ed. Seas, Orrville, O.
L. H. MacDaniels, Ithaca, N. Y.
C. F. Walker, Cleveland, O.
Mr. and Mrs. J. F. Wischhusen, Cleveland, O.
Mr. and Mrs. S. H. Graham, Ithaca, N. Y.
Dr. R. H. Waite, Perrysburg, N. Y.
Kenneth W. Hunt, Yellow Springs, O.
J. F. Wilkinson, Rockport, Ind.
William S. Clarke, Jr., State College, Penna.
Ira M. Kyhl, Sabula, Io.
Edwin W. Lemke, Detroit, Mich.
William C. Hodgson, White Hall, Md.
J. H. Gourley, Wooster, O.
H. R. Gibbs, McLean, Va.
Mr. and Mrs. S. Bernath, Poughkeepsie, N. Y.

Mr. and Mrs. Carl Weschcke, St. Paul, Minn.
Joseph M. Masters, Wooster, O.
George L. Slate, Geneva, N. Y.
George H. Corsan, Toronto, Ont.
Mrs. Katherine Cinadr, 13514 Coath Ave., Cleveland 20, O.
O. D. Diller, Wooster, O.
Emmet Yoder, Smithville, O.
F. L. O'Rourke, E. Lansing, Mich.
R. E. McAlpin, E. Lansing, Mich.
G. J. Korn, Kalamazoo, Mich.
L. W. Sherman, Canfield, O.
H. H. Corsan, Hillsdale, Mich.
J. L. Smith, and daughter, Chester, N. Y.
A. J. Metzger, Toledo, O.
A. W. Weaver, Toledo, O.
S. Shessler, Genoa, O.
A. A. Bungart, Avon, O.
Sterling A. Smith, Vermilion, O.
C. P. Stocker, Lorain, O.
Dr. and Mrs. John E. Cannaday, Charleston, W. Va.
Andres Cross
Mr. and Mrs. R. B. Best, Eldred, Ill.
G. M. Brand, Lincoln, Nebr.
Wm. M. Rohrbacher, Iowa City, Io.
D. C. Snyder, Center Point, Io.
Wm. N. Neff, Martel, O.
E. P. Gerber, Kidron, O.
Geo. Kratzer, Dalton, O.
A. G. Hirschi, Oklahoma City, Okla.
Mr. and Mrs. Harry R. Weber, Cincinnati, O.

Mr. Ford Wallick, Peru, Ind.
Carl Prell, S. Bend, Ind.
Albert B. Ferguson, Center Point, Io.
E. F. Huen, Eldora, Io.
John B. Longnecker, Orrville, O.
Percy Schaible, Upper Black Eddy, Penna.
Ruth Schaible, Upper Black Eddy, Penna.
Mr. and Mrs. Blaine McCollum, White Hall, Md.
Mrs. H. Negus, Mt. Ranier, Md.
Dr. Elbert M. Shelton, Lakewood, O.
H. M. Oesterling, Harrisburg, Penna.
Frank M. Kintzel, Cincinnati, O.
Dr. J. W. McKay, Plant Industry Station, Beltsville, Md.
Dr. A. S. Colby, University of Illinois, Urbana, Ill.
E. C. Soliday, Lancaster, O.
L. E. Gauly, Cleveland, O.
Mrs. Reuben Bixler, Apple Creek, O.

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